

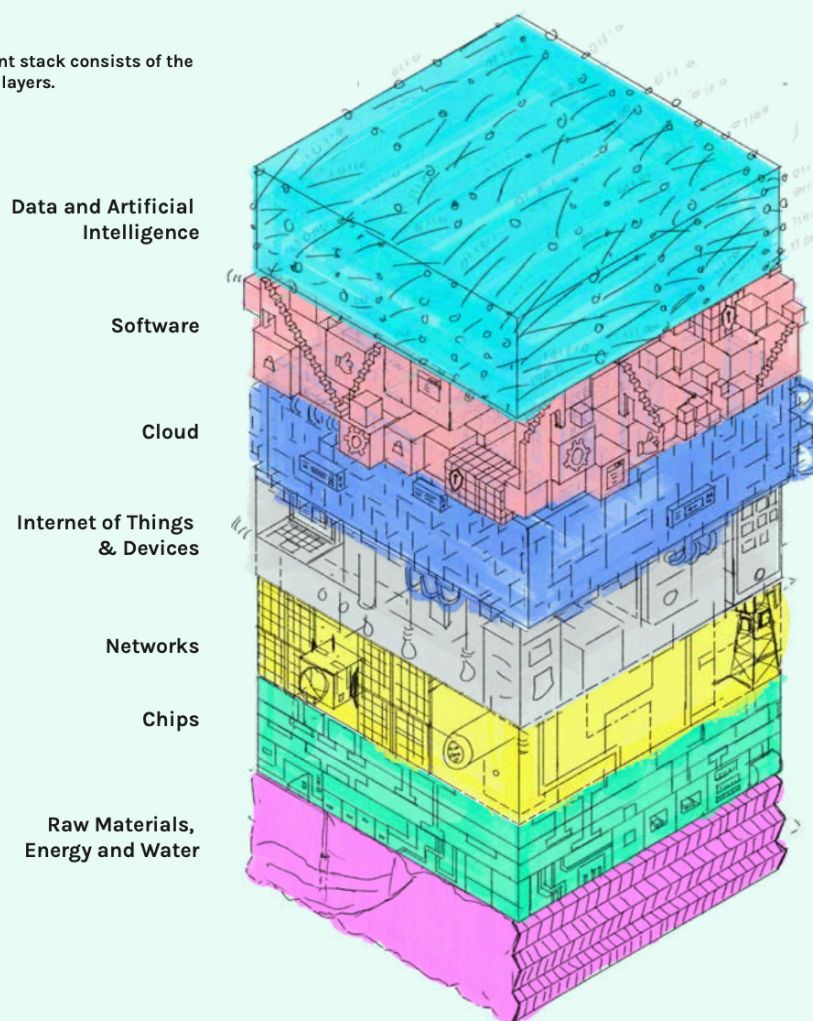
EuroStack

A European Alternative for Digital Sovereignty

The Economics of Digital Strategic Autonomy

The current stack

The current stack consists of the following layers.



February 2025

Commissioned by:

Bertelsmann Stiftung

Supported by:

Mercator Stiftung, CEPS, UCL IIPP

Project Leadership and Coordination: Francesca Bria, Senior Fellow, Mercator Stiftung; Hon. Professor, UCL IIPP

Lead Authors:

Francesca Bria, Senior Fellow, Mercator Stiftung, Hon. Professor, UCL IIPP

Paul Timmers, Professor, KU Leuven, WeltWert®, Research Associate Oxford Internet Institute

Fausto Gernone, UCL IIPP

Credits for the EuroStack Paper

Main Contributors:

Andrea Renda, CEPS
Haroon Sheikh, Fellow at the Netherlands Scientific Council for Government Policy (WRR)
Pierre Bitard, ANRT - Association Nationale de la Recherche de la Technologie
Cecilia Rikap, UCL IIPP
Derk Oldenburg, WeltWert®
Georg Serentschy, WeltWert®
Christopher Fischer, CEPS
Olesya Grabova, CEPS

Infographic and Visuals: Studio Dirma Janse

Reviewers:

Prof. Henning Kagermann, former President of Acatech; former CEO, SAP
Cristina Caffarra, Competition and Antitrust Expert
Pierre Pezziardi, Interministerial Digital Directorate (Direction interministérielle du Numérique)
Prof. Rainer Kattel, Deputy Director, UCL IIPP
Prof. Luc Soete, Director, United Nations University
Rafael Laguna and Jörg Resch, SPRIND (German Federal Agency for Disruptive Innovation)
Stefan Heumann, Director, Agora Digitale Transformation (ADT)
Prof. George Danezis, Information Security Group, Imperial College London
Prof. Daniele Archibugi, Research Director, Consiglio Nazionale delle Ricerche (CNR)
Frank Rieger, Chaos Computer Club
Udbhav Tiwari, Mozilla
Prof. Cristian Hesselman, Director SIDN Labs; University Twente
Volker Stocker, Weizenbaum Institute for the Networked Society, TU Berlin
Axel Voss, MEP, EPP Group
Kai Zenner, EPP Group
Alexandra Geese, MEP, The Greens
Matthias Ecke, MEP, S&D Group

Contributors:

Alberto Marti, Open Nebula, IPCEI Next Generation Cloud Infrastructure and Services (CIS)
Alek Tarkowski, Director of Strategy, Open Future
Paul Keller, Director of Policy, Open Future
Adriana Groh, Co-founder, Sovereign Tech Agency
Sophie Bloemen, Commons Network
Vittorio Bertola, Open-Xchange
Francesco Bonfiglio, Dinamo Cloud
Sebastiano Toffaletti, Secretary General (CEO), European DIGITAL SME Alliance

INTRODUCTION	4
Building the EuroStack – European Alternative for Digital Sovereignty.....	4
The Case for the EuroStack: From Vulnerability and Dependency to Competitiveness and Autonomy.....	6
PART 1 The Eurostack	13
A Vision Tailored for Europe	16
Principles of the EuroStack.....	17
Europe’s Common Digital Stack	20
The EuroStack Industrial Challenge: Minimum Viable Products (MVPs) as Highly Scalable Digital Services.....	24
A Growing Innovation Ecosystem.....	28
PART 2 The Political Economy of Digital Sovereignty: Unpacking Power in the Stack.....	30
Dynamics of the Stack	34
Raw Materials, Energy and Water: The Hidden Backbone of Technology.....	39
Chips: The Brain of the Stack.....	41
Networks: The Lifelines of Connectivity.....	43
Internet of Things: The Connective Tissue of the Digital Ecosystem	45
Cloud: The Operational Core of the Digital Economy	49
Software: The Command Centre of Platforms and Applications	55
Data and Artificial Intelligence: Catalysts for Innovation and Strategic Autonomy.....	61
Cybersecurity: The Cornerstone of Digital Resilience and Autonomy	67
Defence: Strengthening Strategic Capabilities	68
Quantum Technologies: Unlocking Tomorrow’s Potential.....	69
Financial Power: The Driver of Digital Dominance.....	71
The Geopolitical Dimension: Shaping Europe’s Multipolar Vision.....	76
PART 3: European Digital Industrial Policy Blueprint	79
What Worked, What Failed, What’s Next	79
Policy recommendations.....	87
Implementation Roadmap.....	94
PART 4: Readings	98
Annex A – Strategies of Dominance Among Selected Big Tech	100
Annex B Ongoing EU Policy actions and lead stakeholders	103

INTRODUCTION

Building the EuroStack – European Alternative for Digital Sovereignty

Europe is at a crossroads in a world undergoing rapid and dramatic change. Technological innovation is accelerating, global supply chains are fragmenting, and economic dependencies are being weaponized in geopolitical rivalries.

The global digital technology ecosystem is fiercely competitive and profoundly techno-political. Digital technologies define geopolitical power, economic competitiveness, and scientific excellence. Europe faces significant technological dependencies and vulnerabilities in this landscape. Addressing these challenges is critical to securing resilient supply chains, safeguarding energy systems, protecting critical infrastructures, and strengthening cybersecurity. Europe must respond not merely by addressing dependencies but by unifying its fragmented efforts under a cohesive and ambitious vision akin to the Common Market initiative of the past¹.

In this report, we outline the **EuroStack Initiative** as a key policy strategy for Europe. Coined by one of the authors of this report², the term has since united a coalition of European stakeholders during a Brussels event, backed by a multi-party initiative in the European Parliament³.

The **EuroStack Initiative** represents such a vision—not simply a technological program but a political framework designed to ensure Europe’s relevance and leadership in the global digital economy. Just as the **Common Market** galvanized Europe’s economic integration, the EuroStack seeks to provide a bold, unifying strategy. The initiative builds upon Europe’s values of shared governance, subsidiarity, and solidarity to modernize and reorient its approach to digital sovereignty and urges policymakers to take bold action to realize this vision, address today’s evolving threat landscape and secure Europe’s digital strategic autonomy.

This report seeks to analyse Europe’s strategic opportunities in an era of technological disruption, proposing the development of a **Common Digital Stack** as the cornerstone of Europe’s competitiveness in a rapidly evolving global economy. This initiative aligns with the principles of the **Competitiveness Compass**, providing provides a cohesive framework to unify existing initiatives, set clear standards and principles, and chart a path for future actions. By integrating technological infrastructure with Europe’s core political values and industrial capabilities, the Common Digital Stack will serve as a catalyst to enhance economic resilience, close innovation gaps, and ensure global leadership in critical sectors.

The Need for a Common Digital Stack: Beyond Dependency, Toward Integration

The European Union’s current position in the global technological landscape is characterized by significant dependencies. **Over 80% of Europe’s digital technologies are imported.**⁴ These

¹ Enrico Letta, ‘Much More than a Market Report’ (European Council, 10 April 2024), https://single-market-economy.ec.europa.eu/news/enrico-lettas-report-future-single-market-2024-04-10_en.

² Francesca Bria, ‘Open, Sovereign, Independent AI: Europe’s Greatest Challenge?’, *Medium* (blog), 10 December 2023, <https://medium.com/@francescabria/open-sovereign-independent-ai-europes-greatest-challenge-6c8a899041ec>.

³ Francesca Bria, Cristina Caffarra, and Meredith Whittaker, ‘Toward European Digital Independence Brussels 24th September 14.30-18.30’ (Brussels, 24 September 2024), <https://digitalindependenceeu.wordpress.com/>.

⁴ Mario Draghi, ‘The Future of European Competitiveness – A Competitiveness Strategy for Europe’, 9 September 2024, https://commission.europa.eu/topics/strengthening-european-competitiveness/eu-competitiveness-looking-ahead_en.

dependencies primarily involve the United States and China, as well as major tech companies based in these countries:

- **Digital Infrastructure:** Three US-based companies—Amazon, Microsoft, and Google—dominate almost 70% of the European cloud computing (IaaS) market, while Europe's largest cloud provider holds a mere 2% share.⁵
- **Research & Development (R&D) and Patenting:** EU firms represent just 7% of R&D expenditure in software and internet, with the US at 71% and China at 15%. In electronics, EU firms account for 12% of R&D, compared to 40% for the US, and 19% each for China, Japan and South Korea^{6 7}.
- **Critical Raw Materials:** Essential resources such as lithium, cobalt, nickel, gallium, graphite, and tungsten are vital for batteries, semiconductors, aerospace, and other industries. These markets are highly concentrated, with China controlling about 90% of global rare earth refining capacity.
- **Microchips:** Europe consumes about 20% of the world's microchips but manufactures only 9%. The European Chips Act aims to double the EU's global advanced semiconductor market share to 20% by 2030, addressing shortages and reducing reliance on foreign suppliers.⁸ However, this goal faces significant challenges, exemplified by Intel's delayed €30 billion chip factory in Germany due to financial and market conditions.⁹
- **Artificial Intelligence (AI):** Since 2017, 70% of foundational AI models have been developed in the United States, with this figure continuing to grow.¹⁰

While these dependencies expose vulnerabilities, the challenge is compounded by fragmented and undercoordinated efforts in Europe's digital ecosystem. Unlike the economic coherence provided by the Common Market, Europe's growing R&D spending and innovation efforts are not aligned by an overarching vision that leverages synergies or amplifies impact.

This contributes to Europe's **innovation gap**¹¹, which led to slower productivity growth relative to the US, reflecting weaknesses in translating R&D into market success and lower levels of investment. Corporate R&D investment remains concentrated in 'mid-tech' industries, particularly automotive. Of the world's 50 leading technology companies, only 4 are European, and none of these were created in the past fifty years. The digital economy's rapid growth has shifted the global R&D landscape, with the EU lagging compared to the United States and China¹².

This **EU underinvestment** is mirrored by the difficulties start-ups face in scaling within Europe. Between 2008 and 2021, nearly 30% of Europe's "unicorns" relocated their headquarters abroad,

⁵ Synergy Research Group, 'Cloud Market Gets Its Mojo Back; AI Helps Push Q4 Increase in Cloud Spending to New Highs', accessed 7 January 2025, <https://www.srgresearch.com/articles/cloud-market-gets-its-mojo-back-q4-increase-in-cloud-spending-reaches-new-highs>.

⁶ European Investment Bank, 'Chapter 5, Progress on Digital Transformation', in *Resilience and Renewal in Europe*, Investment Report 2022/2023 (Luxembourg: European Investment Bank, 2023), 175–215.

⁷ European Commission. Joint Research Centre., *The 2021 EU Industrial R&D Investment Scoreboard*. (Luxembourg: Publications Office, 2022), <https://data.europa.eu/doi/10.2760/559391>.

⁸ European Commission, 'A Chips Act for Europe', 8 February 2022, <https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX%3A52022DC0045&qid=1664190455505>.

⁹ Christopher Rauwald and al., 'Intel's German Plant Delay Lands a Blow to EU's Chip Ambitions', *Bloomberg.Com*, 17 September 2024, <https://www.bloomberg.com/news/articles/2024-09-17/germany-faces-fresh-setback-as-intel-halts-planned-chip-factory>.

¹⁰ LEAM.AI, 'Large AI Models for Germany – Feasibility Study 2023', accessed 11 January 2025, <https://leam.ai/feasibility-study-leam-2023/>.

¹¹ Mario Draghi, 'The Future of European Competitiveness – A Competitiveness Strategy for Europe'.

¹² European Commission (2022b). *Science, research and innovation performance of the EU 2022: Building a sustainable future in uncertain times*. Luxembourg: publications Office of the European Union.

predominantly to the United States, highlighting structural barriers to retaining high-growth companies. Acquisitions by non-EU players are eroding Europe's digital platform presence. Non-EU entities account for 19% of global online platform acquisitions of EU companies, while EU residents acquire only 6% of companies outside the bloc¹³.

Europe is **geopolitically and geo-economically highly exposed**. COVID-19, the Ukraine war, and the semiconductor shortages and rising energy prices demonstrate dependencies on global supply chains. Disruptions severely impact industries like automotive manufacturing¹⁴ and healthcare¹⁵.

Europe's vulnerabilities and critical foreign dependencies in the digital world lead to:

- The erosion of Europe's ability to compete, innovate, and act autonomously, loss of domestic job opportunities, brain drain, and underinvestment, falling behind in technological breakthroughs. This translates into a lack of competitiveness, ultimately leading to a general deterioration of standards of living.
- The imposition of unwanted rules of foreign countries, shaped by their national security requirements, affecting control of Europeans over sensitive information of citizens, companies, and governments.
- IP theft and espionage create critical vulnerabilities, including reduced resilience and the risk of prepositioning within critical infrastructures for potential future disruption.
- Market dominance with cases of abuse of market power, the need to spend efforts of regulators and competition authorities on keeping the EU Single Market functioning properly.
- A lack of the ability to protect democracy or even facilitating the undermining of democracy.

The Case for the EuroStack: From Vulnerability and Dependency to Competitiveness and Autonomy

Digital autonomy is no longer a luxury—it is the foundation of sovereignty. Europe must act decisively to reduce reliance on external providers for critical digital infrastructure, safeguarding its competitiveness, strategic autonomy, and democratic governance. The European Commission's **Digital Compass** outlines the Union's digital ambitions for 2030, while the **Digital Decade Program** provides a roadmap to achieve these goals. Together with the upcoming **Competitiveness Compass**, these frameworks guide Europe in building a resilient digital ecosystem that aligns with its values, secures technological leadership, and ensures long-term economic strength.¹⁶

EuroStack presents a **bold and balanced vision** for Europe's digital future. This initiative positions digital sovereignty not as an isolationist pursuit but as a cornerstone of deeper integration, ensuring

¹³ Mario Draghi, 'The Future of European Competitiveness – A Competitiveness Strategy for Europe'.

¹⁴ Roberto A. De Santis et al., 'Motor Vehicle Sector: Explaining the Drop in Output and the Rise in Prices', 10 November 2022, https://www.ecb.europa.eu/press/economic-bulletin/focus/2022/html/ecb.ebbox202207_02~5bde8eeff0.en.html.

¹⁵ OECD and European Commission, *Health at a Glance: Europe 2024: State of Health in the EU Cycle*, Health at a Glance: Europe (OECD, 2024), <https://doi.org/10.1787/b3704e14-en>.

¹⁶ European Commission (2022a). "Towards a green, digital and resilient economy: Our European Growth Model." COM(2022) 83 final.

Europe's collective strength in the face of shared challenges. Just as the **Common Market** laid the foundation for economic interdependence and stability, the **EuroStack** will enable Europe to modernize its economy and digital infrastructures while safeguarding democratic principles. The EuroStack is not merely about reducing dependencies but about **enhancing the interconnectedness and competitiveness of the European economy**. The integration of digital infrastructure will ensure that the Single Market remains robust and adaptive to 21st-century challenges. Complete self-sufficiency is neither feasible nor desirable in an interconnected world, but by building the capabilities and control necessary to protect its interests and those of its Member States, Europe can create a resilient digital ecosystem that still benefits from global exchanges.

Europe faces mounting risks from external dependencies and internal fragmentation. Job losses, diminished value creation, and a weakening ability to shape its own destiny under more grim geopolitical conditions underscore the *existential* nature of the challenge. **The need to move from urgency to action is clear**. The EuroStack strategy begins by leveraging existing strengths and policy instruments, while laying the groundwork for a long-term commitment that will take over a decade to fully realise. It balances immediate action with structural sustainability.

Achieving greater digital sovereignty through the EuroStack isn't just about control—it's about boosting competitiveness and narrowing the innovation gap with other regions. By reducing reliance on imported technologies and integrating common tech solutions with Europe's industrial core, the EuroStack aims to lower adoption costs, streamline operations, and accelerate the spread of high-impact technologies. Just as the creation of the Euro required a vast technical infrastructure, the EuroStack necessitates a similarly comprehensive approach to digital systems. This will boost efficiency, expands Europe's capabilities and encourages indigenous innovation.

The EuroStack promotes broader participation addressing issues of concentration and **fostering genuine value creation**. As smaller firms gain access to advanced tools, and as data and AI insights flow freely across sectors, the entire economy benefits. This more balanced power dynamic translates into smarter production processes, more responsive public services, richer product offerings, and powerful knowledge spillovers – addressing the innovation gap with US and China. In short, the EuroStack model transforms a system skewed toward value extraction into a vibrant ecosystem built for creating value in Europe—ultimately strengthening Europe's autonomy, resilience, and sustainable growth.

Strategic autonomy cannot be achieved in isolation. The resources, expertise, and networks required to build EuroStack are beyond the capacity of any single region. Moreover, isolationism and protectionism are counterproductive to innovation, sustainable and inclusive growth. Instead, EuroStack champions openness, knowledge sharing, and inclusivity, forging alliances to co-develop and co-govern the technologies of the future, in the framework of digital public goods and with full respect for sovereignty in Europe. These partnerships should be grounded in shared values and aspirations. Specifically, they must rest on a foundation of mutual agreement regarding economic empowerment, social rights, and democratic principles.¹⁷

In order to pursue this vision of strategic autonomy and promote opportunities for Europe, and given the current serious unbalances and dependencies, **EuroStack considers that a “Europe-first”**

¹⁷ This approach aligns with the European Pillar of Social Rights, which underscores the importance of social inclusion and fairness, See: European Commission, *The European Pillar of Social Rights Action Plan* (Luxemburg: Publications Office, 2021), <https://data.europa.eu/doi/10.2767/111056>; as well as with the inclusive prosperity across the Union, as put forward in the Letta report, see: Enrico Letta, 'Enrico Letta's Report on the Future of the Single Market - European Commission', 10 April 2024, https://single-market-economy.ec.europa.eu/news/enrico-lettas-report-future-single-market-2024-04-10_en.

approach is necessary, that is, as a matter of principle purchasing European solutions, notwithstanding exceptions that can only be justified if realistic and objective assessment is provided of the risks for strategic autonomy in the short and long run.

To realize this vision, Europe needs a **joined-up industrial strategy**¹⁸ that integrates all relevant policy domains, spanning market access, standardization, research and development, public procurement, investment, trade, international cooperation, and both inbound and outbound investment control. Coordination and alignment must come with a mandate to drive synergy within the block. Learning from the global competitors China and the United States and from countries such as Brazil and India, EuroStack seeks to demonstrate how comprehensive and coordinated industrial policies can drive progress and respond to Europe's specificities. This also means learning from past failures and setting a modern, proactive industrial policy that avoids fragmentation, fosters strategic collaboration, and ensures long-term resilience and competitiveness in the global digital economy. Past industrial policies in Europe have often lacked the coherence and strategic focus needed to drive competitiveness, leading instead to inefficiencies and a failure to close the innovation gap, as highlighted by Draghi in his competitiveness report.

Inclusivity, paired with speed and agile execution, is central to the EuroStack strategy. By uniting industry, Member States, the tech sector, and civil society, it strengthens both national and EU sovereignty. A successful industrial strategy requires recognizing the shifts it will create, with new winners and losers across countries, regions, sectors, and individuals. To harmonize this transformation, Europe must incentivize today's leaders to support change while investing in lagging regions through cohesion funds, skills programs, and training initiatives. Engaging trade unions as partners is essential to empower workers as drivers of change, ensuring a just and inclusive transition. This approach balances economic competitiveness with social equity, reinforcing Europe's digital

EuroStack Summarized: Investment and Strategic Impact

The **EuroStack Initiative** represents Europe's ambition to achieve digital strategic autonomy through a **total investment of €300 billion over ten years**.

This effort begins with the creation of a **European Sovereign Tech Fund**, starting with an initial **€10 billion** dedicated to the development of **Digital EuroStack demonstrators**. These demonstrators—selected via open competition through the **EuroStack Challenge**—will serve as minimum viable products (MVPs) to showcase Europe's ability to innovate **and scale foundational digital technologies**.

By consolidating and aligning ongoing efforts across public and private sectors, the EuroStack Initiative underscores Europe's commitment to reducing dependencies, fostering industrial champions, and driving competitiveness in critical technologies such as AI, semiconductors, cloud infrastructure, and IoT. The EuroStack strategy supports:

- **Economic Resilience:** Creating high-skilled jobs, reducing dependencies, and supporting critical industries like healthcare, manufacturing, and energy.
- **Strategic Leadership:** Positioning Europe as a leader in the world in values-driven, citizen-centric digital innovation.
- **Technological Independence:** Building a sustainable, interoperable digital ecosystem rooted in European values of privacy, transparency, and trust.

¹⁸ Francesca Bria, 'X. European Digital Independence: Building the EuroStack', *AI Now Institute* (blog), 15 October 2024, <https://ainowinstitute.org/publication/x-european-digital-independence-building-the-eurostack>.

sovereignty and shared progress. Even with a robust industrial strategy and strong partnerships, critical dependencies will remain. These can be weaponized in times of geopolitical tension, posing risks to Europe's stability and security. Technological disruptions further complicate the landscape. To mitigate these risks, the EuroStack strategy incorporates **proactive monitoring and contingency planning**, and the agility to adapt to unforeseen challenges. This also reinforces democratic accountability about EU resources.

Ultimately, **EuroStack is designed to strengthen Europe's competitiveness, security, and democracy**. By addressing the needs of industry and investing in foundational public digital infrastructure, the strategy ensures that Europe is well-positioned to compete in global markets while safeguarding its citizens and democratic values.

This is more than a policy framework—it **is a call to action**. It reflects Europe's commitment to shaping a digital future aligned with its democratic ideals, economic aspirations, and global responsibilities. Through unified, principled action, EuroStack positions Europe as a leader in the digital age, prepared to confront challenges and seize opportunities on the global stage. The actions required are vast and complex, but they are possible and necessary.

Summary of Main Actions: A Radical Vision for Europe's Digital Competitiveness

The **EuroStack Initiative** represents a bold and transformative strategy to secure Europe's digital sovereignty and competitiveness. By addressing structural weaknesses that have historically hindered Europe's digital economy, EuroStack aims to turn the continent's unique strengths into a cohesive, scalable, and innovative digital ecosystem, starting from the strengths present in Europe's key industrial sectors, such as manufacturing, biotechnology, and healthcare. Recognizing the increasing returns to scale in digital products and services—where dominance in unified markets like the US and China has outpaced Europe—EuroStack proposes coordinated actions to make Europe a global leader in technology within the next ten years.

EuroStack recognizes that incremental reforms are insufficient. Europe's past policies failed to fully capitalize on the potential of the Single Market. For instance, the liberalization of national telecom markets did not create a unified European telecom market. Similarly, Europe's early leadership in mobile telephony did not translate into success in digitalization or platform development due to fragmented governance and misaligned priorities. Recent challenges, like the limited success of Gaia-X in achieving cloud sovereignty, highlight the persistent issues of competing national agendas, unnecessary red tape, lack of scalability, and lack of common vision. These failures mirror earlier struggles in coordinating microelectronics and enabling platform firms to thrive, leaving Europe vulnerable to external dominance.

To address these structural shortcomings, EuroStack proposes **radical reforms** including:

- **Reimagining Market Harmonization:** Move beyond fragmented national efforts to create a truly unified European digital market with a common digital stack that integrates platforms, innovation, procurement, and investment.
- **Championing Strategic Coordination:** Establish governance frameworks that align member states, industries, and investors toward long-term digital sovereignty, prioritizing collective goals over short-term national interests.
- **Scaling Through Federation:** Embrace a federated approach that balances regional autonomy with shared objectives, leveraging the benefits of global supply chains while retaining European control and independence.

Seizing Opportunities Through European Digital Sovereignty

Europe stands at a pivotal moment, with the potential to become a global leader in the digital age by embracing digital sovereignty. The EuroStack initiative offers a blueprint for building a resilient, competitive, and inclusive digital ecosystem, tailored to European values and strengths. Rather than focusing solely on risks, this summary highlights the significant opportunities that arise from achieving greater digital autonomy.

- **Unlocking Innovation and Growth:** By reducing reliance on imported technologies and fostering a vibrant open-source community, the EuroStack can ignite innovation across Europe. Investing in key areas like AI, cloud infrastructure, and quantum technologies will create new markets, high-quality jobs, and long-term economic growth. The EuroStack is not just about control; it is about boosting competitiveness and narrowing the innovation gap with other regions.
- **Strengthening European Industry:** The EuroStack provides a unique opportunity to revitalize European industries, particularly in manufacturing, healthcare, and biotechnology. By integrating digital solutions with Europe's industrial core, the EuroStack can create more efficient processes, secure supply chains, and new business models. The initiative promotes the development of "Made in Europe" digital products and services, fostering regional value creation and strengthening Europe's position as a global manufacturing hub.
- **Leading in Ethical and Sustainable Technology:** Europe has the opportunity to lead the world in the development and deployment of ethical and sustainable technologies. By prioritizing data sovereignty, transparency, and environmental responsibility, the EuroStack will create a trustworthy digital ecosystem that aligns with European values. The focus on renewable energy sources, water efficiency, and responsible data governance will ensure that Europe's digital transformation is both innovative and sustainable.
- **Empowering Citizens and Businesses:** Through the development of secure, interoperable digital public infrastructure, the EuroStack empowers citizens and businesses. Digital identity systems, secure payments, and data platforms will create a more inclusive and efficient society. By promoting digital skills and fostering a culture of innovation, the EuroStack will enable all Europeans to benefit from digital opportunities.
- **Shaping a Digital Foreign Policy:** Europe's unique approach, balancing strategic autonomy with international partnerships, allows it to set international standards for ethical, inclusive, and sustainable digital leadership. By fostering cooperation with like-minded countries and promoting open-source technologies, and fair technology transfer, Europe can contribute to a global digital order that prioritizes public interest and resilience. The EuroStack is a model for a decentralized and collaborative digital future.
- **Leveraging Open Source and Collaboration:** The EuroStack champions an open-source innovation ecosystem. By leveraging Europe's strengths in open-source software and fostering a collaborative community of developers, researchers, and innovators, the

EuroStack will drive rapid innovation and long-term sustainability. The focus on open APIs, data sharing, and digital commons empowers European stakeholders and promotes collective growth.

- **Strategic Investments in Key Technologies:** The EuroStack involves strategic investments in key technologies, such as AI, cloud computing, quantum computing, and next-generation semiconductors. These investments, coupled with targeted public-private partnerships, will ensure Europe's leadership in high-impact fields.

Core Strategic Actions

1. Develop a European Common Digital Stack

Build interoperable, cyber-secure platforms for AI and cloud computing, eID, data access, and digital currencies like the Digital Euro. Unlike fragmented efforts of the past, this approach focuses on harmonized European platforms that rival global competitors in scale and impact. Focus on infrastructure includes also investment in public computing capacity (EuroHPCs) and next generation chips for AI development and uptake across key sectors.

2. Deploy High-Impact Digital Services

Launch Minimum Viable Products (MVPs) as scalable digital solutions in strategic areas such as AI-driven biotech, smart manufacturing, robotics, mobility, and healthcare. These unified, pan-European deployments will demonstrate Europe's capacity for high-value innovation.

3. Foster Sovereign AI and Federated Data Spaces

Invest in sovereign AI cloud, platforms and interoperable data ecosystems to support innovation and adoption in industrial sectors while maintaining ethical oversight and control. These initiatives will address past market fragmentation and promote scalability.

4. Lead in Next-Generation Technologies

Strengthen Europe's capabilities in edge computing, quantum technologies, industrial AI, space technologies, and biotechnology. Strategic investments will align with industrial priorities to secure leadership in cutting-edge fields and next-generation digital infrastructure.

5. Scale Innovation Through Strategic Investments

Establish a European Sovereign Technology Fund to close investment gaps and prevent foreign acquisitions of critical firms like ARM and DeepMind. This fund will ensure a robust financial ecosystem that sustains innovation and long-term competitiveness.

The EuroStack is not just about reducing dependencies; it is a proactive strategy to create a thriving digital future for Europe. By focusing on opportunities, embracing collaboration, and prioritizing European values, Europe can position itself as a global leader in the digital age. This strategy will unlock new possibilities for growth, innovation, and social progress, making Europe a strong and influential actor on the world stage.

Hitting the ground running

The **EuroStack Initiative** is well-positioned for a rapid start by leveraging Europe's existing assets, which align closely with its overarching objectives. Europe is fortunate to possess a wealth of resources that, while often fragmented and subscale, are both impressive and innovative.

These assets include a range of **world-class European companies** driving innovation across the tech stack. A representative sample of these companies is illustrated in Figure 5. Additionally, the initiative will capitalize on the extensive network of **current partnerships**, including public, private, and industrial partnerships, as well as ongoing initiatives. Further details can be found in Figure 5 and Annex B. By consolidating and scaling these strengths, the EuroStack Initiative can overcome fragmentation, maximize impact, and accelerate progress toward achieving Europe’s digital sovereignty.

Report Structure

This report outlines the EuroStack vision and provides a comprehensive roadmap for advancing Europe’s digital sovereignty, and it is organised as follows:

Part 1: The EuroStack Initiative

The first part outlines the vision and strategic objectives of the EuroStack initiative, focusing on leveraging Europe’s industrial strengths to create a common digital stack. It emphasizes the transformative potential of Minimum Viable Products (MVPs) as highly scalable digital services, selected for high-impact sectors such as advanced manufacturing, biotech, mobility, and healthcare. These sectors and their use cases are presented as illustrative examples, with a focus on identifying services or products that can drive joint European industrial missions. The initiative prioritizes fostering open-source innovation and agile development to accelerate collaboration and ensure the efficient execution of projects. It also introduces a governance framework designed to guarantee effective implementation, balancing flexibility with alignment to Europe’s values through a clearly defined execution model.

Part 2: The Political Economy of Digital Sovereignty

The second part situates EuroStack within the broader geopolitical and macroeconomic context, analysing the tech stack as a battleground where technology, strategy, and power converge. It examines how Big Tech firms are reshaping the stack through vertical integration and dominance strategies, while governments compete to control technologies critical to economic power and geopolitical influence. The analysis highlights Europe’s dependencies and bottlenecks in raw materials, semiconductors, networks, cloud computing, and AI, emphasizing the interplay between value creation and value capture. Building the EuroStack is presented as a strategic necessity—not only to integrate and strengthen the European Common Market but also to address global challenges and secure Europe’s competitiveness, security, and technological sovereignty in a rapidly evolving world.

Part 3: European Digital Industrial Policy Blueprint

The third part evaluates Europe’s industrial policy successes and failures, translating the EuroStack vision into actionable recommendations. It proposes a forward-looking digital industrial policy centred on coordinated investment, harmonized regulations, and public procurement that prioritizes European-made digital solutions. The blueprint emphasizes partnerships and robust governance frameworks, culminating in an agile implementation roadmap to position Europe as a global tech leader within the next decade. Crucially, EuroStack rejects more bureaucracy and instead advocates for an agile, innovative approach. This pragmatic strategy ensures efficiency, adaptability to geopolitical and technological shifts, and a focus on delivering tangible results that bolster Europe’s strategic autonomy and competitiveness.

Part 4: References and Annexes

The report concludes with references and two annexes. Annex A examines how major tech firms shape the digital stack to assert dominance and extract rents, while Annex B maps ongoing EU policy actions and key stakeholders, aligning these efforts with the EuroStack initiative.

PART 1 The EuroStack

An Infrastructural Layered Framework for Digital Sovereignty

Carlota Perez's techno-economic paradigms framework¹⁹ highlights that each technological revolution progresses through phases, from early innovation to widespread deployment. In the current ICT revolution, we are in the deployment phase, where digital technologies are becoming integral to all aspects of society and industry. These drive a growing demand for robust infrastructure like broadband, cloud computing, and data centres. Emerging technologies further amplify this need, making digital infrastructure a cornerstone of economic, industrial, and societal transformation.

The EuroStack framework organizes digital infrastructure into a cohesive system of interconnected layers, ranging from foundational technologies to advanced applications. It follows the stack model as a conceptual framework²⁰, where each layer builds upon the capabilities of the one below it, while all layers are dynamically interconnected, forming a network of interdependencies and interactions. This design allows for better visualization and targeting of specific areas of action.

Specialized "sub-stacks," such as cloud and AI, address specific needs but remain integral to the unified EuroStack. The framework prioritizes interoperability, resilience, scalability, and adaptability to ensure seamless functionality across the system. These layers also underpin sectoral applications in biotechnology, advanced manufacturing, and public services, helping Europe remain competitive and self-reliant in the global digital landscape. Moreover, the EuroStack includes cross-stack topics such as cybersecurity, emerging technologies and competitive shifts that can reshape its structure and create new strategic autonomy opportunities.

The layers of the Stack are represented as follows:

1. **Resources:** This foundational layer includes critical materials like rare earth elements, energy sources, and skilled workers, forming the bedrock for all digital infrastructure.
2. **Chips:** Processors and memory technologies, essential for powering digital infrastructure and ensuring secure supply chains, GPUs and emerging quantum communication.
3. **Networks:** Physical and digital connections, such as cell towers, fiber-optic networks, undersea cables, that link Europe to the global digital ecosystem.
5. **Connected Devices & IoT:** From smartphones and laptops to IoT devices, enabling real-time information processing and data collection.
4. **Cloud Infrastructure:** Secure data storage and computational power, distributed computing infrastructures which are central to data sovereignty and autonomy.
6. **Software Platforms, Applications, and Algorithms:** Operating systems, applications, and cybersecurity frameworks that drive digital interactions.
7. **Artificial Intelligence and Data:** This layer processes data and generates insights, positioning Europe to build and control core AI capabilities for a competitive edge.

The EuroStack, presented here in a layered framework, should also be envisioned as a dynamic circular ecosystem. Each element functions as a node, highlighting the interdependencies and interconnected nature of digital technologies. This approach emphasizes the need for adaptability, allowing the model to evolve as new technologies emerge or priorities shift. It also visualizes the strengths of European capabilities in each area while identifying potential vulnerabilities or investment needs to strengthen the ecosystem further.

¹⁹ Carlota Perez, 'Technological Revolutions and Techno-Economic Paradigms', *Cambridge Journal of Economics* 34, no. 1 (2010): 185–202.

²⁰ Haroon Sheikh, 'European Digital Sovereignty: A Layered Approach', *Digital Society* 1, no. 3 (18 November 2022): 25, <https://doi.org/10.1007/s44206-022-00025-z>.

The Economics of Digital Strategic Autonomy

The current stack

The current stack consists of the following layers.

Data and Artificial Intelligence

Software

Cloud

Internet of Things & Devices

Networks

Chips

Raw Materials,
Energy and Water

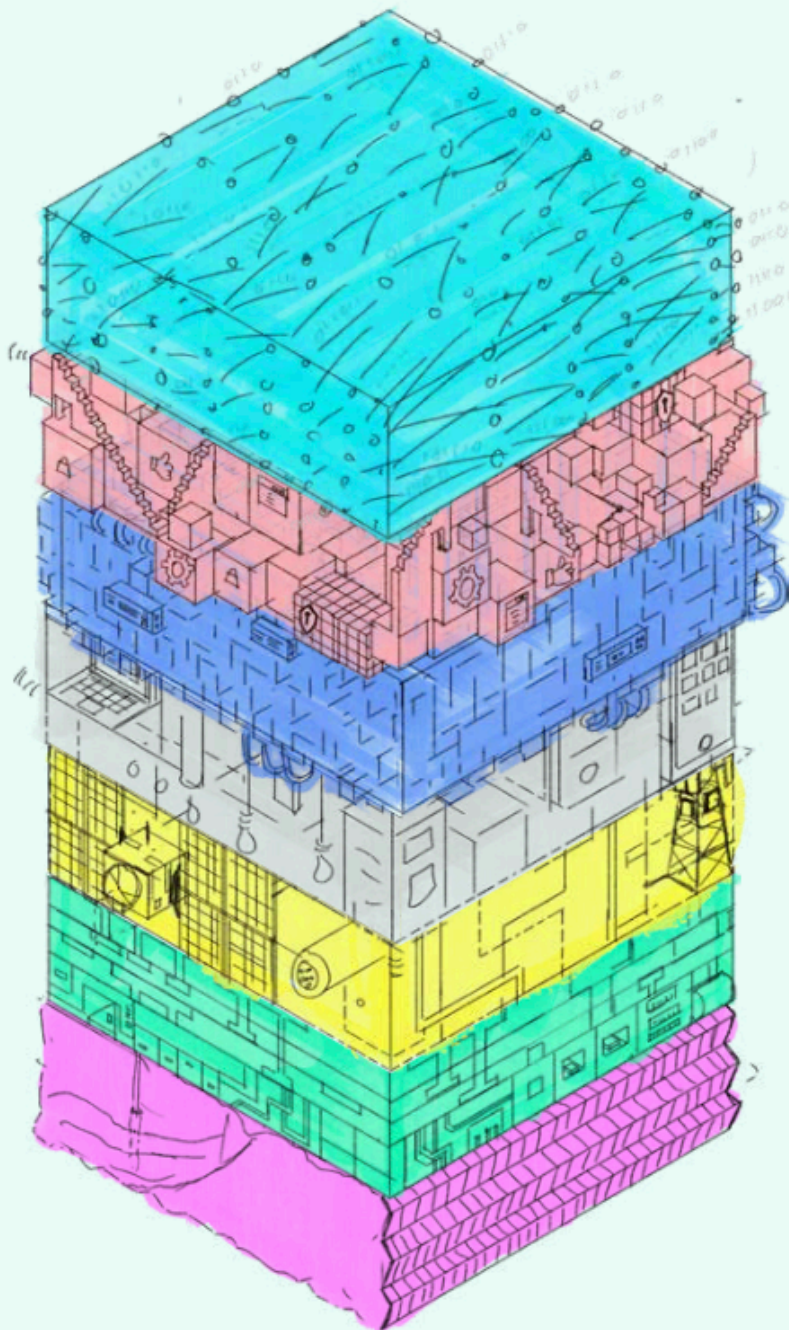


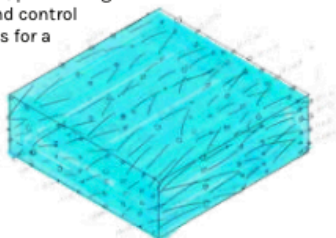
Figure 1. Technology Stack Model

The separate layers explained

Data and Artificial Intelligence

Catalysts for Innovation and Strategic Autonomy

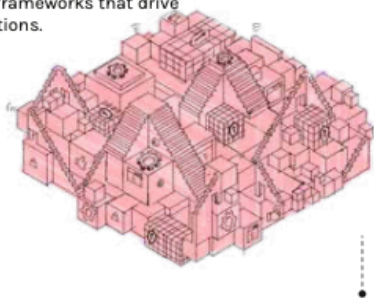
This layer processes data and generates insights, positioning Europe to build and control core AI capabilities for a competitive edge.



Software

The Command Center of Platforms and Applications

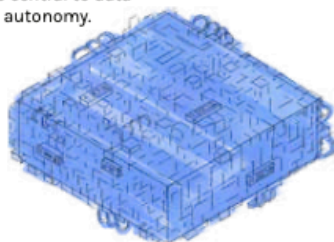
Operating systems, applications, and cybersecurity frameworks that drive digital interactions.



Cloud

The Operational Core of the Digital Economy

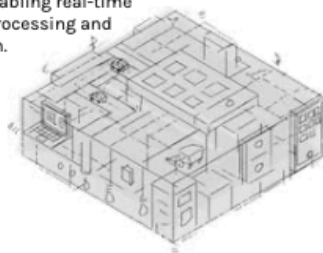
Secure data storage and computational power, which are central to data sovereignty and autonomy.



Internet of Things & Devices

The Physical Interface of the Stack

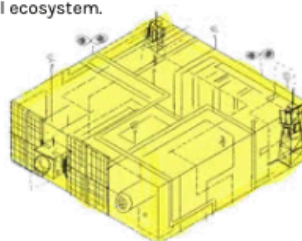
From smartphones and laptops to IoT devices, enabling real-time information processing and data collection.



Networks

The Lifelines of Connectivity

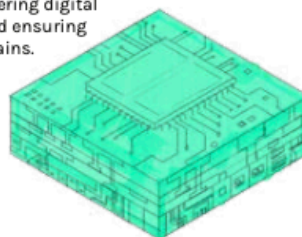
Physical and digital connections, such as cell towers, undersea cables, and emerging quantum communication, that link Europe to the global digital ecosystem.



Chips

The Brain of the Stack

Processors and memory technologies, essential for powering digital infrastructure and ensuring secure supply chains.



Raw Materials, Energy and Water

The Hidden Backbone of Technology

This foundational layer includes critical materials like rare earth elements, energy sources, and skilled workers, forming the bedrock for all digital infrastructure.

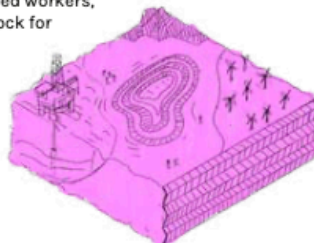


Figure 2. Separated Layers of the Stack

A Vision Tailored for Europe

The EuroStack offers a bold and pragmatic vision for Europe's digital future, grounded in its industrial strengths, innovative capabilities, and unwavering commitment to democratic values. In a world marked by increasing geopolitical fragmentation, **the EuroStack champions a federated and decentralized approach.** This model not only reflects Europe's diversity but also ensures resilience by fostering collaboration and aligning efforts across the Single Market.

Unlike the centralized dominance of US Big Tech or China's state-driven strategies, the EuroStack leverages Europe's unique position within global supply chains. By embracing federation, Europe can maintain the benefits of international cooperation and interconnected markets while building the autonomy needed to safeguard its strategic interests. This approach balances regional autonomy with shared objectives, enabling Europe to navigate a multipolar world while driving innovation and sustaining its competitive edge.

Europe has solid assets to build upon. Global leaders like ASML and SAP exemplify industrial and technological excellence, while innovative firms like ARM and DeepMind showcase Europe's potential for ground-breaking advancements—despite their acquisition by foreign investors. Advanced research centres and industries with strengths in advanced manufacturing, networking, robotics, chips manufacturing, and biotech provide a solid foundation for Europe to reclaim leadership in science and technology.

The EuroStack is designed to leverage these strengths to reduce reliance on foreign technologies, nurture homegrown innovation, and revitalize Europe's industrial ecosystem. By focusing on interoperability, security, and sustainability, it aligns technological progress with societal needs while **fostering inclusive growth and global competitiveness.**

This vision is actionable and grounded in **clear principles.** Joining-up initiatives like EuroHPC, IPCEIs, and the Quantum Flagship within a cohesive framework can streamline innovation and accelerate the development of scalable platforms. The EuroStack prioritizes open-source technologies and collaboration across member states, ensuring that Europe's technological backbone reflects its values of transparency, accountability, and inclusivity.

The EuroStack Initiative represents a distinctly European approach to achieving digital sovereignty. It provides the structure and governance needed to unify industry, policymakers, and investors, driving the development of critical technologies and reducing market fragmentation.

By addressing vulnerabilities across the tech stack and building on its industrial capabilities, the EuroStack positions Europe to compete globally. This is not about excessive regulation but about empowering Europe's industrial champions and fostering innovation, ensuring that the continent leads in a changing global economy while maintaining its commitment to democracy and sustainability.

At its core, the EuroStack Flagship is built around **six key pillars**:

1. **A Vision Tailored for Europe:** A bold yet achievable vision to increase EU digital strategic autonomy, that draws on Europe's unique strengths—its diversity, democratic governance, regulatory leadership, and emphasis on industrial innovation and competitiveness. This vision is grounded in clear principles that ensure feasibility and alignment with Europe's long-term priorities.
2. **A joined-up modern industrial policy:** investment, market regulation, R&D, standardisation, trade and competition policy, international partnerships and other relevant policies are designed to consistently reinforce each other.
3. **The EuroStack Digital Infrastructure:** The backbone of Europe's digital services, hardware, and software, designed to connect citizens, businesses, and governments through secure and interoperable systems, make European innovation viable and democratize access to computing. It integrates critical elements like networks, chips, cloud, IoT, data platforms, and AI, built with an emphasis on sustainability, resilience, and sovereignty.
4. **Minimum Viable Products (MVPs):** A set of scalable, **interoperable digital services and apps** designed to succeed in the Single Market. These are operational trailblazers and will meet the immediate needs of citizens and businesses while demonstrating the feasibility and value of Europe's digital sovereignty strategy.
5. **A Growing Open Source Community:** At the heart of the EuroStack is an open ecosystem of developers, researchers, SMEs, industry, and innovators across **software, hardware, and AI**. This community will collaborate with industry, member states, European institutions, and investors to build the EuroStack from the ground up, ensuring transparency, innovation, and adaptability.
6. **A Sustainable Governance Model:** To ensure long-term impact, the EuroStack Initiative will adopt an accountable and independent governance framework. This framework will bring together **public and private stakeholders** to harmonize policies, monitor progress, and secure investments, ensuring the sustainability and growth of this initiative.

Principles of the EuroStack

The EuroStack is grounded in seven guiding principles:

1. **Sovereignty and Security:** Guaranteeing that Europe's critical digital infrastructure is under European jurisdiction, protected by robust security-by-design and privacy-by-design.
2. **De-proprietarisation and Interoperability:** Promoting integration across an open source, federated tech stack, while reducing reliance on Big Tech proprietary solutions.
3. **Sustainability:** Building energy-efficient and resource-resilient systems to meet Europe's Environmental and Climate objectives.
4. **Data as a Public Good:** Treating data as a shared resource to unlock innovation while safeguarding societal interests and fundamental rights.
5. **Decentralized Sovereign Infrastructure:** Combining edge computing and centralized systems for improved efficiency and data sovereignty.
6. **Inclusive Governance:** Ensuring harmonized regulations, accountability, and a balance between short-term resilience and long-term autonomy.
7. **Strong Democracy:** Digital technologies that do no harm, but fundamentally strengthen democratic societies.

These principles ensure that the EuroStack is **ethical, resilient, and adaptable**, creating long-term value for stakeholders.

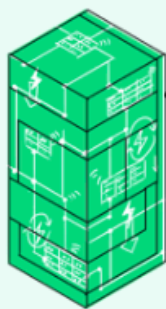
The Eurostack

Core principles



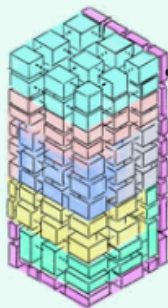
Sovereignty and Security

Guaranteeing that Europe's critical digital infrastructure is under European jurisdiction, protected by robust security-by-design and privacy-by-design.



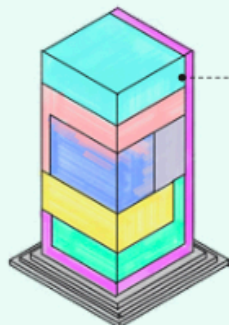
Sustainability

Building energy-efficient and resource-resilient systems to meet Europe's Environmental and Climate objectives.



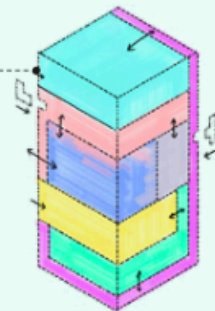
Decentralized Sovereign Infrastructure

Combining edge computing and centralized systems for improved efficiency and data sovereignty.



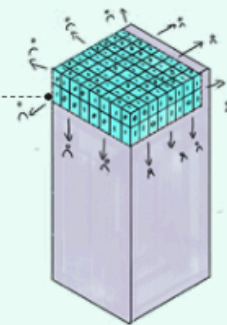
Strong Democracy

Digital technologies that not only do no harm, but fundamentally strengthen democratic societies



De-Proprietarization and Interoperability

Promoting integration across an open source, federated tech stack, while reducing reliance on Big Tech proprietary solutions.



Data as a Public Good

Treating data as a shared resource to unlock innovation while safeguarding societal interests and fundamental rights.



Inclusive Governance

Ensuring harmonized regulations, accountability, and a balance between short-term resilience and long-term autonomy.

Figure 3. Core Principles of the EuroStack

The Eurostack

Key components/common services?

This robust infrastructure is built around key components designed to strengthen Europe's autonomy and drive innovation.

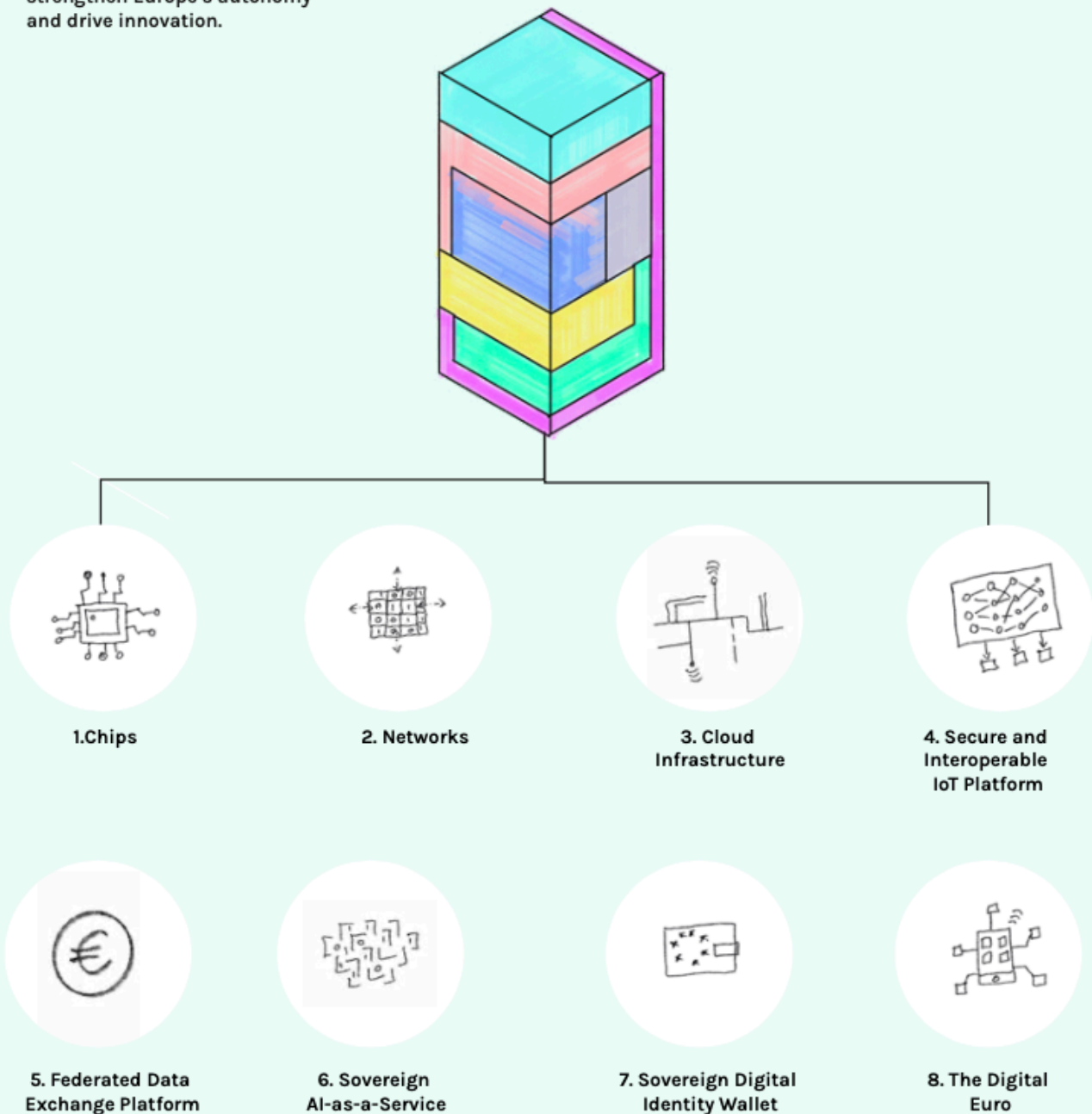


Figure 4. Elements of the EuroStack

Europe's Common Digital Stack

At the core of EuroStack is its infrastructure, a foundational digital backbone that integrates Europe's capabilities across **semiconductors, networks, cloud, IoT, data platforms, AI, and digital identity**. Designed to ensure Europe's digital sovereignty, this infrastructure guarantees a **resilient, interoperable, and scalable** ecosystem capable of supporting both public and private services, all in alignment with EU regulatory and ethical standards.

This robust infrastructure is built around key components designed to strengthen Europe's autonomy and drive innovation:

1. Chips: Powering the EuroStack

Semiconductors are the basic building block of Europe's digital infrastructure. Through the European Chips Act, Europe is making significant investments in cutting-edge fabrication facilities and next-generation chip designs, with a strong focus on energy-efficient, low-power semiconductors. These efforts are further strengthened by the adoption of **RISC-V**, an open-standard architecture that reduces reliance on proprietary technologies, fosters innovation, and enhances Europe's technological independence. To ensure resilience, the EuroStack initiative will expand support for RISC-V development, establish a robust software platform around it, and build secure, sovereign supply chains. These steps are essential to guarantee access to critical components, safeguard uninterrupted production, and position Europe as a leader in the global semiconductor ecosystem.

2. Networks: The Backbone of Connectivity

EuroStack's network infrastructure is to deliver secure, seamless communication across borders, enabling the real-time data exchange essential for Europe's digital economy. The initiative promotes the development of standalone 5G networks while adopting a forward-looking approach toward 6G technology, ensuring Europe remains globally competitive. Decentralized and localized edge operations reduce latency, enhance resilience and optimize performance, particularly for critical sectors such as healthcare, manufacturing, and smart cities. By prioritizing energy-efficient network technologies, EuroStack also minimizes environmental impact, aligning with Europe's sustainability goals. To ensure resilience, the infrastructure incorporates redundancy and cybersecurity, and prepares for quantum technologies. Additionally, anticipating future needs, EuroStack promotes the adoption of SCION architecture that redefines how data moves across the Internet, offering enhanced scalability, robust control, and improved isolation; and active EU engagement in management of the future public core of the Internet, as well as resilience of undersea cables and satellite connectivity to strengthen Europe's digital autonomy.

3. Cloud Infrastructure: The Operational Core

Leveraging initiatives like 8ra and the IPCEI-CIS, EuroStack is advancing **decentralized cloud and edge infrastructure** to reduce reliance on foreign providers, ensuring sovereignty and enhancing operational flexibility. The disruption in the AI cloud space presents a pivotal opportunity for Europe to lead by developing a **sovereign AI cloud infrastructure**, purpose-built mid-sized data centers, and AI factories—offering the critical infrastructure needed for advanced AI development and deployment. By integrating and expanding **public High-Performance Computing (HPC)** centers and **optimizing them for AI**, EuroStack aims to establish a scalable, unified cloud infrastructure operating entirely under European jurisdiction. This infrastructure will serve diverse sectors such as healthcare, energy management, and manufacturing, delivering reconfigurable, tailored, and scalable cloud services. A strong focus on sustainability and energy efficiency aligns with Europe's climate

goals, while redundant systems enhance resilience and ensure robust performance under high demand.

4. Secure and Interoperable IoT Platform

The EuroStack IoT platform enables the large-scale deployment of certified devices and services, driving innovation across smart cities, industrial automation, renewable energy management, advanced manufacturing and robotics (e.g., Industry 5.0). Designed to comply with EU regulations like the **Cyber Resilience Act** and **NIS-2 Directive**, it ensures the highest standards of privacy and security. Built on EU-developed specifications like **FIWARE** and the work in the **IPCEI-CIS**, the platform fosters interoperability, allowing seamless integration of IoT solutions across sectors. The platform is scalable, adapting to diverse applications while supporting Europe's regulatory principles and sustainability goals.

5. Federated Data Exchange Platforms: Industry and Public Interest Data

Industry Data Exchange: Accelerating Industrial Innovation

For industrial sectors, platforms like Manufacturing-X and Catena-X create secure ecosystems for sharing industry-specific data, fostering collaboration, and driving innovation. These platforms enable manufacturers and supply chain partners to share data efficiently while maintaining data sovereignty and ensuring compliance with European regulations. They prioritize interoperability, allowing integration of data across businesses and sectors. By fostering trust and transparency, these platforms enhance Europe's competitive edge in advanced manufacturing and industrial ecosystems.

Public Interest Data as a Public Good

Public interest data platforms treat data as a collective resource managed for societal benefit. Drawing inspiration from Finland's Suomi.fi and Estonia's X-Road, these platforms enable secure and ethical cross-border data sharing among public and private entities. Focused on public interest use cases such as healthcare, urban planning, and environmental monitoring, they ensure that data sovereignty and privacy compliance are central to their operation. By treating data as a public good, these platforms uphold European values, facilitating transparent, ethical exchanges that prioritize citizen rights and societal well-being.

Both frameworks emphasize interoperability, enabling data to flow seamlessly within and across sectors, while safeguarding trust, security, and compliance. Together, they form the backbone of Europe's data-driven economy, balancing innovation with ethical stewardship to harness data as a strategic asset for industry and the public good.

6. Software: the command centre

Software runs the digital infrastructure, spanning operating systems, application platforms, and algorithmic frameworks. It powers critical functions such as identity management, electronic payments, transactions, and document delivery, forming the foundation of digital public infrastructures. In this domain, US companies dominate foundational tools, with Microsoft, Apple, and Google controlling over 90% of the European market for operating systems across desktops, mobile devices, and embedded systems. While Europe boasts global leaders in enterprise software, its presence in algorithmic frameworks, vital for innovation and developer ecosystems, remains limited and often tied to US-based platforms. Programs like the **Next Generation Internet (NGI)** have developed alternative software solutions in Europe, but their scale remains insufficient to compete globally.

The **Sovereign Digital Identity Wallet** provides secure, **privacy-first** authentication for citizens and businesses, ensuring seamless access to both public and private services across Europe. By overcoming the limitations seen in models like **India Stack**, which rely on **centralized biometric IDs** and foreign cloud infrastructure, the EuroStack offers a **federated, privacy-preserving platform**. With a focus on **privacy by design, cross-border interoperability, and user empowerment**, the wallet ensures that citizens control their data, sharing only what is necessary while maintaining full compliance with **EU privacy standards**. As the gateway to Europe's digital infrastructure, the wallet enables seamless access while safeguarding privacy and sovereignty. Governments should protect against coercive enrolment in digital ID programs by ensuring voluntary participation with opt-out options, and safeguards against service denial for non-participants. Clear procedures for withdrawing consent and deleting data, along with regular audits, can ensure accountability.

The **Digital Euro**, to be issued by the European Central Bank, anchors trust and stability in Europe's monetary system as a **central bank-backed digital currency**. It offers **universal access**, ensuring all citizens and businesses, including those underserved by traditional banking, can fully participate in the digital economy. **Fee-free transactions** foster financial inclusion, reducing costs and promoting **economic equity**. With **GDPR compliance** at its core, the Digital Euro ensures secure transaction data processing exclusively within **European jurisdiction**, supporting seamless **cross-border transactions** and integration with both public and private digital systems. This ensures the Digital Euro strengthens Europe's financial **sovereignty**, supporting innovation and sustainable economic growth across the EU.

7. Sovereign AI-as-a-Service

Sovereign AI in the EuroStack is to power critical sectors such as **mobility, healthcare, education, and climate monitoring** across Europe. Combining the scalability of large AI models with the precision of bespoke solutions, Europe is leveraging its growing ecosystem of **AI labs, companies, and public HPC centres**. Initiatives like **AI Factories** and **OpenGPT-X** utilize Europe's public compute infrastructure to develop **large, multilingual AI models** that reflect European values. At the same time, smaller, bespoke AI models are tailored to niche sectors, ensuring localized, sector-specific solutions. By embracing composite learning, a hybrid approach combining decentralized, distributed, and federated learning, Europe is building privacy-preserving, accountable AI systems that align with public-interest goals, driving innovation while ensuring data sovereignty and ethical governance. A sound approach for Europe's AI development involves investing in specialized large-scale models that address its unique linguistic and cultural diversity, as well as domain-specific needs. Where possible, these models should be open source to foster transparency and collaboration. At the same time, agile application development is essential, leveraging a mix of European-developed technologies and globally available models to accelerate deployment. Strategic funding and partnerships between government, industry, and academia will be crucial to sharing the costs and benefits of foundational AI investments, leading to robust, innovative, and competitive outcomes.

The EuroStack represents Europe's commitment to building a secure, resilient, and sovereign digital ecosystem. Its goal is to **develop European alternatives across the technology stack**, starting from Europe's core assets and industrial capabilities. Achieving this requires the creation of an industry coalition of European tech champions to drive innovation and reduce reliance on foreign technologies. A critical step is mapping Europe's existing assets in a dynamic, evolving joint catalogue, which will link and align various industrial policy initiatives (see overview in Figure 5 below). By integrating these efforts and scaling existing alternatives, the EuroStack can strengthen Europe's industrial competitiveness, foster innovation, and uphold digital rights, sustainability, and ethical governance—positioning the EuroStack as an alternative in the global digital economy.

The Eurostack

Key companies and Alliances and networks per layer

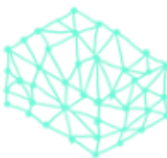
KEY COMPANIES		ALLIANCES AND NETWORKS	
Data and Artificial Intelligence	- Mistral (France) - Hugging Face (France) - Aleph Alpha (Germany) - Siemens (Germany) - ABB (Switzerland)		European AI Alliance: Facilitates dialogue on AI policies and developments in Europe. AIMEU Platform: Provides resources and tools for AI development and collaboration. Big Data Value Association (BDVA): Promotes big data technologies and services in Europe.
Software	- SAP (Germany) - Nextcloud (Germany) - LibreOffice (Germany) - Dassault Systèmes (France) - Thales (France) - Bitdefender (Romania)		OpenForum Europe: Advocates for open source and open standards in Europe. European Software Strategy Alliance (ESSA): Promotes strategic initiatives in the European software industry. EU Cybersecurity Certification Framework: Establishes cybersecurity certification schemes for ICT products and services.
Internet of Things & Devices	- Siemens (Germany) - Schneider Electric (France) - Bosch (Germany) - ABB (Switzerland) - Philips (Netherlands)		IoT European Platforms Initiative (IoT-EPI): Supports the development of IoT platforms in Europe.
Cloud	- OVHcloud (France) - Scaleway (France) - Deutsche Telekom (Germany)		Gaia-X: Develops a federated and secure data infrastructure for Europe. European Alliance for Industrial Data, Edge, and Cloud: Promotes the development of next-generation cloud and edge technologies.
Networks	- Nokia (Finland) - Ericsson (Sweden) - Alcatel Submarine Networks (France) - Orange (France) - Airbus Defence and Space (Germany/France)		5G Infrastructure Public Private Partnership (5G PPP): Collaborative initiative to advance 5G development in Europe.
Chips	- ASML (Netherlands) - STMicroelectronics (France/Italy) - Infineon Technologies (Germany) - Fraunhofer Society (Germany) - IMEC (Belgium)		Important Projects of Common European Interest (IPCEI) on Microelectronics: Supports large-scale transnational projects in microelectronics. Silicon Saxony: A leading microelectronics cluster in Germany. European Semiconductor Industry Association (ESIA): Represents the semiconductor sector's interests in Europe. Industrial Alliance on Processors and Semiconductor Technologies: Aims to strengthen Europe's capabilities in processor and semiconductor technologies
Raw Materials, Energy and Water	- Umicore (Belgium) - Boliden (Sweden)		European Raw Materials Alliance (ERMA): Facilitates secure and sustainable access to critical raw materials essential for the digital industry.

Figure 5. European companies and Alliances

The EuroStack Industrial Challenge: Minimum Viable Products (MVPs) as Highly Scalable Digital Services

The EuroStack Initiative introduces **Minimum Viable Products (MVPs)**—sovereign, interoperable, and cross-border trailblazers of **digital services** that showcase the scalability, flexibility, market potential, and societal value of Europe's **Digital Infrastructure Stack**. These MVPs align with Europe's principles of **openness, privacy, and sovereignty**, proving the EuroStack's capacity to support transformative services in mobility, healthcare, manufacturing, and biotech.

To foster innovation and accelerate adoption, the EuroStack will provide developers with **open APIs, open reference implementations, and open-source development environments** to build and execute applications, tailored AI models, and services. These resources will ensure an inclusive developer experience, enabling startups, researchers, and enterprises to harness the full capabilities of the EuroStack infrastructure without barriers to entry.

The **EuroStack MVP Challenge** will further catalyse innovation by inviting Europe's brightest developers, researchers, and entrepreneurs to create new services leveraging these tools. This initiative is designed to unlock Europe's talent pool, foster collaboration, and validate the infrastructure's ability to solve real-world challenges while expanding its ecosystem of open-source solutions.

By providing interoperable, privacy-first platforms and the tools to enable rapid innovation, the EuroStack creates the foundation for a **resilient, unified European digital ecosystem**. It empowers developers to contribute to Europe's digital future while securing long-term **digital sovereignty**, ensuring services that meet the needs of both citizens and businesses.

Comprehensive **pilot testing** will be conducted across diverse geographic and demographic contexts to ensure system resilience, including stress testing under various operational conditions and integration with existing infrastructure. Clear success criteria, combining quantitative metrics and qualitative user experience assessments, must be defined before launch, and regularly updated based on feedback and technological advancements. A robust feedback mechanism should gather input from users, technical evaluations, and stakeholder consultations, with transparent procedures for implementing improvements and reporting adjustments throughout the pilot phase.

MVP Examples

MVP 1: Citizen-Centric Public Services (Mobility and Healthcare)

Clara, a Barcelona resident, visits Berlin for a music festival, seamlessly accessing **EuroStack-enabled mobility and healthcare services**.

Using her **Sovereign Digital Identity Wallet**, Clara logs into a Berlin mobility app, which verifies her identity securely. Tailored discounts for eco-friendly transport options, such as e-scooters, are offered. Payments are instant and fee-free via **Digital Euro**, while **Federated AI** provides real-time updates for low-carbon travel routes.

Later, Clara visits a pharmacy for an emergency prescription. The pharmacist accesses her medical history through the **Federated Data Exchange**, ensuring GDPR compliance and safeguarding her privacy. Clara pays securely with Digital Euro, completing the transaction effortlessly. The payment is automatically registered with her home healthcare insurance.

This MVP illustrates how EuroStack creates **interoperable, privacy-first solutions**, enhancing citizens' experiences, saving them hours of administrative hassle, while advancing sustainability and cross-border collaboration in mobility and healthcare.

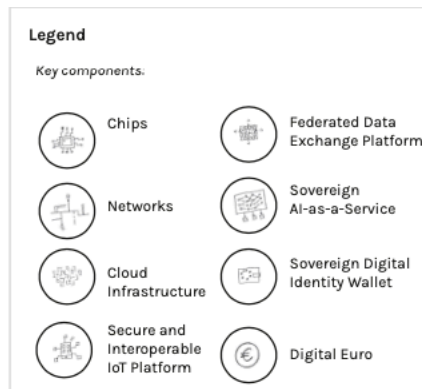
The Eurostack

Minimum Viable Products (MVPs),

What are MVPs?

A set of scalable, interoperable digital services and apps designed to succeed in the Single Market. These are operational trailblazers and will meet the immediate needs of citizens and businesses while demonstrating the feasibility and value of Europe's digital sovereignty strategy. They use the key components of the EuroStack

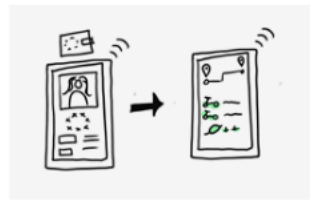
User Story Examples



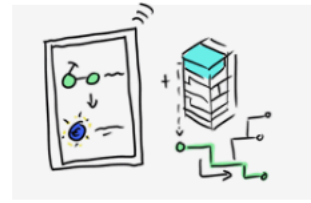
MVP Example 1 - Citizen-Centric Public Services (Mobility and Healthcare)



Clara, a Barcelona resident, visits Berlin for a music festival, seamlessly accessing EuroStack-enabled mobility and healthcare services.



Using her **Sovereign Digital Identity Wallet**, Clara logs into a Berlin mobility app, which verifies her identity securely. Tailored discounts for eco-friendly transport options, such as e-scooters, are offered.



Payments are instant and fee-free via **Digital Euro**, while **Federated AI** provides real-time updates for low-carbon travel routes.



Later, Clara visits a pharmacy for an emergency prescription. The pharmacist accesses her medical history through the **Federated Data Exchange**, ensuring GDPR compliance and safeguarding her privacy.



Clara pays securely with Digital Euro, completing the transaction effortlessly. The payment is automatically registered with her home healthcare insurance.

What does this MVP illustrate?

How EuroStack creates interoperable, privacy-first solutions, enhancing citizens' experiences, saving them hours of administrative hassle, while advancing sustainability and cross-border collaboration in mobility and healthcare.

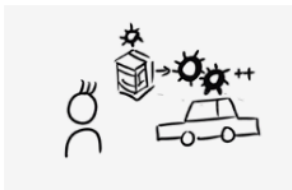
MVP 2: Advanced Manufacturing – Transforming Supply Chains

Markus, an automotive manufacturer in Munich, optimizes production with a Dutch supplier using EuroStack-enabled tools.

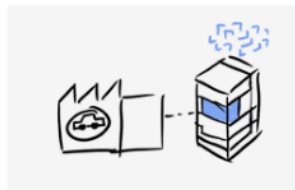
Markus's factory connects to the **EuroStack Decentralized Cloud**, securely processing real-time production data locally. **Federated AI** analyses data across EU factories, providing actionable insights to reduce waste and energy use in line with the EU Green Deal.

Supplier credentials are verified through the **Sovereign Digital Identity Wallet**, accelerating contract approvals and ensuring compliance. Payments are processed instantly via **Digital Euro**, avoiding delays and fees. Material quality updates and delivery schedules are shared securely via the **Federated Data Exchange**, enabling predictive maintenance, and preventing disruptions. Markus becomes a leader in productivity growth in their sector. This MVP highlights EuroStack's ability to enhance efficiency, sustainability, and resilience in Europe's manufacturing sector.

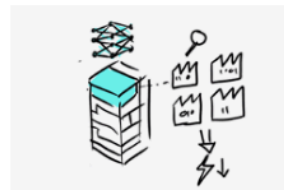
MVP Example 2 - Advanced Manufacturing – Transforming Supply Chains



Markus, an automotive manufacturer in Munich, optimizes production with a Dutch supplier using EuroStack-enabled tools.



Markus's factory connects to the **EuroStack Decentralized Cloud**, securely processing real-time production data locally.



Federated AI analyses data across EU factories, providing actionable insights to reduce waste and energy use in line with the EU Green Deal.



Supplier credentials are verified through the **Sovereign Digital Identity Wallet**, accelerating contract approvals and ensuring compliance.



Payments are processed instantly via **Digital Euro**, avoiding delays and fees.



Material quality updates and delivery schedules are shared securely via the **Federated Data Exchange**, enabling predictive maintenance, and preventing disruptions. Markus becomes a leader in productivity growth in their sector.

What does this MVP illustrate?

EuroStack's ability to enhance efficiency, sustainability, and resilience in Europe's manufacturing sector.

MVP 3: Cross-Border Genomic Innovation for Precision Medicine

Maria, a biotech product manager in Ljubljana, collaborates with teams in Bologna and Finland to develop a precision medicine drug.

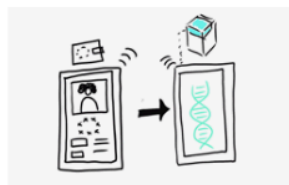
Using her **Sovereign Digital Identity Wallet**, Maria securely accesses sensitive genomic datasets stored in EU data infrastructures. **Federated AI** analyses distributed data without transferring raw information, identifying genetic markers while maintaining privacy.

Simulations are run on Bologna's **Leonardo** and Finland's **LUMI** supercomputers via the **EuroStack Decentralized Cloud**, ensuring compliance and security. Anonymized findings are shared through the **Federated Data Exchange**, allowing a healthcare provider in Denmark to pilot the drug ethically and cross-border. Maria's Ljubljana company consistently beats international competition in lab-to-market time. This MVP demonstrates EuroStack's role in accelerating innovation in precision medicine, enabling secure collaboration while safeguarding patient privacy.

MVP Example 3 - Cross-Border Genomic Innovation for Precision Medicine



Maria, a biotech product manager in Ljubljana, collaborates with teams in Bologna and Finland to develop a precision medicine drug.



Using her **Sovereign Digital Identity Wallet**, Maria securely accesses sensitive genomic datasets stored in EU data infrastructures.



Federated AI analyses distributed data without transferring raw information, identifying genetic markers while maintaining privacy.



Simulations are run on Bologna's **Leonardo** and Finland's **LUMI** supercomputers via the **EuroStack Decentralized Cloud**, ensuring compliance and security.



Anonymized findings are shared through the **Federated Data Exchange**, allowing a healthcare provider in Denmark to pilot the drug ethically and cross-border. Maria's Ljubljana company consistently beats international competition in lab-to-market time.

What does this MVP illustrate?

EuroStack's role in accelerating innovation in precision medicine, enabling secure collaboration while safeguarding patient privacy.

A Growing Innovation Ecosystem

The EuroStack is more than infrastructure. It is Europe's platform for talent, innovation, and collaboration, designed to address Europe's critical talent gap and empower its brightest minds to build transformative digital solutions. At its heart is a vibrant, growing ecosystem of **open-source developers, hardware designers, AI experts, entrepreneurs, and researchers** working in tandem across public and private sectors.

This ecosystem aims to create an environment where talent can thrive. Innovators are encouraged to create **open-source software, hardware solutions, and sovereign AI models**, strengthening Europe's leadership in ethical and accountable digital development. By connecting research centres, SMEs, large industries, and member states, the EuroStack fosters a culture of innovation and collaboration that anchors Europe in the global technology landscape.

With transparency, adaptability, and community ownership as guiding principles, this ecosystem drives **long-term sustainability and impact**. By mobilizing talent and aligning diverse stakeholders, the EuroStack ensures Europe builds an innovation-driven future that reflects its democratic values and economic goals.

A Governance Framework to Deliver the EuroStack

The EuroStack Initiative relies on a governance model that combines **disruption with shared accountability**, designed to coordinate and scale Europe's digital infrastructure in an era of rapid technological change. Rooted in the principles of **digital commons**—shared resources managed collectively for public benefit—and inspired by dynamic initiatives like the **8ra model**, the proposed **European Digital Infrastructure Consortium (EDIC)** acts as a **transformative, industry-led framework**. The proposed EDIC should be **industry-led** and based on identifying **key highly scalable use cases** (what we call MVPs) to drive value. It represents a bold departure from traditional governance models, functioning not as a bureaucratic oversight body but as a **disruption engine**, accelerating Europe's path toward digital sovereignty. It leverages **collaborative governance and open innovation** to ensure that Europe's digital infrastructure is not only world-class but also reflects the values and needs of its citizens.

The successful implementation and sustainability of the EuroStack initiative require a robust and adaptive governance framework that ensures safety, security, and alignment with European values. This section proposes a multi-layered governance model balancing strategic oversight, operational efficiency, and ethical considerations while embracing agility to adapt to emerging technologies.

Strategic Oversight and Leadership

The **EuroStack Steering Committee**, comprising representatives from EU institutions, member states, industry leaders, academia and civil society, will oversee strategic direction, approve major policies, and make critical decisions. Meeting quarterly, this committee will ensure that the initiative remains aligned with Europe's values of inclusivity, transparency, and sovereignty, while fostering equitable access to innovation and resisting monopolization.

A **Chief EuroStack Officer (CEO)**, appointed by the Steering Committee, will oversee day-to-day operations, and ensure alignment with strategic goals. Supporting the CEO, a **EuroStack Advisory Board** of independent experts will provide guidance on technical, ethical, and policy matters.

Operational Coordination and Implementation

A **EuroStack Program Office** will manage the initiative's implementation, coordinate stakeholders, and prepare progress reports for the Steering Committee. To maintain focus on measurable outcomes, the Program Office will drive systemic change through **Key Performance Indicators (KPIs)**, such as adoption rates of cross-border digital services, energy efficiency improvements, and development of scalable interoperable systems. Layer-specific **Working Groups** will focus on components of the EuroStack, bringing together experts to develop detailed implementation plans and technical standards. A strategic emphasis will be placed on empowering startups, scale-ups, and unconventional innovators through open innovation frameworks and funding mechanisms, directly addressing critiques of over-funding entrenched incumbents.

Agility and Rapid Response

To stay ahead of emerging trends, the governance framework incorporates **agile governance principles**, allowing for rapid adaptation to technological advancements in areas like AI, quantum computing, and cybersecurity. Transparent technical audits and security assessments should use standardized methodologies with independent third-party verification and risk-based frameworks, ensuring clear remediation timelines and accountability. To prevent mission and function creep, systems must have explicit scope limitations, regular reviews to align usage with original mandates, and strict protocols for managing any expansion of capabilities or data use.

A **Rapid Response Team** will address urgent security or operational challenges, while a **Policy Adaptation Mechanism** ensures that policies and standards evolve to reflect new developments and lessons learned.

Transparency, Accountability, and Collaboration

Accountability is ensured through **radical transparency**, with publicly available progress reports tracking KPIs such as innovation milestones and cost reductions. An annual **EuroStack Conference** will present achievements and gather feedback from stakeholders, while a **Stakeholder Engagement Platform** will foster ongoing dialogue with industry, civil society, and the public.

Leadership within the EuroStack governance model will be **decentralized and collaborative**, reflecting Europe's diversity and shared commitment to sovereignty and inclusivity. Member states, public institutions, industry leaders, and innovators will co-create and govern the initiative to ensure collective ownership of its success.

Global Partnerships and Knowledge Sharing

Recognizing the global nature of digital technologies, a **Global Partnerships Office** will manage relationships with international partners, ensuring alignment with global standards while upholding European values. A **Technology Transfer and Knowledge Sharing Unit** will facilitate responsible sharing of EuroStack innovations and promote the adoption of European standards internationally.

Ensuring Long-Term Success

This governance framework integrates strategic oversight, operational efficiency, and robust safeguards while prioritizing outcomes and accountability. The EuroStack execution strategy should incorporate sophisticated success metrics beyond adoption rates, focusing on service quality through benchmarks for latency, uptime, and response times, complemented by user satisfaction surveys and

real-time monitoring. Inclusion metrics should track adoption across demographics like the elderly, disabled, and rural populations, alongside improvements in digital literacy and support for underserved communities. Privacy and security metrics must include breach incidents, transparency reports, consent management effectiveness, penetration tests, and incident response times. Economic impact measures should analyse job creation, innovation, ecosystem growth, local technology development, and the expansion of European digital service providers.

By emphasizing agility, transparency, and collaboration, the EuroStack initiative positions Europe to lead in digital sovereignty and innovation, ensuring it remains aligned with its values and long-term objectives in a rapidly evolving digital landscape.

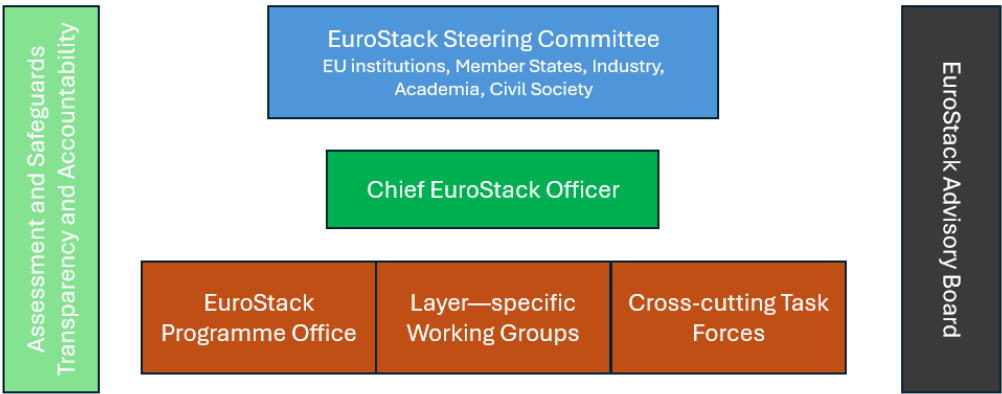


Figure 6. A Governance Framework for the EuroStack

PART 2 The Political Economy of Digital Sovereignty: Unpacking Power in the Stack

Strategies of Value Capture in the Digital Economy

Digital strategic autonomy, or ‘digital sovereignty’, is a cornerstone of Europe’s strategic vision for its future. Digital strategic autonomy consists of the capabilities, capacities and control in the digital domain that are necessary to safeguard sovereignty.²¹ Digital infrastructures—data centres, cloud platforms, semiconductors, software ecosystems, and AI—have reshaped global industries. The current economic landscape sees a concentration of power in the hands of a few dominant players that are not European—particularly in cloud infrastructure, semiconductors, and AI platforms. This hampers Europe’s ability to innovate, compete, and be in control in the digital economy, society, and democracy.

Reclaiming digital sovereignty requires Europe to address structural dependencies, tackle systemic bottlenecks and other control points, and leverage its regulatory and industrial strengths to build a resilient, competitive digital ecosystem. Bottlenecks—essential inputs with restricted supply—highlight vulnerabilities across the digital stack. These can get weaponized in geo-politics and geo-economics.

Market concentration and structural dependencies on technology giants transcend individual layers. Typically, dominant companies seek to extend their business across the stack through vertical integration, data monopolization, and control of key markets from digital advertising to semiconductor manufacturing and AI computing.²² These strategies *de facto* increase the risk of abuse of market power, imposition of unwanted rules of foreign regimes, lack of responsibility to protect democracy, loss of domestic job opportunities and brain drain.

At the foundation of the digital stack, China exerts near-total control over the refining of certain rare earth elements (REEs), essential for manufacturing electronics, magnets, and other high-tech components. China accounts for approximately 90% of global REE refining capacity,²³ a dominance maintained through deliberate state policies and tightly integrated supply chains. Further up the stack, Taiwan Semiconductor Manufacturing Company (TSMC) dominates advanced chip production, controlling over 90% of the global market for cutting-edge semiconductors²⁴.

By contrast, Europe produces less than 10% of the world’s semiconductors and relies heavily on TSMC and South Korea’s Samsung for advanced chips.²⁵ This dependence creates a significant strategic vulnerability, especially in the face of geopolitical tensions, such as between China and Taiwan, or disruptions to global supply chains as we have witnessed during and after the COVID-19 pandemic, which significantly affected industries like automotive, consumer electronics, and telecommunications. However, one key strength Europe possesses is its dominance in the machines

²¹ Paul Timmers, ‘How Europe Aims to Achieve Strategic Autonomy for Semiconductors’, *Brookings Tech Stream*, 9 August 2022, <https://www.brookings.edu/techstream/how-europe-aims-to-achieve-strategic-autonomy-for-semiconductors/>.

²² Fausto Gernone, ‘Moore’s Death and the Rebirth of Vertical Monopolies’, *UCL IIPP* (blog), 27 April 2023, <https://medium.com/iipp-blog/moores-death-and-the-rebirth-of-vertical-monopolies-befa9ce5b892>.

²³ IEA, ‘Energy Technology Perspectives 2023 – Analysis’, IEA, 12 January 2023, <https://www.iea.org/reports/energy-technology-perspectives-2023>.

²⁴ The Economist, ‘Taiwan’s Dominance of the Chip Industry Makes It More Important’, 6 March 2023, <https://www.economist.com/special-report/2023/03/06/taiwans-dominance-of-the-chip-industry-makes-it-more-important>.

²⁵ European Commission, ‘A Chips Act for Europe’.

that make chips, with ASML leading the global market in advanced lithography equipment essential for semiconductor manufacturing, giving Europe a critical foothold in the semiconductor value chain.

Many global tech companies have mastered "ecosystem power" integrating several products into cohesive offerings, including cloud infrastructure, hardware, software, and user interfaces.²⁶ Central to this dominance is their "**platform play**"²⁷, a strategy that leverages programmability, software, APIs, and infrastructure services to create powerful ecosystems. This model allows a small, highly specialized (and often expensive) team of experts to build and maintain a platform, enabling vast networks of developers and tech workers to create applications for end users. By leveraging dominance in one layer, these companies control others, dictating terms for downstream applications, extracting rents through fees or commissions, and capturing valuable data streams. This entrenches market power, raises barriers to entry, and fosters dependency, reshaping competition across the digital stack.²⁸

These companies, typically based in the US, also benefit from robust venture capital ecosystems and public markets that reward innovation and scale. Nvidia, with a market capitalization exceeding \$1 trillion in 2023, exemplifies this advantage, reinvesting heavily in research, development, and strategic acquisitions to consolidate its leadership in AI and GPU technologies. Sovereign wealth funds, such as Saudi Arabia's Public Investment Fund (PIF), have further reshaped the AI and semiconductor landscapes by backing transformative initiatives like SoftBank's Vision Fund, which channels vast resources into AI and semiconductor startups. These financial networks enable dominant firms to acquire promising startups, further consolidating their power.

The same holds for China. Moreover, China²⁹ and recently also the US³⁰ have industrial policy programs with massive financial support – to the tune of hundreds of billions – for the digital sector and both identified a wide range of digital technologies as strategic for economic and national security and to be under national control³¹. Global technology giants—ranging from US heavyweights like Amazon, Microsoft, Google, Apple, Meta, Nvidia, Tesla/SpaceX, and Musk's associated ventures, to Chinese powerhouses like Huawei³², Alibaba, Tencent, and ByteDance—each pursue integrated ecosystems that span critical infrastructure, platforms, and services³³. Rather than simply offering discrete products, they embed their services deeply into essential layers of the digital economy—from cloud computing and data centres to software, operating systems, chips, AI models, mobile platforms, fulfilment logistics, advertising networks, and social media communities.

²⁶ Cristina Caffarra, Annabelle Gawer, and Michael Jacobides, 'Mapping Antitrust onto Digital Ecosystems', *CPI Antitrust Chronicle*, 25 October 2024.

²⁷ Annabelle Gawer, 'Digital Platforms and Ecosystems: Remarks on the Dominant Organizational Forms of the Digital Age', *Innovation* 24, no. 1 (2 January 2022): 110–24, <https://doi.org/10.1080/14479338.2021.1965888>.

²⁸ Vili Lehdonvirta, *Cloud Empires: How Digital Platforms Are Overtaking the State and How We Can Regain Control* (Cambridge, Massachusetts London, England: The MIT Press, 2022).

²⁹ García-Herrero, A. and R. Schindowski (2024) 'Unpacking China's industrial policy and its implications for Europe', *Working Paper* 11/2024, Bruegel

³⁰ Aurelia Glass and Karla Walter, 'How Biden's American-Style Industrial Policy Will Create Quality Jobs', *Center for American Progress* (blog), 27 October 2022, <https://www.americanprogress.org/article/how-bidens-american-style-industrial-policy-will-create-quality-jobs/>.

³¹ Alessandro Gili and Davide Tentori, 'The Comeback of Industrial Policy. The Next Geopolitical Great Game' (ISPI, n.d.), accessed 11 January 2025.

³² Xiaoying Dong, Mengling Yan, and Yanni Hu, *Huawei: From Catching Up To Leading* (Singapore: Springer Nature, 2023), <https://doi.org/10.1007/978-981-19-4078-1>.

³³ Michael G. Jacobides, 'How to Compete When Industries Digitize and Collide: An Ecosystem Development Framework', *California Management Review* 64, no. 3 (May 2022): 99–123, <https://doi.org/10.1177/00081256221083352>.

By controlling “stacks” of intertwined technologies, these companies generate powerful **lock-in effects**. Firms like Amazon, Microsoft, and Google blend cloud, AI, and productivity tools, ensuring steady revenue streams and discouraging customers from migrating away due to the complexity and cost involved. Apple’s similarly closed ecosystem tightly couples hardware, software, and its App Store, generating high user loyalty while capturing revenue from third-party developers. Meanwhile, Meta’s social and VR platforms rely on user data and open-source AI frameworks to encourage developers to build within its orbit, and Nvidia’s hardware-software alignment in GPUs fosters a similar kind of dependency.

On another front, Tesla has revolutionized electric vehicles with its advanced software ecosystem and AI-powered Autopilot, while SpaceX leads global satellite communications through Starlink, operating over 60% of active satellites.³⁴ Meanwhile, Musk’s AI venture, xAI, enhances Tesla’s autonomous driving and explores broader AI applications, leveraging data from X, which he aims to transform into a unified hub for communication, payments, and AI services.

In China, well-funded government policies, also through the use of Government Venture Capital³⁵, support firms like Huawei, Alibaba, Tencent, and ByteDance in replicating these strategies, leveraging dominance in one domain—be it telecom, e-commerce, social networks, or video-sharing—to expand into cloud, AI, and other crucial layers.

Taken together, **this model of infrastructural integration** amplifies the power and resilience of a few dominant players, undermining domestic competitiveness, complicating regulatory oversight, and shaping information and commerce landscapes in ways that raise serious concerns about fairness, consumer choice, and democratic governance. Moreover, in both China and the US, demand from defence and security sectors plays a significant role in driving technological advancement. In contrast, the EU’s decentralized defence policies and fragmented markets leave a critical tool absent from its industrial policy arsenal.

Dominance is often fuelled through **data monopolization**. Consumer-facing platforms collect vast amounts of data, which feed AI models. User data is commodified to predict and influence behaviour³⁶. Acting both as an input and as an output for many digital products, data creates feedback loops that entrench the dominance of existing players. For instance, Google’s extensive data collection not only helps the company build better products, but it fuels advertising algorithms that dominate the digital advertising market, capturing approximately 39% of global digital ad revenues in 2023. Meta holds an additional 18%³⁷. These advertising revenues reinforce the financial strength of these companies, which in turn finances their expansion into digital ecosystems.

This has far-reaching societal consequences. In their quest to increase their data harvest, platforms amplify polarizing content to maximize engagement, creating fertile ground for fake news, conspiracy theories, and extremist ideologies. Algorithmic prioritization of sensational content fosters division, undermines democratic discourse, and supports the rise of populism, which exploits these dynamics

³⁴ Debra Werner, ‘Want to Challenge Starlink in the Satcom Market?’, *SpaceNews*, 16 September 2024, <https://spacenews.com/want-to-challenge-starlink-in-the-satcom-market/>.

³⁵ Martin Beraja et al., ‘Government as Venture Capitalists in AI’ (Cambridge, MA: National Bureau of Economic Research, July 2024), <https://doi.org/10.3386/w32701>.

³⁶ This practice is called “surveillance capitalism” by Shoshana Zuboff, ‘The Age of Surveillance Capitalism: The Fight for a Human Future at the New Frontier of Power’, *Yale Law Journal* 129, no. 5 (2020): 1460–1515, <https://doi.org/10.26522/brooked.v29i2.849>.

³⁷ Statista, ‘Digital Ad Revenue Share by Company 2023’, 3 December 2024, <https://www.statista.com/statistics/290629/digital-ad-revenue-share-of-major-ad-selling-companies-worldwide/>.

to spread misinformation and erode trust in institutions³⁸. To protect democracy, regulatory action seeks to enforce transparency and pluralism in algorithms, limit data monopolies, and regulate targeted advertising and addictive design³⁹.

If Europe fails to create the EuroStack and establish digital sovereignty, the global digital economy and its value capture mechanisms are likely to become even more concentrated and dominated by non-European players. Looking ahead 10 years, the consequences of inaction would be severe. Without the EuroStack, Europe risks becoming a "digital colony," where critical technologies, data, and digital services are almost entirely controlled by external powers. The Internet of Things (IoT) and edge computing will likely be dominated by non-European ecosystems, leaving Europe as a passive consumer rather than an innovator in these transformative technologies. The lack of European alternatives in cloud services and AI platforms will make it nearly impossible for EU companies to compete globally in data-driven industries. This digital dependency will have profound implications for Europe's economic sovereignty, potentially constraining policy choices and leaving the continent vulnerable to economic coercion through control of vital digital infrastructure.

The absence of a strong European digital ecosystem will likely result in a significant loss of high-skilled jobs and economic value creation within the EU. The data generated by European citizens and businesses will increasingly be processed and monetized outside Europe, leading to a massive transfer of wealth and strategic assets to foreign entities. This scenario not only threatens Europe's economic prosperity but also its ability to uphold its values of privacy, transparency, and democratic governance in the digital realm.

Dynamics of the Stack

The "stack" is not a static, layered framework; it is an increasingly dynamic battleground where technology, strategy, and power converge. Big Tech firms are reshaping the stack through vertical integration, uniting hardware, software, and cloud infrastructure into cohesive ecosystems. Simultaneously, governments worldwide are racing to control the technologies underpinning economic power and geopolitical influence. Today, the stack is a living, evolving system, with every layer—hardware, cloud, AI, edge computing, and quantum technologies—being transformed by relentless innovation and strategic intent of both governments and large companies.

Vertical integration is illustrated well in AI but not limited to this. The surge in generative AI has propelled demand for Nvidia's advanced GPUs, particularly the H100, catapulting the company to a trillion-dollar valuation. This demand has also pushed companies like Microsoft, Meta, OpenAI, Amazon, and Google to develop custom AI processors, signalling a shift away from dependence on external suppliers. Examples such as Google's Axion chip and Amazon's Trainium3 illustrate how bespoke hardware is driving performance, cutting costs, and enabling domination in AI workloads.⁴⁰ AI-specific chips is set to account for a growing share of global semiconductors revenue, as AI applications proliferate across industries.

Infrastructure is also being rapidly transformed to support the AI boom. Microsoft, in partnership with asset managers like BlackRock, is building AI-dedicated data centres powered by renewable

³⁸ Steven Livingston, 'The Rise of Right-Wing Populism: Diagnosing the Disinformation Age', *Media@LSE* (blog), 8 November 2023, <https://blogs.lse.ac.uk/medialse/2023/11/08/the-rise-of-right-wing-populism-diagnosing-the-disinformation-age/>.

³⁹ European Parliament, 'Addictive Design of Online Services and Consumer Protection in the EU Single Market - Tuesday, 12 December 2023', 12 December 2023, https://www.europarl.europa.eu/doceo/document/TA-9-2023-0459_EN.html.

⁴⁰ Gernone, 'Moore's Death and the Rebirth of Vertical Monopolies'.

energy and exploring micro-nuclear reactors to meet the growing energy demands of advanced AI systems. Amazon is experimenting with on-site energy generation to reduce dependency on external grids. Musk's xAI, powered by its Colossus data centre with 100,000 Nvidia GPUs, has taken aggressive steps to scale its infrastructure, doubling chip capacity and straining Nvidia's supply chain⁴¹. These moves underscore how infrastructure innovation is not only about efficiency but also about integrating environmental resources imperatives into business strategies while consolidating market power. The stack is currently dominated by centralization, as exemplified by OpenAI and Microsoft's collaboration on custom AI supercomputers, which further consolidates chip design and manufacturing under a few global players. Without decisive action to enhance its own capabilities, this centralization risks deepening Europe's dependency on non-European systems, particularly in critical sectors like AI and cloud infrastructure.

The evolution of the tech stack is as much **geopolitical as it is technological**. Nations are prioritizing control over critical technologies to secure economic competitiveness and national security. The US-China rivalry for technological dominance has become a defining contest of the 21st century, reshaping global supply chains and alliances.

The United States has intensified efforts to curb China's advances in semiconductors, artificial intelligence (AI), and quantum computing. In October 2022, the Biden Administration imposed aggressive export controls to block Chinese firms from accessing advanced chips and manufacturing tools like the EUV lithography machines made by Dutch firm ASML. These measures are part of a broader US strategy to limit China's technological rise, pressuring allies such as the Netherlands and Japan to restrict key exports.

The US has also targeted Huawei, a global leader in 5G, citing national security concerns. These actions have significantly disrupted Huawei's supply chain. At the same time, Washington is investing heavily in AI and quantum technologies through initiatives like the CHIPS and Science Act, which allocates billions to strengthen domestic semiconductor production and R&D.

China is countering with a two-pronged strategy: ramping up domestic production and leveraging its dominance in critical raw materials. Beijing has committed over €140 billion under its "Made in China 2025" plan to achieve self-sufficiency in advanced technologies. Despite restrictions, China's SMIC has developed 7-nanometer chips, showcasing resilience, though limitations remain. Additionally, China has restricted exports of rare earth elements like gallium and germanium, critical for semiconductors and advanced technologies, using these as leverage in the global supply chain.

Europe finds itself navigating this rivalry with a pivotal but delicate role. ASML, the world's sole producer of EUV lithography machines, faces US pressure to curb sales to China while balancing its economic interests—14% of its 2022 revenue came from China.⁴² The European Union is advancing its digital sovereignty with the €43 billion European Chips Act, aiming to double its share of global semiconductor production by 2030. Initiatives like EuroStack will further emphasize Europe's push for a secure, independent digital infrastructure aligned with democratic values.

However, challenges remain. The semiconductor supply chain spans multiple countries and processes, making enforcement of export controls complex. Economic dependencies on China further complicate efforts to decouple. As the US seeks to maintain dominance and China accelerates its self-sufficiency goals, Europe has the opportunity to assert its leadership by investing in innovation,

⁴¹ Stephen Morris and Tabby Kinder, 'Elon Musk Plans to Expand Colossus AI Supercomputer Tenfold - FT', *Financial Times*, 4 December 2024, sec. xAI, <https://www.ft.com/content/9c0516cf-dd12-4665-aa22-712de854fe2f>.

⁴² Adam Levine, 'The U.S. Has 2 Choices on China Chip Policy. Neither Are Good for Stocks.', *barrons*, accessed 13 January 2025, <https://www.barrons.com/articles/chip-stocks-us-china-trade-policy-78acc6e4>.

protecting strategic industries, and charting a path toward autonomous yet interconnected digital sovereignty.

The stakes are clear: this tech rivalry will shape the future of digital capitalism, determining global power dynamics for decades to come. However, escalating trade wars fueled by these technological tensions risk deepening economic fragmentation, disrupting supply chains, and undermining the collaborative frameworks needed to address global challenges, ultimately making the path to stability and innovation more uncertain for all parties involved.

Regarding technological development, key paradigm shifts are underway, such as decentralization through edge-cloud computing, quantum technologies, and deep integration of physical and digital realms. **Decentralization** is emerging as a countertrend to centralized paradigms. The rise of edge-cloud computing and next-generation networks is reshaping how data is processed and managed. Europe is well-positioned to lead in this space, leveraging its expertise in industrial IoT, federated cloud systems, and regulatory frameworks favouring data sovereignty.

Initiatives like IPCEI-CIS (Important Projects of Common European Interest in Cloud Infrastructure and Services) aim to build decentralized, federated infrastructures that process data closer to the source, enhancing resilience, reducing latency, and improving security. Edge computing is no longer theoretical; it is actively being implemented in sectors such as autonomous driving, smart factories, and healthcare, positioning Europe at the forefront of these transformative technologies. However, blockchain technology, which underpins much of the current decentralization movement and the so-called Web 3.0 paradigm, remains dominated by US players. Europe has yet to reclaim its footing in this critical domain, presenting a challenge to its broader decentralization ambitions.

The cutting edge of the Stack's transformation lies in quantum, photonics, and neuromorphic technologies which promise to disrupt traditional computing and communications paradigms. Quantum processors, such as IBM's 127-qubit Eagle and Pascal's 100-qubit quantum computer⁴³, may be advancing encryption and simulation capabilities. Meanwhile, European hubs in Germany and the Netherlands are leading in photonics, using light for data processing and transmission. Photonics offers significant advantages in speed and energy efficiency over traditional semiconductors, positioning Europe as a leader in this sustainable technology innovation.

The profound convergence of the physical and digital worlds is creating integrated ecosystems of AI, IoT, robotics, and AR/VR. Technologies such as Amazon's warehouse robotics, Apple's Vision Pro AR headset, and Bosch's industrial IoT systems exemplify this deep integration, blurring the lines between real and virtual environments. Major players in the stack such as Meta (AR/VR) and Nvidia (robotics) are investing heavily in this paradigm shift⁴⁴. Europe's strong presence in areas like industrial IoT and robotics offers opportunities to lead in this domain, but competition from China and the US is fierce.

The stack has become a battleground for economic power and digital sovereignty. Big Tech is consolidating control through bespoke innovation and vertical integration, while Europe's initiatives focus on decentralization, resilience, and sustainability. The future stack will evolve as a dynamic interplay of centralized and decentralized systems, shaped by technological breakthroughs and geopolitical priorities. For Europe, this transformation presents both challenges and opportunities.

⁴³ insideHPC, 'Pasqal 100-Qubit Quantum Computer Shipped to Jülich Supercomputing Centre', *High-Performance Computing News Analysis* (blog), 2 December 2024, <https://insidehpc.com/2024/12/pasqal-100-qubit-quantum-computer-shipped-to-julich-supercomputing-centre/>.

⁴⁴ Michael Acton and Cristina Criddle, 'Nvidia Bets on Robotics to Drive Future Growth', *Financial Times*, 29 December 2024.

The growing centralization of AI and cloud infrastructure by non-European players underscores vulnerabilities. The future stack is not just about technology—it is about shaping the systems that will define economic, societal, and political power for decades to come.

The Economics of Digital Strategic Autonomy

Geopolitical dimension

Most important countries of each stack layer and their key firms

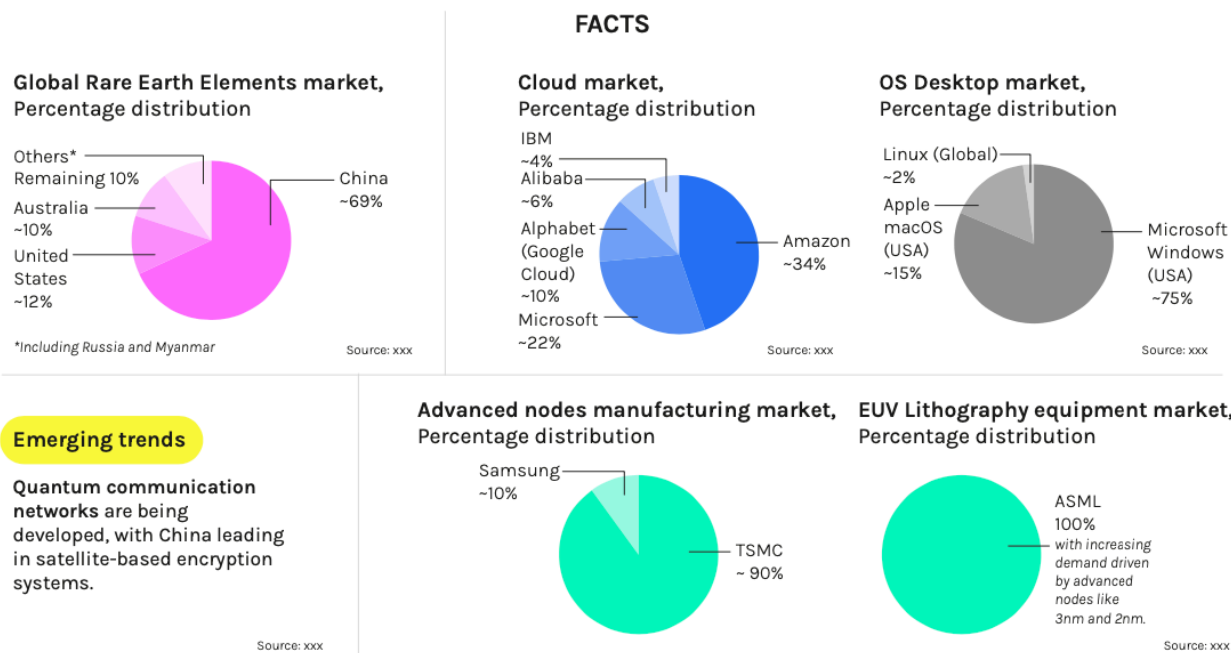
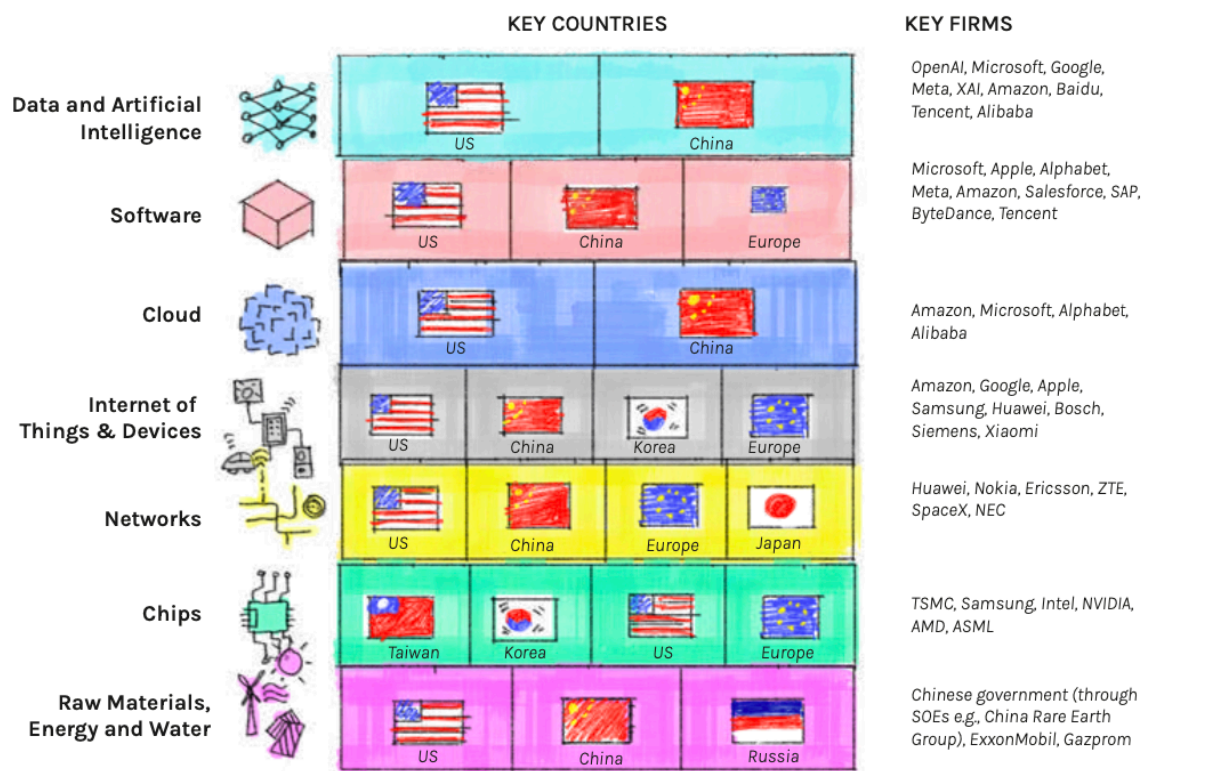


Figure 7. The Geopolitical Dimension of the Stack

Raw Materials, Energy and Water: The Hidden Backbone of Technology

At its foundation, the digital stack relies on critical raw materials such as rare earth elements (REEs), lithium, and cobalt. These materials are indispensable for producing core components of modern digital infrastructure, such as semiconductors, batteries, displays, and high-performance magnets. However, their supply chains are overwhelmingly controlled by foreign actors, particularly China. China dominates the refining of REEs, accounting for over 90% of global capacity⁴⁵. This dominance, achieved through state-backed policies and strategic acquisitions, extends to key battery materials like lithium and cobalt, where China's influence is reinforced by its investments in the Democratic Republic of Congo (DRC). Companies like China Northern Rare Earth Group and China Minmetals Corporation lead the sector, giving China unparalleled leverage, including the power to impose export restrictions during geopolitical disputes.

Europe's dependence on critical raw materials is significant. Over 68% of its cobalt and 78% of its lithium are imported, leaving the continent vulnerable to geopolitical risks and supply chain disruptions⁴⁶. While countries like Sweden and Portugal, as well as Serbia—geographically close but outside the EU—possess promising reserves, large-scale extraction has yet to materialize. Notably, Serbia holds an estimated 1.2 million tonnes of lithium, representing a potential opportunity for reducing Europe's reliance on imports if sustainable extraction practices are developed and implemented. Europe also lacks sufficient refining infrastructure, further compounding its vulnerabilities. Resource dependency can not only be countered by more access, but also by technological innovation: Germanium, one resource that China is limiting the export of, can be made from zinc by Europe's Nyrstar. An industrial policy is needed to further stimulate such technological innovation in resources.

The European Union has launched ambitious initiatives to reduce these dependencies. The **Critical Raw Materials Act** aims to secure 15% of Europe's raw material needs through recycling and ensure 10% of critical materials are sourced domestically by 2030⁴⁷. Initiatives like the **European Raw Materials Alliance (ERMA)** are fostering partnerships with resource-rich regions such as Australia and Latin America while promoting investment in advanced recycling technologies. Recycling innovations offer a promising path forward. Companies like Germany's **Aurubis** and Belgium's **Umicore** leaders in electronic waste recycling, and Norway's **Hydrovolt**, which specializes in battery recycling, demonstrate the potential of circular economies to reclaim valuable materials. However, scaling these efforts remains a challenge, requiring substantial investments and alignment with sustainability goals.

The digital economy's reliance on energy is also intensifying, driven by the rapid growth of artificial intelligence (AI) models, whose training requires immense computational power. According to the **International Energy Agency (IEA)**, data centres consume 2–4% of global electricity, with localized impacts even higher⁴⁸. In Ireland, for instance, data centres account for over 20% of electricity use. These figures are likely to increase, as US tech giants are investing heavily in Europe's

⁴⁵ IEA, 'Critical Minerals Market Review 2023 – Analysis', IEA, 11 July 2023, <https://www.iea.org/reports/critical-minerals-market-review-2023>.

⁴⁶ Samuel Carrara et al., 'Supply Chain Analysis and Material Demand Forecast in Strategic Technologies and Sectors in the EU – A Foresight Study', 16 March 2023, <https://doi.org/10.2760/386650>.

⁴⁷ Council of the European Union, 'Critical Raw Materials Act', Consilium, 18 March 2024, <https://www.consilium.europa.eu/en/press/press-releases/2024/03/18/strategic-autonomy-council-gives-its-final-approval-on-the-critical-raw-materials-act/>.

⁴⁸ IEA, 'What the Data Centre and AI Boom Could Mean for the Energy Sector – Analysis', IEA, 18 October 2024, <https://www.iea.org/commentaries/what-the-data-centre-and-ai-boom-could-mean-for-the-energy-sector>.

data economy⁴⁹. While these investments create opportunities, they also pose challenges, as many projects explore energy-intensive solutions, including nuclear power, which clash with Europe's renewable energy priorities. The widespread use of renewable energy certificates (RECs) by major providers often masks the true environmental cost, underscoring the need for stricter transparency and oversight.

Europe's **energy** landscape is evolving. Renewable sources like wind, solar, and hydropower are expanding, with leaders such as Sweden and Finland attracting energy-intensive industries due to their clean energy production. However, countries like Germany and Poland face challenges balancing coal and gas reliance with renewable integration. The 2022 energy crisis, triggered by Russian supply disruptions, accelerated Europe's adoption of renewables but also exposed its energy vulnerabilities. Energy prices broadly increased, raising running costs for businesses, and potentially reducing investment. Addressing these challenges requires Europe to prioritize energy-efficient, renewables-based solutions for AI-centric infrastructure. Innovations like Finland's **LUMI supercomputer**, powered entirely by hydropower from **Vattenfall**, provide a model for sustainable high-performance computing (HPC). Transparent emissions reporting and stricter oversight of renewable energy certificates (RECs) are critical to ensuring alignment with Europe's carbon neutrality goals.

The security and resilience of the electricity grids is another prerequisite for digital sovereignty and strategic autonomy. The security of ICT and SCADA components for controlling the power grids is an additional factor that has received so far only little attention in this context. These components must be subjected to a 360° risk assessment. The following risk categories, for example, are the focus of such an investigation:

- **Geopolitical:** disruption of supply chains, distortion of competition as a means of hybrid warfare, foreseeable future trade restrictions, impact on economic security, etc.
- **Strategic:** Economic dependence, technological dependence, influence on standards, loss of sovereignty, reduced market diversity, etc.
- **Technical:** Manipulation of data, targeted infiltration (keyword "backdoors"), hidden "kill switches", etc.

Water is another critical resource, particularly for cooling high-performance computing systems. Google's hyperscale data centres used an average of 2.1 million litres of water daily in 2023 for cooling⁵⁰. With over half the global population projected to live in water-stressed areas by 2050, these demands present long-term risks, including in drought-prone regions such as Southern Europe. While advances in water recycling and desalination technologies offer potential, their adoption across Europe remains inconsistent. Strengthening these systems is vital to mitigate the environmental impact of expanding digital infrastructure.

Europe's digital sovereignty depends on scaling its mining, refining, and recycling capabilities while aligning with sustainability goals. Partnerships with resource-rich regions and advanced recycling technologies can reduce import reliance. Integrating renewable energy and efficient water systems into AI and data centres is also key to meeting climate targets and ensuring competitiveness.

⁴⁹ United Nations Conference on Trade and Development, 'Digital Economy Report 2024', 2024, https://unctad.org/system/files/official-document/der2024_en.pdf.

⁵⁰ Mary Zhang, 'Data Center Water Usage: A Comprehensive Guide', Dgtl Infra, 17 January 2024, <https://dgtlinfra.com/data-center-water-usage/>.

Chips: The Brain of the Stack

The Forces Shaping Semiconductor Dominance

Semiconductors are the cornerstone of the digital world. They include mundane components like resistors, sensors of all sorts, memories and microprocessors. The latter were once limited to general-purpose tasks but have evolved into highly specialized processors tailored for applications such as artificial intelligence (AI), high-performance computing (HPC), and machine learning (ML). Chips range from general-purpose CPUs for PCs and servers to GPUs and task-specific processors like Tensor Processing Units (TPUs) for AI and HPC workloads. With increasing demands for efficiency, scalability, and energy optimization in cloud environments, the global microprocessor market has become a critical enabler of technological innovation. The global semiconductor market is estimated to grow to from over €600 billion in 2024 to over €1 trillion by 2030⁵¹.

The semiconductor industry is shaped by powerful economic dynamics that concentrate control among a few dominant players yet have also led to very complex, globally spread supply chains.⁵² The manufacturing stage of semiconductors is characterised by significant **economies of scale**, requiring billions of euros in investment, and huge concentration of tacit knowledge and talent creating formidable barriers for new entrants. Dominated by Asian foundries, particularly in Taiwan and South Korea, the manufacturing landscape operates at the cutting edge, producing chips at 5nm and 3nm process nodes. TSMC alone commands over 50% of the global semiconductor market and 90% of advanced chip production (when adding source always check whether data is overall or of the foundry market). Companies like TSMC and Samsung leverage their scale to reduce per-unit costs, ensuring efficiency and reinforcing their market dominance. Technological progress is exceptional and the foundational driver of this industry ('Moore's law').

Another critical factor is **the network effect in chip design**, where established architectures like x86 (Intel and AMD) and ARM gain dominance through widespread adoption and robust developer ecosystems. ARM's architecture, owned by Japan's SoftBank, underpins many custom chip designs, including AWS's Graviton and Google's TPUs. This network effect not only drives innovation but also locks developers into specific ecosystems, consolidating market control.

Co-specialization plays an important role in the chips industry. Intel's integrated design-manufacturing (IDM) exemplifies the traditional model based on the co-specialization between design and production, allowing seamless optimization of chip performance and cost efficiency. More recently, new business models harness the specialization between chip design and end use.⁵³ Nvidia, for example, controls over 80% of the chips specialised for AI—so-called Graphics Processing Units (GPUs)—and provides a programming platform tightly coupled with its chips, enhancing performance while fostering ecosystem dependence. The company generated €26.3 billion in data centre revenue in the second quarter of the 2024 calendar year⁵⁴. Similarly, cloud providers like AWS, Google Cloud, and Microsoft Azure increasingly integrate chip design and cloud-specific applications, creating proprietary processors tailored to their platforms. Innovations like AWS's Graviton processors, Google's TPUs, and Nvidia's proprietary GPU architectures exemplify this trend, optimizing performance and energy efficiency while locking customers into their ecosystems.

⁵¹ PwC, 'State of the Semiconductor Industry', PwC, 28 November 2024,

<https://www.pwc.com/gx/en/industries/technology/state-of-the-semicon-industry.html>.

⁵² Chris Miller, *Chip War* (Scribner, 2022), <https://www.christophermiller.net/semiconductors-1..>

⁵³ Gernone, 'Moore's Death and the Rebirth of Vertical Monopolies'.

⁵⁴ Statista, 'Data Center/AI Chip Revenue of Nvidia, AMD, and Intel 2024', 29 August 2024, <https://www.statista.com/statistics/1425087/data-center-segment-revenue-nvidia-amd-intel/>.

Nvidia CUDA further illustrates this model by providing a programming platform tightly coupled with Nvidia's GPUs, enhancing performance while fostering ecosystem dependence.

These economic forces often work in tension, shaping the industry's evolution. Specialization fosters the development of narrow-purpose chips tailored to specific applications, but this clashes with the benefits of standardization, which supports network effects by enabling broad applicability across platforms. Similarly, the tailored processes required for specialized designs conflict with the economies of scale that favour high-volume production of standardized chips. These tensions highlight the industry's complexity⁵⁵.

Europe's Path to Semiconductor Resilience

Europe plays a limited role in the semiconductor industry, producing only 10% of the world's supply, with very small capacity in advanced chip manufacturing. The continent relies heavily on foreign suppliers for leading-edge chips⁵⁶. As said, this reliance exposes the region to geopolitical risks.

However, Europe is not without strategic assets. Europe has approximately a 25% market share in semiconductor equipment, with key players such as **ASML**, **Bosch**, and **STMicroelectronics** driving the industry forward⁵⁷. The Netherlands-based ASML, a global leader in the production of semiconductor equipment, holds a critical position in the semiconductor ecosystem. ASML's extreme ultraviolet (EUV) lithography technology is indispensable for manufacturing advanced chips, a monopoly that gives Europe leverage in the global value chain. Yet despite ASML's present dominance, it is subject to US export controls on national security grounds and possibly also for economic reasons⁵⁸. Moreover, Europe lacks the downstream production capacity to capitalize on this advantage, particularly in areas like high-performance computing (HPC) and AI-specific chips.

To address these vulnerabilities, Europe acted decisively. The European Chips Act, with its €43 billion investment plan, aims to double Europe's share of global advanced semiconductor production to 20% by 2030 and leverage strengths in specialized equipment and areas like automotive and industrial electronics. This fits in a pattern of global efforts: the US CHIPS Act allocates €51 billion to domestic semiconductor production, while China is pursuing self-sufficiency with a €140 billion investment over five years. Nevertheless, Europe's contribution to the semiconductor industry remains uneven. While companies like STMicroelectronics, NXP Semiconductors, and Infineon excel in automotive and industrial electronics, they have limited presence in sectors driving the next wave of digital innovation. Furthermore, despite significant investments, Europe's semiconductor industry faces challenges in securing subsidies, diversifying production, and attracting talent, exemplified by the delays and uncertainties surrounding the Intel factory in Magdeburg⁵⁹.

Initiatives like SiPearl's Rhea processor and the Barcelona Supercomputing Center's investment in open-source RISC-V architecture, reflect Europe's shift toward reducing dependence on proprietary technologies like ARM. RISC-V's flexibility and free licensing can enable Europe to develop tailored

⁵⁵ W. Edward Steinmueller, 'The Economics of Flexible Integrated Circuit Manufacturing Technology', *Review of Industrial Organization* 7, no. 3/4 (1992): 327–49.

⁵⁶ European Commission, 'European Chips Act: Staff Working Document | Shaping Europe's Digital Future', 12 May 2022, <https://digital-strategy.ec.europa.eu/en/library/european-chips-act-staff-working-document>.

⁵⁷ Credence Research, 'Semiconductor Equipment Market Size, Growth and Forecast 2032', 7 October 2024, <https://www.credenceresearch.com/report/semiconductor-equipment-market>.

⁵⁸ Reuters, 'ASML CEO Says US Desire to Restrict Exports to China "Economically Motivated"', *Reuters*, 4 September 2024, sec. Technology, <https://www.reuters.com/technology/asml-ceo-says-us-motivation-restricting-equipment-exports-china-is-economically-2024-09-04/>.

⁵⁹ Politico, 'The EU's Chips Plan Implodes as Intel Pauses Investments', *POLITICO*, 17 September 2024, <https://www.politico.eu/article/intel-germany-chips-plant-competitiveness-eu-ambition/>.

software ecosystems for AI, advanced manufacturing, and high-performance computing, fostering innovation while reducing reliance on foreign intellectual property. The RISC-V is projected to grow at a compound annual growth rate (CAGR) of 33.1%, reaching €2.6 billion by 2027⁶⁰. Photonic chips offer another area of promise, leveraging light for faster processing speeds and lower energy consumption. The Spanish Institute of Photonic Sciences (ICFO) in collaboration with institutions from Ireland, the Netherlands, Finland, Belgium, Portugal, Poland, Austria, Italy, and France, is leading a €380 million project in this field. These efforts showcase Europe's ability to lead in next-generation semiconductor technologies.

Networks: The Lifelines of Connectivity

The network layer is the backbone of Europe's digital infrastructure, connecting devices, services, and systems across telecommunications and cloud environments. The public core of the Internet including the domain name management system (DNS) and Internet Exchange Points (IPX) are located here. With the rise of 5G and the ongoing evolution toward standalone 5G networks—which integrate 5G radio and core systems, and the growing role of private networks supported by satellite systems and undersea cables—this layer is undergoing a profound transformation. These networks enable low-latency, high-speed, and redundant connections that are critical for cloud platforms as well as for emerging technologies such as autonomous vehicles, industrial automation, and smart cities. Virtualization is now a defining feature of these networks, as they increasingly rely on cloud services, blurring the lines between traditional telecommunications providers and cloud providers.

Network strategic autonomy (or network sovereignty) has become a pressing issue for Europe, impacting its competitiveness, security, and resilience. This manifests especially in network equipment, undersea cables, and satellite systems, but may also become more prominent in DNS. Globally, the communications equipment market is dominated by a handful of players, namely Huawei, Ericsson, and Nokia, followed by ZTE, Cisco, Ciena and Samsung⁶¹. European manufacturers, particularly Ericsson and Nokia, excel in technological innovation, such as 6G research and pre-standardization. However, they face stiff competition from Chinese vendors, whose state-backed vendor and adoption support enables aggressive pricing strategies and rapid global market penetration. Europe's fragmented telecom market, with hundreds of mobile operators compared to the much more concentrated markets in the US and China, further complicates its ability to compete⁶². Internal market fragmentation including on security requirements reportedly limits economies of scale and reduces investment efficiency, constraining the sector's ability to innovate and deploy advanced networks⁶³.

The telecom ecosystem is shaped by equipment manufacturers, telecom operators (telcos), cloud providers, and value-added service providers (VAS) such as entertainment and e-commerce platforms. European telcos, including Deutsche Telekom, Telefónica, and Vodafone, maintain large global footprints, particularly in the US and Latin America. Despite this, European telcos face significant financial constraints. Average Revenue Per User (ARPU) in Europe is far lower than in other regions, which limits the resources available for investing in advanced networks. This financial

⁶⁰ BCC Publishing, 'RISC-V Technology Market Size, Share & Growth Analysis Report', 1 December 2022, <https://www.bccresearch.com/market-research/semiconductor-manufacturing/global-risc-v-technology-market.html>.

⁶¹ 2023 data as reported in Mario Draghi, 'The Future of European Competitiveness – A Competitiveness Strategy for Europe'.

⁶² Mario Draghi, 69.

⁶³ ETNO, 'ETNO - Future of Electronic Communications Networks in Europe - Fact-Pack' (ETNO, September 2023), <https://etno.eu/library/reports/116-future-of-electronic-communications-networks-in-europe.html>.

pressure is one factor exacerbating delays in 5G deployment and take-up, leaving Europe lagging behind the US and China.

Dominant foreign cloud and platform providers play a growing role in the network layer, providing virtualized services critical for 5G while increasingly competing with telcos. These companies are not bound by EU telecom regulations (as these address public communication providers) such as universal broadband provision, giving them an advantage, and further disrupting market dynamics. Their proprietary virtualized network solutions allow them to control critical parts of the network layer. Moreover, several of the platform providers have become main players in the telecoms backbone including undersea cables (such as Alphabet, Meta) and emerging low-earth satellite networks both for satellite launchers (SpaceX) as for satellite communications operations (Starlink), leading to further concerns about weaponizable dependencies. In the undersea cable business, there is an additional concern about the limited and concentrated capacity to repair undersea cables. The EU has initiated coordinated action to better protect undersea cables⁶⁴ and ESA to bring satellite launch capacity on-stream⁶⁵ but there is no assessment yet that this will deliver strategic autonomy as aspired.

The EU must remain vigilant in safeguarding its strategic autonomy in the evolution of the public core of the internet—this includes critical elements like DNS management (e.g., ICANN), undersea cables, and satellite networks. Discussions on proposals such as new-IP at the ITU underscore the risks of influence from major powers, which could undermine European interests. Emerging technologies like AI and blockchain are set to transform domain name management, potentially shifting the balance of control. To protect its digital sovereignty, the EU must actively engage in these developments, ensuring its values and priorities remain central in shaping the global digital landscape.

Security concerns add another dimension to the challenges facing Europe's network layer. The reliance on Chinese equipment has prompted the European Commission to issue a 5G Security Recommendation⁶⁶ leading some operators to remove Huawei and ZTE equipment from their networks. However, adoption of these measures has been uneven. Large telcos such as Deutsche Telekom and Telefónica have been slower to replace Chinese components.

To ensure digital sovereignty in 6G, comprehensive network security is needed through an architecture that allows to separate security concerns (such as for core-of-government information, learning lessons from the 5G security challenge), and adapt to emerging business models and regulatory frameworks⁶⁷. This also holds for potential new Internet designs such as SCION⁶⁸.

The security risks extend beyond 5G hardware to encompass undersea cables and – increasingly – satellite connection and AI for network management. They also go beyond cyber-threats and include physical sabotage (cutting undersea communications cables such as in the Baltic Sea, arson attacks on 5G base stations), hybrid warfare, and geopolitical disruption. This is exacerbated by climate-related risks such as the 2024 Boris flooding, disrupting both electricity and telecoms networks which are mutually highly interdependent. These evolving threats necessitate the integration of

⁶⁴ European Commission, 'Commission Recommendation on the Security and Resilience of Submarine Cable Infrastructures | Shaping Europe's Digital Future', 21 February 2024, <https://digital-strategy.ec.europa.eu/en/library/recommendation-security-and-resilience-submarine-cable-infrastructures>.

⁶⁵ Ariane 6, <https://esas.int>.

⁶⁶ European Commission, 'COMMISSION RECOMMENDATION (EU) 2019/ 534 - of 26 March 2019 - Cybersecurity of 5G Networks', *Official Journal L*, no. 88/42 (26 March 2019): 6..

⁶⁷ Paul Timmers, 'There Will Be No Global 6G Unless We Resolve Sovereignty Concerns in 5G Governance', *Nature Electronics* 2020 3:1 3, no. 1 (24 January 2020): 10–12, <https://doi.org/10.1038/s41928-020-0366-3..>

⁶⁸ For more information, see SCION's website: <https://www.scion.org>.

cybersecurity and defence policies into telecom strategies as well as addressing resilience in cross-sectoral cooperation⁶⁹.

The rise of *Open Radio Access Network (OpenRAN) technologies* offers a potential pathway for Europe to reduce dependencies and strengthen its position. OpenRAN decouples hardware and software in the access (radio) network, allowing telcos to source components from multiple vendors and fostering competition. Countries such as Canada have successfully replaced Chinese equipment with OpenRAN solutions, highlighting its potential as a secure and cost-effective alternative. For Europe, OpenRAN presents an opportunity to promote domestic innovation and diversify supply chains.

Europe's strengths in 5G, 6G R&D and pre-standardization provide a solid foundation for future competitiveness if adoption is accelerated. Supported by leading academic institutions and research organizations, European manufacturers are well-positioned to shape the next generation of network technologies. Initiatives like the announced Digital Networks Act (DNA) may address market fragmentation, enhance security, and facilitate telco consolidation, fostering a more competitive telecom sector while safeguarding consumer interests. Revising competition policies to prioritize public interests such as security and sustainability could further bolster the industry⁷⁰.

Finally, integrating advanced technologies into telecom infrastructure is essential for Europe's resilience. Quantum cryptography, including quantum key distribution (QKD) and post-quantum cryptography (PQC), will be critical for securing next-generation networks against emerging cyber threats. Additionally, the telecom sector's heavy reliance on electricity highlights the need for closer integration of energy and network strategies to ensure resilience and sustainability, as emphasized previously. Europe has an opportunity to redefine its telecom sector. With strategic investments in 6G, OpenRAN, and quantum technologies, combined with policies aimed at fostering innovation, adoption and consolidation, Europe can build a robust and secure network layer that supports its digital sovereignty, competitiveness and economic growth.

Internet of Things: The Connective Tissue of the Digital Ecosystem

The Internet of Things (IoT) bridges the physical and digital worlds by embedding sensors, processors, and connectivity into objects, enabling real-time data exchange and automation across a broad range of sectors. Connected devices consist of self-contained equipment connected to a digital communication, from heart monitors and other medtech to mobile phones. We do not consider here more complex equipment such as connected e-vehicles. The number of IoT devices is forecast to grow to 39 billion by 2033 from 18 billion in 2024⁷¹. IoT-generated data is projected to generate over 79.4 zettabytes of data annually by 2025⁷². IoT global markets are projected to reach €1.3 trillion in value by 2030⁷³. High growth sectors include healthcare (expected €60 billion IoT-driven growth by 2030) and a significant role of industrial manufacturing.

⁶⁹ Richard Feasey et al., 'Ideas for the Future of European Telecommunications Regulations | CERRE' (CERRE, 12 September 2024); Georg Serentschy, 'Digital Networks Resilience and Security, Policy Implications and Mitigation Measures - Summary', 13 January 2024.

⁷⁰ Feasey et al., 'Ideas for the Future of European Telecommunications Regulations | CERRE'.

⁷¹ Transforma, 'Current IoT Forecast Highlights - Transforma Insights', 9 December 2024, <https://transformainsights.com/research/forecast/highlights>.

⁷² Statista, 'IoT Devices Installed Base Worldwide 2015-2025', 27 November 2016, <https://www.statista.com/statistics/471264/iot-number-of-connected-devices-worldwide/>.

⁷³ Transforma Insights, 'Global IoT Market to Grow to \$1.5trn Annual Revenue by 2030 | IoT Now News & Reports', *IoT Now News - How to Run an IoT Enabled Business* (blog), 20 May 2020, <https://www.iot-now.com/2020/05/20/102937-global-iot-market-to-grow-to-1-5trn-annual-revenue-by-2030/>.

The scope of IoT and its transformative potential spans key industrial sectors such as manufacturing (smart factories), agri-food, smart energy, and mobility as well as health and environment. For instance, IoT enhances grid efficiency by enabling real-time data analysis and decision-making.⁷⁴ IoT technical system often rely on orchestration software that allows to identify, authenticate and link IoT and devices and aggregate IoT-generated data, usually in a cloud service.

Smart factories (Industry 4.0) that integrate IoT, AI/machine learning and 5G, are poised to enhance productivity and efficiency. By fostering AI-driven industrial innovation, Europe can strengthen its manufacturing base while promoting ethical and trustworthy AI applications. While Industry 4.0 focuses on automation, interconnectivity, and data exchange in manufacturing technologies, Industry 5.0 represents the next evolutionary step in industrial development (see Box Advanced Manufacturing and Robotics). Industry 5.0 emphasizes the **collaboration between humans and machines**, aiming to combine the speed and precision of automation with human creativity, critical thinking, and problem-solving skills.

Companies like Bosch and Siemens are at the forefront of deploying industrial IoT for industrial efficiency and scalability. Their initiatives demonstrate how integrating IoT with AI and data analytics can lead to significant improvements in operational performance, resource management, and energy consumption. Overall, Germany's Industrial IoT (IIoT) market is projected to grow by 10.54% annually from 2024 to 2029, reaching €12.5 billion by 2029⁷⁵. Germany's industrial IoT capabilities are bolstered by robust market growth, active academic research, and strong public-private partnerships. The strategy of Bosch and Siemens reflects a commitment to fostering open ecosystems that support widespread adoption of IoT technologies across various industries.

Europe, home to two of the world's top three telecom vendors, has a strong foundation in 5G and 6G, which enables to further step up IoT, both at larger scale and with more reconfigurability.

The EU's **Next Generation Internet of Things** initiative highlights IoT's transformational role in the digitization of the economy, society, and environment⁷⁶. As connected devices proliferate, IoT integrates data, computing, and connectivity, enabling intelligent systems that collect, process, and act on information in real time. Trusted IoT and edge computing platforms, along with effective orchestration mechanisms, are widely recognized as essential for this next phase of digitization. The European Commission, through **Horizon Europe**, is funding research and innovation to connect digital technology joint undertakings, AI and robotics partnerships, and initiatives like sector-specific data spaces and cloud-edge federations. The **Digitizing European Industry** initiative prioritizes ecosystem development, interoperability, technology integration, and standardization⁷⁷. Large-scale pilots and Technology and Experimentation Facilities (TEFs) target key sectors including agriculture, energy, manufacturing, mobility, healthcare, and smart communities, advancing Europe's IoT innovation.

⁷⁴ Sunkari Pradeep et al., 'Analysis and Functioning of Smart Grid for Enhancing Energy Efficiency Using Optimization Techniques with IoT', in *2023 IEEE 5th International Conference on Cybernetics, Cognition and Machine Learning Applications (ICCCMLA)*, 2023, 316–21, <https://doi.org/10.1109/ICCCMLA58983.2023.10346767>.

⁷⁵ Statista, 'Industrial IoT - Germany | Statista Market Forecast', June 2024, <https://www.statista.com/outlook/tmo/internet-of-things/industrial-iot/germany>.

⁷⁶ European Commission, 'The next Generation Internet of Things | Shaping Europe's Digital Future', 29 November 2024, <https://digital-strategy.ec.europa.eu/en/policies/next-generation-internet-things>.

⁷⁷ European Commission, 'The Digitising European Industry Initiative in a Nutshell | Shaping Europe's Digital Future', 16 March 2018, <https://digital-strategy.ec.europa.eu/en/library/digitising-european-industry-initiative-nutshell>.

Open-source initiatives, such as RIOT OS⁷⁸ for low-power IoT devices and the EEBus Initiative⁷⁹ for smart home and energy communication, exemplify efforts to foster collaboration, reduce reliance on proprietary platforms, and build a resilient IoT ecosystem. Meshtastic⁸⁰ is an open-source, community-driven project using inexpensive LoRa radios to create long-range, off-grid communication platforms in areas without reliable infrastructure. Arduino is another leading open-source initiative, offering microcontroller boards widely used in IoT projects to connect sensors, actuators, and other components, commercialised by companies such as Lithuania/Poland-based Cogniteq⁸¹. FIWARE defines open standards for IoT, data management, and Big Data, enabling businesses to build scalable, interoperable, and sustainable solutions free from vendor lock-in⁸². Libelium, a standout European IoT company focuses on applying IoT solutions to enhance productivity and reduce costs while minimizing environmental impact⁸³.

Challenges for Europe in IoT and connected devices.

Tightly integrated IoT ecosystems: while being strong in industrial IoT, Europe faces significant challenges in consumer IoT and connected devices, where global tech giants like Apple, Google, Huawei, and Samsung dominate and Chinese companies increasingly enter medtech. These companies often leverage vertically integrated ecosystems that combine hardware, software, and cloud services, allowing them to set *de facto* global standards and capture significant market value. Smartphone operating systems are increasingly the backbone of their IoT strategies.

Cybersecurity has become a critical issue as IoT and connected devices massively expand what is called the cyber-attack surface. The EU's Cybersecurity Resilience Act will be pivotal in shaping IoT's security regulatory environment. This is expected to drive demand for integrated hardware and software security, and data minimization (see also the data layer).

Data governance, privacy and ethical use all pose challenges in the world of IoT and connected devices. Especially in the consumer world, massive amounts of data are collected by the vendors that pursue integrated digital ecosystems (for more context, see the data layer). Despite strong EU legislation on personal data protection (GDPR), severe privacy infringements continue to happen. With the expanding presence of IoT in the home, there is a need for strong reassurance such as by voluntary certification and trust labelling. This can be turned into an opportunity for competitive differentiation if the path is eased for EU vendors to apply data protection law and ethical standards.

⁷⁸ 'RIOT - The Friendly Operating System for the Internet of Things', accessed 11 January 2025, <https://www.riot-os.org/>.

⁷⁹ 'Eebus - Homepage', EEBus - Empowering the digitalisation of Energy transition., accessed 11 January 2025, <https://www.eebus.org/>.

⁸⁰ 'Meshtastic - Introduction', accessed 11 January 2025, <https://meshtastic.org/docs/introduction/>.

⁸¹ 'Building IoT Ecosystems with Open-Source Tools | Cogniteq', 20 August 2024, <https://www.cogniteq.com/blog/building-iot-ecosystems-open-source-tools>.

⁸² European Commission, 'FIWARE – a European Success Story | Shaping Europe's Digital Future', 30 March 2017, <https://digital-strategy.ec.europa.eu/en/news/fiware-european-success-story>.

⁸³ Susana Asin, 'We were pioneers in providing IoT services', *Go Aragón* (blog), 28 September 2022, <https://www.goaragon.eu/alicia-asin-ceo-of-libelium-iot-we-were-pioneers-in-providing-iot-services/>.

Standards: related to the points above is standardisation. Europe faces strong Chinese presence in international standardization as developed in and promoted by the oneM2M industry cooperation⁸⁴. The EU currently holds a limited share of leadership roles in IoT standardization. Despite initiatives like AIOTI and StandICT.eu, Europe's influence in global IoT standards remains constrained, emphasizing the need for further engagement.⁸⁵ Unless Europe takes a proactive stance on technical interoperability and responsible international IoT governance—addressing critical issues like security and privacy—a recurrence of challenges similar to the 5G security debacle, including potential import bans, is highly likely⁸⁶.

Box 1: Advanced Manufacturing and Robotics: resilience and strategic autonomy

Advanced manufacturing and robotics are integral to Europe's competitive advantage, driving key industries such as automotive, aerospace, and industrial machinery. Europe's leadership in precision engineering, exemplified by Germany's automotive sector and its complex industrial products, underscores its potential as a global manufacturing hub. Recent disruptions in global supply chains have highlighted the importance of a robust manufacturing base for resilience and strategic autonomy. In the context of advanced manufacturing and robotics, Industry 5.0 will revolutionize production by fostering human-robot collaboration, with cobots working alongside humans to combine creativity and problem-solving with machine precision and efficiency. Robotics will become increasingly intelligent and adaptive, leveraging advanced AI, machine learning, edge computing, and 5G networks for real-time decision-making and autonomous operations. This will enable highly flexible and customizable production lines, supporting mass customization without sacrificing efficiency. Industry 5.0 robotics will optimize energy use, reduce waste, and incorporate bionic design and soft robotics to handle delicate objects safely and efficiently. Augmented reality will enhance human-robot interaction, providing intuitive visualization and control of processes.

Within this vertical we find specialized materials, such as advanced composites and alloys, supplied by European firms like BASF and ArcelorMittal. However, raw materials often come from global sources, exposing the sector to geopolitical risks. The integration of industrial IoT, which connects sensors and machines with digital systems, creates responsive and intelligent manufacturing ecosystems. European companies like Siemens lead in creating digital twins and industrial metaverses, transforming factory processes through virtual planning and resource management.

Robotics and factory automation are also areas of European strength. Swiss ABB Robotics is a global leader but faces stiff competition from Japanese companies like Fanuc and Yaskawa Electric. While the integration of AI-driven software for process automation is an emerging field, companies like Romanian UiPath showcase Europe's potential. Here too, however, the reliance on US cloud providers for real-time analytics and automation remains a vulnerability.

⁸⁴ 'oneM2M Sets Standards For The Internet Of Things & M2M', accessed 11 January 2025, <https://www.onem2m.org/>.

⁸⁵ https://standict.eu/news/iot-standards-landscape-report?utm_source=chatgpt.com

⁸⁶ The White House, 'FACT SHEET: Protecting America from Connected Vehicle Technology from Countries of Concern', The White House, 23 September 2024, <https://www.whitehouse.gov/briefing-room/statements-releases/2024/09/23/fact-sheet-protecting-america-from-connected-vehicle-technology-from-countries-of-concern/>.

Cloud: The Operational Core of the Digital Economy

Cloud infrastructure has become far more than a tool for data storage or application hosting; it is now the critical backbone for digital economy. Serving as the ‘power grid’ of the new economy, it constitutes a foundational production environment, to transform industries, public services, and operational models into tightly integrated ecosystems⁸⁷. This shift has entrenched the dominance of a few global tech giants who now wield extraordinary influence over critical digital infrastructure. The Draghi Report (September 2024) highlights the vital role of cloud infrastructure in ensuring Europe’s competitiveness and strategic autonomy. Cloud is not mentioned in the mere context of a digital strategy, but as holistic economy enabler. The report addresses the region’s gap in cloud technology provisioning and adoption, which has deepened the dependency from foreign providers, and weakened data sovereignty. To counter this, the report proposes a unified strategy through a New EU Cloud and AI Development Act, focusing on high-performance computing (HPC), artificial intelligence (AI), and quantum technologies, alongside harmonized standards, and procurement processes⁸⁸.

Recent data highlights the increasing reliance on cloud services. In 2023, European spending on public cloud services was projected to reach €139 billion, with expectations to grow to €285 billion by 2027⁸⁹. Thirty percent of companies now store 41–60% of their data in the cloud, while 22% rely on the cloud for more than 60%.⁹⁰ Despite the adoption of multi-cloud strategies to mitigate risks, these efforts have not disrupted the dominance of leading cloud providers. Amazon Web Services (AWS), Microsoft Azure, and Google Cloud control almost 70% of the global Infrastructure-as-a-Service (IaaS) market, with AWS holding 31%, Azure 24%, and Google Cloud 11% as of Q4 2024.⁹¹ On the other side, European Cloud Service Providers (CSPs) have cumulatively seen a loss of market share, from the 26% of 2017, to a 10%, in a constant decline that will reach the point of no return in 3 to 5 years, around 2030.

As of 2021, EU cloud providers held less than 16% of the market, with Deutsche Telekom—the largest—capturing just 2%⁹². Most European players focus on basic IaaS offerings and often rely on reselling PaaS solutions from dominant US providers like Amazon, Microsoft, and Google, limiting their competitiveness. Non-EU providers such as IBM, Oracle, and Alibaba Cloud also claim portions of the market but lack the scale to challenge US giants. European firms like OVHcloud, IONOS, and Aruba Cloud face significant constraints in competing on resource capacity, portfolio breadth, and innovation speed.

In response, some European providers have begun challenging unfair market practices. For example, in July 2024, the Cloud Infrastructure Services Providers in Europe (CISPE) and Microsoft resolved a dispute over licensing practices, with Microsoft committing to enhance Azure Stack HCI the hybrid, multi-tenant version of the public Azure cloud stack) for European providers as part of the settlement. While this settlement seeks to balance the playing field in the European cloud market, major providers

⁸⁷ BEREC, ‘Draft BEREC Report on Cloud and Edge Computing Services’, 12 March 2024, <https://www.berec.europa.eu/en/document-categories/berec/reports/draft-berec-report-on-cloud-and-edge-computing-services>.

⁸⁸ Mario Draghi, ‘The Future of European Competitiveness – A Competitiveness Strategy for Europe’.

⁸⁹ https://www.idc.com/getdoc.jsp?containerId=prEUR151144823&utm_source=chatgpt.com

⁹⁰ IDC, ‘Worldwide Quarterly Enterprise Infrastructure Tracker: Buyer and Cloud Deployment’, IDC: The premier global market intelligence company, 2024, https://www.idc.com/getdoc.jsp?containerId=IDC_P31615.

⁹¹ Synergy Research Group, ‘Cloud Market Gets Its Mojo Back; AI Helps Push Q4 Increase in Cloud Spending to New Highs’.

⁹² Digitalisation World, ‘European Cloud Providers Double in Size but Lose Market Share’, Digitalisation World, 28 September 2021, <https://digitalisationworld.com/news/62320/european-cloud-providers-double-in-size-but-lose-market-share>.

like Google and AWS were excluded, and fresh complaints highlight persistent concerns. If Microsoft fails to fulfil its commitments within nine months, CISPE reserves the right to refile its case.⁹³ But, without putting in place proactive actions and an industrial plan to protect, reinforce, and increase the size, capacity, and competitiveness of individual European providers in the market, they will not be considered a viable alternative by the end users.

In the SaaS market, businesses increasingly rely on tailored applications, with recent data showing organizations use an average of 130 SaaS applications as of 2022. This reflects the growing specialization of tools to enhance productivity and efficiency, with some companies relying on over 100 applications to meet diverse operational needs⁹⁴. Despite its fragmented nature, the SaaS market remains heavily integrated into the ecosystems of dominant IaaS providers, reinforcing their control over the digital stack. To address this, the open-source software ecosystem (outlined in the next chapter) offers a strategic alternative to reduce dependency on Big Tech's proprietary solutions. However, significant private investments are needed to support re-engineering programs, which are challenging to secure due to the uncertainty of their business benefits.

The Dual Influence of Cloud Providers

Leading cloud providers dominate the digital economy by combining physical infrastructure with advanced software platforms, creating tightly integrated ecosystems. This integration simplifies tasks for software development teams through seamless hardware/software compatibility and powerful CI/CD (Continuous Integration/Continuous Deployment) tools. Computational infrastructures, i.e., cloud plus end devices, are not products but *production environments* for digital services⁹⁵. At the same time, business decision-makers are incentivized to choose these global platforms for their reliability and widespread adoption, often overlooking smaller local providers perceived as less scalable or reliable. Whether organizations avoid big cloud providers due to trust concerns, bypass smaller providers for scalability reasons, adopt off-the-shelf solutions, or re-engineer applications for cloud migration, the influence and control of Big Tech in the market remain substantial.

At the physical and computational layer, providers like Amazon Web Services (AWS) and Google offer the raw computational power for high-demand workloads, including AI. Leading cloud providers do not rely on standard commercial computing technologies (e.g., compute servers, storage servers, or network routers). Instead, they design custom hardware to optimize costs and performance while reducing dependencies on third-party suppliers. The rapid growth of AI technology has, however, created significant dependencies even for Big Tech, particularly on single-source chip suppliers like NVIDIA, due to a lack of viable alternatives. This reliance has driven a new industrial strategy, focusing on the development of proprietary technologies encompassing both software stacks and chips to protect margins and retain customer loyalty.

Building on the foundational computing infrastructure, the **software and platform layer** transforms how ordinary businesses operate daily. These platforms are one-stop-shop solutions offering application development tools, integrated marketplaces, and advanced machine learning capabilities. For example, services like Google's TensorFlow and AWS's SageMaker provide not just the underlying computing power, but also pre-configured development environments, specialized

⁹³ Ben Maynard, 'CISPE and Microsoft Agree Settlement in Fair Software Licensing Case', accessed 11 January 2025, <https://cispe.cloud/cispe-and-microsoft-agree-settlement-in-fair-software-licensing-case/>.

⁹⁴ Statista, 'Average Number of Software as a Service (SaaS) Applications Used by Organizations in the US from 2015 to 2024', n.d., https://www.statista.com/statistics/1233538/average-number-saas-apps-yearly/?utm_source=chatgpt.com.

⁹⁵ Agathe Balayn and Seda Gürses, "Misguided: AI Regulation Needs a Shift in Focus," *Internet Policy Review* 13, no. 3, September 30, 2024, <https://policyreview.info/articles/news/misguided-ai-regulation-needs-shift/1796>

software libraries, and automated workflows. As a result, businesses rely on cloud providers to create and deploy applications for everything from sales forecasting to customer service chatbots, embedding a firm's most essential operations directly into the provider's ecosystem. Consequently, organizations become deeply dependent on these platforms, since migrating away would mean not only transferring data, but also rebuilding the core architecture of their applications. In addition, customers become dependent on the willingness of platform providers to support 'baseline security-by-default'⁹⁶.

This model of end-to-end integration through the technology stack, from SW applications (SaaS) to SW frameworks (PaaS), down to HW (IaaS), has strengthened the positioning in the market and amplified Big Tech's influence across industries. These include healthcare, finance, manufacturing, and automotive, where the exponentially increasing amount of data, has imposed the need for more integrated, auto-scalable, and therefore cloud-native, platforms for predictive analytics, diagnostics, and operational optimization. Notable examples include Volkswagen's partnership with Microsoft to streamline production data and Renault's collaboration with Google Cloud to develop Software-Defined Vehicles. These alliances underscore the growing reliance on centralized cloud providers.

Partnerships between major telecom providers and Big Tech cloud platforms have driven the market's evolution. Rising operational costs and competition led telecoms to focus on cloud-based value-added services, forming strategic alliances to avoid costly proprietary infrastructure investments. Examples include Deutsche Telekom and Google, Telecom Italia and Google, and Orange and Microsoft with Bleu. These partnerships have deeply tied European businesses to Big Tech platforms, making it challenging to untangle dependencies due to significant business and technical barriers.

The Physical Infrastructure of the Cloud

The rapid expansion of cloud computing and AI technologies underscores the importance of physical infrastructure for Europe's digital future. Data centres, high-performance computing (HPC) clusters, and their energy and cooling systems, long-distance and undersea cables and soon even satellite systems (Starlink of Musk's SpaceX, Kuiper of Amazon) enable these advancements but also expose Europe to critical dependencies, sustainability challenges, and geopolitical risks.

Investment in data centres has surged globally, with private US spending increasing from €1.8 billion in 2014 to €17.8 billion in 2023. The US now hosts one-third of the world's data centres. HPC clusters like those within Europe's EuroHPC Joint Undertaking seek providing a sustainable and sovereign alternative⁹⁷. Finland's LUMI supercomputer, powered by hydropower from Sweden's Vattenfall, demonstrates the potential for green computing. Expanding EuroHPC into a public alternative for computing power could enhance European autonomy, fostering innovation, and aligning with climate goals. However, the significant power demands of AI computing, the rapid obsolescence of AI technologies, and the dominance of the private sector in developing foundational models and inference networks highlight the need for **sustainable, long-term investment strategies** with appropriate governance models. The HPC market will reach approximately €85.56 billion by 2030, with a compound annual growth rate (CAGR) of 7.5% from 2023 to 2030⁹⁸. The performance of the world's fastest compute has grown by a factor of 626 since 2010, with even larger increases in

⁹⁶ See: 'Security By Default - Homepage', accessed 12 January 2025, <https://securitybydefault.org/>.

⁹⁷ 'EuroHPC JU - Homepage', 19 December 2024, https://eurohpc-ju.europa.eu/index_en.

⁹⁸ Grand View Research, 'High Performance Computing Market Worth \$87.31 Billion By 2030', accessed 11 January 2025, <https://www.grandviewresearch.com/press-release/global-high-performance-computing-hpc-market>.

compute requirements⁹⁹. Europe's success in this field is evident, with **three of the world's top 10 supercomputers**—a remarkable leap from having none in 2016¹⁰⁰.

The concentration of data centres in a few countries¹⁰¹ and competition to attract infrastructure often leads to tax breaks, weakened privacy or environmental standards and lack of inclusive growth¹⁰², resulting in a “race to the bottom” that undermines job creation, trade, and long-term economic growth¹⁰³.

Sustainability looms large. Data centres, driven by IoT data growth and AI applications face increasing environmental pressure, with cooling systems accounting for up to **40% of energy consumption** in some facilities. From 2015 to 2020, data centre energy use in Europe rose by **10%**, and by 2022, they consumed **2.7% of Europe's electricity**,¹⁰⁴ including almost **20% of Ireland's national consumption**.¹⁰⁵ Google's hyperscale data centres used an average of **2.1 million litres of water daily** in 2023 for cooling.¹⁰⁶ The widespread use of renewable energy certificates (RECs) often obscures true environmental costs, calling for stricter transparency. The rapid expansion of data centres in Europe has also raised the call for micro-nuclear plants (small modular reactors or SMRs) to meet future AI computing demands. However, these advancements also intersect with Europe's ongoing taxonomy debate, which seeks to define the environmental sustainability of emerging technologies and energy sources. The inclusion of nuclear energy and quantum computing under 'green' or 'sustainable' classifications has sparked contention among policymakers, reflecting broader concerns about balancing innovation with ecological and social objectives¹⁰⁷. Green computing standards, emphasizing renewable energy, efficient cooling, and reduced water use, are essential to align data centre growth with carbon neutrality targets.

Even if still years away, the use of quantum computing with its exponential computing power, can also help reducing power demand of data centres. Quantum computing has the potential to reduce the power demand of data centres indirectly by enabling highly efficient algorithms for certain classes of problems. This efficiency gains are based on algorithmic efficiency, reduced hardware needs and the use of enhanced machine learning and AI.

European Challenges and Opportunities in Cloud and Edge Infrastructure

European cloud service providers are making significant investments in infrastructure and services that emphasize data sovereignty and compliance with EU regulatory standards. By aligning their

⁹⁹ Jaime Sevilla et al., ‘Compute Trends Across Three Eras of Machine Learning’, 2022, <https://doi.org/10.48550/ARXIV.2202.05924>.

¹⁰⁰ TOP500, ‘November 2024 | TOP500’, 64th edition of the TOP500, November 2024, <https://top500.org/lists/top500/2024/11/>.

¹⁰¹ Jukka Ruohonen, *Geospatial Insights on the EuroHPC Supercomputing Ecosystem*, 2024, <https://doi.org/10.31219/osf.io/z94f2>.

¹⁰² Enrico Letta, ‘Enrico Letta's Report on the Future of the Single Market - European Commission’.

¹⁰³ I. Papadakis and M. Savona, ‘The Uneven Geography of Digital Infrastructure: Does It Matter?’, November 2024, <https://leap.luiss.it/publication-research/publications/i-papadakis-m-savona-the-uneven-geography-of-digital-infrastructure-does-it-matter/>.

¹⁰⁴ Directorate-General for Energy, ‘Commission Adopts EU-Wide Scheme for Rating Sustainability of Data Centres - European Commission’, accessed 12 January 2025, https://energy.ec.europa.eu/news/commission-adopts-eu-wide-scheme-rating-sustainability-data-centres-2024-03-15_en.

¹⁰⁵ European Commission. Joint Research Centre., *Energy Consumption in Data Centres and Broadband Communication Networks in the EU*. (LU: Publications Office, 2024), <https://data.europa.eu/doi/10.2760/706491>.

¹⁰⁶ Zhang, ‘Data Center Water Usage’.

¹⁰⁷ Martina Pilloni, ‘The Nuclear Debate and Energy Taxonomy in the European Union | Heinrich-Böll-Stiftung | Tel Aviv - Israel’, 22 March 2022, <https://il.boell.org/en/2023/03/30/nuclear-debate-and-energy-taxonomy-european-union>.

offerings with European data protection regulations, such as GDPR, and operating entirely within European territory, these providers ensure their services remain insulated from non-European jurisdictions, including the implications of legislation like the US CLOUD Act.

France's OVHcloud positions itself as a secure alternative to non-European providers by emphasizing GDPR compliance and interoperability. Outscale, in partnership with Docaposte and Dassault, launched NUMSPOT as a sovereign cloud focused on public administration and defence services. Scaleway has shifted its business model from traditional data centres to exclusively public cloud services, pursuing growth through acquisitions.

In Italy, Aruba Cloud combines retail and enterprise cloud offerings with leading trust services (e.g., digital identity) and proprietary European data centres powered by renewable energy. Germany's IONOS, the largest European cloud service provider (CSP) with over €6 billion in revenue, plans to expand across Europe. StackIT, a subsidiary of Germany's Schwarz Group, has followed Amazon's AWS model by opening its IT infrastructure as a cloud platform, aiming to become the largest EU CSP. Hetzner focuses on cost-effective bare-metal servers, while Leaseweb, based in the Netherlands, offers a diverse portfolio like IONOS and Aruba.

Most European CSPs operate on a small scale: only a few, like IONOS, OVHcloud, have market capitalizations exceeding €1 billion, while others, such as Aruba and Leaseweb, generate revenues of €3–5 million. Scaleway and a handful of others exceed €100 million, but the majority remain below €50 million, with many under €20 million. This fragmented industry is further overshadowed by large European telecom providers offering cloud services based on US hyperscalers like Microsoft Azure, Google Cloud, or Amazon AWS.

Deutsche Telekom and TIM have deepened partnerships with US tech giants like Microsoft and Google to develop cloud and edge solutions. At the same time, proprietary tech vendors like the US-based VMware (now part of Broadcom) have doubled their efforts in Europe to present themselves as providers of "Sovereign Cloud" solutions, as do some of the hyperscalers. The European cloud market has also become a focal point for substantial investments by global technology giants, highlighting both opportunities and challenges for the region's digital sovereignty.

In Germany, AWS announced a €17.8 billion investment through 2040, including €7.8 billion for its European 'Sovereign Cloud' in Brandenburg. In France, Microsoft committed €4 billion to expand AI capabilities, including advanced cloud infrastructure and GPU deployment. Meanwhile, in Spain, Oracle plans to invest \$1 billion in AI and cloud computing, while AWS has committed €15.7 billion to data centres in Aragon over the next decade. A sharpening of the definition of 'sovereign cloud' should counter the trend towards 'sovereign-washing'.

While these examples of collaborations address immediate market needs, they heighten reliance on foreign providers subject to US legislation, such as the Cloud Act. This law allows US authorities to access data stored by American companies, regardless of its physical location, raising significant concerns about data sovereignty and security in Europe. Such dependencies, compliance, and regulatory challenges¹⁰⁸ expose critical infrastructure to external legal jurisdictions and undermine Europe's strategic autonomy, particularly in sensitive sectors such as government, aerospace and defense.

¹⁰⁸ Dhruv Seth, Madhavi Najana, and Piyush Ranjan, 'Compliance and Regulatory Challenges in Cloud Computing: A Sector-Wise Analysis', *International Journal of Global Innovations and Solutions (IJGIS)*, 1 June 2024, <https://doi.org/10.21428/e90189c8.68b5dea5>.

The market control of cloud providers is partially addressed by the Digital Market Act (DMA) where these cloud providers are also gatekeeper platforms providers. The DMA is intended to ensure fair competition and reduce gatekeeper dominance. Under the DMA, companies classified as gatekeepers must adhere to specific obligations and limitations, such as ensuring fair access to their platforms, preventing self-preferencing, and promoting interoperability between different services. While it is early to assess the effectiveness of the regulatory regime, this alone is unlikely to address the competitiveness gap of vertically disintegrated European cloud providers.

To address these structural weaknesses, Europe has launched several ambitious policy initiatives¹⁰⁹. **Gaia-X**, initiated by Germany and France in 2019, was presented as a key component of Europe's strategy to reduce dependence on global hyperscalers and regain digital sovereignty. Designed as a federated cloud infrastructure built on shared European standards, Gaia-X has focused on addressing market fragmentation by fostering interoperability and trust. While it has succeeded in generating extensive awareness about this topic and in establishing a collaborative framework and several pilot projects, its reliance on non-European hyperscalers by many participants has created confusion about its contribution to digital sovereignty. Additionally, its broad scope, its late tendency to overlook the cloud infrastructure layer, and its slow progress have raised questions about its ability to deliver tangible results¹¹⁰. However, other initiatives such as 8ra¹¹¹ and the IPCEI-CIS can take GAIA-X achievements further with clear focus on EU strategic autonomy.

The new Important Project of Common European Interest on Next Generation Cloud Infrastructure and Services (IPCEI-CIS), approved in 2023 with €3 billion in funding (half of which co-investments of private companies), has taken a different approach. Focused on open-source innovation, it aims to create the first interoperable, openly accessible cloud-edge computing continuum in Europe. Backed by over 100 companies and research institutions across 12 EU Member States, the initiative provides €1.2 billion in state aid and is expected to leverage an additional €1.4 billion in private investments. The goal is to develop a decentralized, vendor-neutral platform that integrates data centres, public clouds, and edge computing, emphasizing multi-cloud interoperability, application portability, and low-latency solutions for critical sectors like energy, mobility, manufacturing, and tourism. The IPCEI-CIS initiative does not cover yet hardware or infrastructure investments, still failing to meet expectations for building a truly European cloud infrastructure. Additionally, it would need the creation of a legal entity or federation to commercialize pooled services, which could be addressed by 8ra.com.

Recognizing these gaps, the European Commission is planning new waves of IPCEIs to address infrastructure development and establish a commercial offering. However, the average timeline for such projects—3-5 years from inception to delivery—risks delaying tangible results until 2030, by which time EU CSPs may have lost significant market share. To address scalability challenges new IPCEIs will support the development of a large-scale federated edge computing infrastructure and distributed AI services¹¹². These efforts aim to amplify the impact of “made in Europe” edge cloud technologies developed under IPCEI-CIS, driving innovation and competitiveness across the EU.

¹⁰⁹ European Commission, ‘Cloud Computing | Shaping Europe’s Digital Future’, 2022, <https://digital-strategy.ec.europa.eu/en/policies/cloud-computing>.

¹¹⁰ Francesco Bonfiglio, XI. Why Europe’s Cloud Ambitions Have Failed, 15 October 2024, <https://ainowinstitute.org/publication/xi-why-europes-cloud-ambitions-have-failed>.

¹¹¹ See: ‘8ra - Homepage’, 8ra, accessed 11 January 2025, <https://www.8ra.com/>.

¹¹² Joint European Forum for IPCEIs, ‘Opinion of the Joint European Forum for Important Projects of Common European Interest’, 27 November 2024, https://competition-policy.ec.europa.eu/document/download/3d01ea9f-2c29-4f83-a66f-44f2e345c015_en?filename=JEF-IPCEI_Opinion%20of%20the%20high-level%20meeting_27%20November%202024.pdf.

The initiative's decentralized software infrastructure will reduce technological dependencies and lock-in effects while fostering innovative, data-driven business models in areas like artificial intelligence and Industrial Internet of Things (IIoT). By leveraging decentralization and federation, Europe can transform its heterogeneity into a strategic advantage, interconnecting European cloud providers into a diverse but scalable computing continuum, promoting the emergence of “proximity cloud” service providers across the EU with strong local roots and a sense of social responsibility.

Software: The Command Centre of Platforms and Applications

The software layer is at the core of digital infrastructure, spanning operating systems, application platforms, and algorithmic frameworks. It orchestrates the running of applications and services that use the functions and facilities of these systems, platforms, and frameworks. This includes common building-block functions such as identity and access management, electronic payments, transactions, and document delivery. In essence, software “runs the world”—yet Europe has no comprehensive software industrial strategy to speak of. US companies dominate foundational tools that underpin modern economies and societies, leaving Europe's digital sovereignty heavily contested.

Operating systems for desktops, mobile devices, and embedded systems are monopolised by three US companies—Microsoft, Apple, and Google—controlling over 90% of European markets. Microsoft Windows powers more than 70% of desktops¹¹³, Google's Android commands 72% of Europe's mobile market, and Apple's iOS dominates high-value segments¹¹⁴. Although open-source Linux has a global impact in servers, cloud environments, and the internet's core infrastructure, its adoption in consumer-facing systems remains limited. Efforts to develop Linux-based national operating systems have struggled to achieve scale, yet Linux's flexibility, security, and alignment with European values make it a critical asset for enhancing digital strategic autonomy.

Application platforms present a mixed picture. Europe boasts global leaders like SAP and Dassault Systèmes in enterprise software and industrial tools, with SAP leading the global Enterprise Resource Planning (ERP) software market with a 6.2% share in 2023.¹¹⁵ However, even these platforms depend on foreign-controlled operating systems and middleware, limiting their potential to function as fully sovereign alternatives. Moreover, in many key applications —office productivity, messaging, collaboration—the dominance of US companies is uncontested. However, open-source platforms developed by the private sector (Nextcloud) or through a public-private partnership (openDesk, La Suite Numérique) represent a good model to build European alternatives.

Similarly, Europe's record in developing and adopting **common services** is uneven. Secure eID and secure authentication methods to use government services digitally, successful in several countries such as Estonia and Belgium, sees limited uptake in Germany. Nonetheless, the EU's long-term investments across the Internal Market, such as the EU Connecting Europe Facility (CEF) funding, demonstrate that common services can evolve into scalable digital infrastructures. Examples include the e-invoicing based on the PEPPOL system that is becoming mandatory in ever more EU countries,

¹¹³ StatCounter, ‘Desktop Operating System Market Share Europe’, November 2024, https://gs.statcounter.com/os-market-share/desktop/europe?utm_source=chatgpt.com.

¹¹⁴ Statista, ‘Mobile OS Market Share Europe 2010-2023’, 7 March 2024, https://www.statista.com/statistics/639928/market-share-mobile-operating-systems-eu/?utm_source=chatgpt.com.

¹¹⁵ Albert Pang, Misho Markovski, and Ristik Marija, ‘Top 10 ERP Software Vendors, Market Size and Market Forecast 2023-2028’, Apps Run the World, 10 June 2024, <https://www.appsruntheworld.com/top-10-erp-software-vendors-and-market-forecast/>.

business reporting, document delivery, translation, next to e-signatures and timestamping, with EU-wide legal recognition under regulations like the eIDAS2 Regulation¹¹⁶.

Despite their open-source nature, **algorithmic frameworks** such as TensorFlow and PyTorch, remain tethered to US-based ecosystems, influencing developer communities, aligning innovation with proprietary platforms like Google Cloud or Meta's AI infrastructure, and reinforcing dependencies on foreign systems¹¹⁷. Structural integration across operating systems, application platforms, and algorithmic frameworks creates high switching costs, further entrenching dependency and lock-in. Consequently, much of the value generated by these systems flows out of Europe, weakening influence over global standards and eroding tax revenues that could otherwise fuel local innovation.

Applications are what the end-user interacts with in daily use of mobile phone, laptop, or professional equipment. They span a huge range of use cases, from personal navigation to inventory management, from health monitoring to traffic control systems. There are many well-known European companies such as Spotify and SAP, but generally with relatively low capitalisation compared to foreign competitors. Over the years entrepreneurial EU application start-ups have often expanded globally, thanks for successful capitalization which, however, is often by foreign funds and larger companies.

Europe's Role in Software

Beneath this external dominance lies an “invisible” European strength: Europe has a longstanding presence in open-source software (OSS) and open standards that underpin critical digital infrastructures. From Linux and Python to core internet protocols, Europe's contributions to open source are notable. European companies like SAP have long been global leaders in business systems, and Europe also shows promise in AI software, embedded security, and cybersecurity forensics—albeit on a smaller scale than the US or China.

Open-source software, which constitutes 70-90% of modern codebases, is a cornerstone of software development and a promising pathway to strengthening Europe's digital sovereignty.¹¹⁸ OSS enables decentralised, interoperable online services that can replace today's centralised and surveillance-based platforms. Beyond its technical advantages, OSS aligns with Europe's vision of transparency, collaboration, and interoperability. Public investments in OSS have shown astonishing returns: according to a 2021 European Commission study, €1 billion in OSS investment can yield up to €95 billion in economic impact.¹¹⁹ This makes it not only a technological enabler but also a strategic economic driver, capable of fostering innovation, reducing dependencies, and strengthening Europe's global competitiveness. By leveraging and investing in OSS, Europe can accelerate its digital sovereignty agenda while fostering an open and sustainable technological landscape.¹²⁰

¹¹⁶ European Commission, 'eGovernment and Digital Public Services | Shaping Europe's Digital Future', 28 November 2024, <https://digital-strategy.ec.europa.eu/en/policies/egovernment>.

¹¹⁷ Amba Kak, Sarah Myers West, Meredith Whittaker, 'Make No Mistake—AI Is Owned by Big Tech', MIT Technology Review, 5 December 2023, <https://www.technologyreview.com/2023/12/05/1084393/make-no-mistake-ai-is-owned-by-big-tech/>.

¹¹⁸ Jameson Perlow, 'A Summary of Census II: Open Source Software Application Libraries the World Depends On', Linux Foundation, 7 March 2022, <https://www.linuxfoundation.org/blog/blog/a-summary-of-census-ii-open-source-software-application-libraries-the-world-depends-on>.

¹¹⁹ OpenForum Europe and Fraunhofer ISI, 'Open Source Study', OpenForum Europe, 2021, <https://openforumeurope.org/open-source-impact-study/>.

¹²⁰ Content and Technology European Commission: Directorate-General for Communications Networks et al., *The Impact of Open Source Software and Hardware on Technological Independence, Competitiveness and Innovation in the EU Economy – Final Study Report* (Publications Office, 2021), <https://doi.org/10.2759/430161>.

However, the sustainability and governance of Europe's OSS ecosystem need bolstering. Funding remains fragmented, and reliance on non-European platforms like GitHub compromises sovereignty. Public procurement often overlooks open-source solutions in favour of proprietary ones, while Big Tech companies tactically leverage OSS to reinforce their dominance¹²¹—acquiring platforms like GitHub, mixing proprietary code into open environments (as with Android or Chromium), and profiting from community-driven innovations without adequately giving back. The European Commission's **Open-Source Software Strategy** has laid the groundwork for addressing this, emphasizing the need for equal treatment of OSS in procurement and fostering a level playing field. Additionally, efforts such as the **Open-Source Observatory (OSOR)** and the **Interoperable Europe Act** underscore Europe's dedication to open-source integration.

The success of these initiatives depends on sustained investment, effective governance, and alignment with market realities to counteract the dominance of proprietary solutions. Power dynamics within open-source settings are often obscured, with some actors disproportionately reaping the benefits. For instance, Microsoft, once a vocal opponent of open-source,¹²² has dramatically shifted its stance over the years, culminating in its acquisition of open-source platform GitHub.

Big Tech companies increasingly take advantage of open-source communities by sharing portions of their code while keeping their core business proprietary. By doing so, they gain access to cost-effective, high-quality feedback and improvements, leveraging the community's collaborative efforts to their advantage. For example, Google's management of Android and Chromium, mixing proprietary components and features into the open environment, allows them to entrench dominance in key markets and shape browser development and web standards¹²³. Amazon's forking of Elasticsearch allowed them to avoid contributing to the original project while profiting from its functionality, raising concerns about the sustainability of open-source initiatives. Oracle's contentious stewardship of Java, particularly its legal battles over Java APIs in Android, underscores the conflicts that arise when proprietary interests intersect with widely used open-source technologies. This trend is also evident in the rise of so-called open AI models, where foundational research and tools are released as open source, yet the resulting implementations are frequently embedded into proprietary platforms, reinforcing control by dominant players¹²⁴.

These practices underscore the need for policies that ensure open-source principles are not diluted by corporate interests, preserving the integrity and accessibility of digital commons. To counter this trend, Europe must build robust, independent open-source ecosystems that prioritize decentralization, interoperability, and alignment with its regulatory and ethical standards.

Successful examples exist. Europeana institutionalises data sharing in cultural heritage, and the European Open Science Cloud (EOSC) provides a trusted, sovereign framework for open science. Initiatives like Decidim thrive because they are not only open source but supported by long-term institutions ensuring their growth¹²⁵. Future opportunities include investing in Linux-based systems for public administration, leveraging open-source solutions in key sectors like healthcare and smart cities, and championing federated AI models. Decentralised networks based on open protocols can

¹²¹ Alice Pannier, 'How Big Tech Is Shaping the Global Open Source Ecosystem | SovereignEdge.EU', *SovereignEdge* (blog), 22 May 2023, <https://sovereignedge.eu/blog/how-big-tech-is-shaping-the-global-open-source-ecosystem/>.

¹²² Then Microsoft CEO Steve Ballmer famously declared that "*Linux is a cancer that attaches itself in an intellectual property sense to everything it touches.*" See: Thomas C. Greene, 'Ballmer: "Linux Is a Cancer"', *The Register*, 2 June 2001, https://www.theregister.com/2001/06/02/ballmer_linux_is_a_cancer/.

¹²³ David McCabe and Nico Grant, 'What's Next for Google's Search Monopoly', *The New York Times*, 21 December 2024, sec. Technology, <https://www.nytimes.com/2024/12/20/technology/google-antitrust-case-chrome.html>.

¹²⁴ David Gray Widder, Meredith Whittaker, and Sarah Myers West, 'Why "Open" AI Systems Are Actually Closed, and Why This Matters', *Nature* 635, no. 8040 (November 2024): 827–33, <https://doi.org/10.1038/s41586-024-08141-1>.

¹²⁵ See: 'Decidim, Free Open-Source Democracy - Homepage', n.d., <https://decidim.org/>.

offer privacy-first, transparent alternatives in messaging and social media, while regulatory tools such as the Digital Markets Act, Digital Services Act, Media Freedom Act, and the Interoperable Europe Act help ensure that competition and innovation can flourish within a trusted, pluralistic digital ecosystem. Initiatives such as the **EDIC (European Digital Infrastructure Consortium)** offer a pathway for building open-source solutions in areas of public interest, from access to public services to key intermediation technologies.

Aligning digital commons with industrial strategies and leveraging public institutions as strategic supporters and early adopters can consolidate these gains. Europe can scale funding for OSS-based AI projects, digital infrastructure consortia, and common services for eID and e-invoicing. Europe's public institutions lack capabilities to understand and address critical dependencies in the digital stack. Yet, institutions like the proposed **Sovereign Tech Agency** and initiatives such as **FOSSEPS** and bug bounty programs represent important steps forward. The German **Sovereign Tech Agency**¹²⁶ is a promising model for funding the maintenance of digital commons while strategically driving their development. However, the related Sovereign Tech Fund, with a total allocation of only EUR 11.5 million for 2023, highlights the challenge of limited funding. This modest budget underscores the need for more substantial financial commitments to ensure the sustainability and impact of open-source software. In the 2024 German Federal budget, EUR 17 millions were allocated to the Sovereign Tech Fund. Public institutions must ensure long-term support and governance for open-source ecosystems, reducing reliance on external partners and safeguarding Europe's technological independence.

On the other hand, the French Ministry of Public Transformation and Civil Service has launched an ambitious "Free Software and Digital Commons" action plan, spearheaded by the newly established Free Software Unit within the Inter-ministerial Directorate for Digital Affairs (DINUM). This initiative aims to promote the adoption of free software and digital commons across public administration, encourage the release and publication of source codes, and enhance the State's appeal as a digital employer by recognizing public contributions to open-source projects.

Using agile methodologies, the French government is developing collaborative products such as *La Suite* and *Albert AI*. *La Suite* is now actively utilized by approximately 300,000 public servants daily. The ambitious goal is to scale these solutions to serve millions while fostering the reuse of their modular building blocks—each representing an active open-source community—within the private sector as well. This initiative exemplifies the critical role of public administrations as orchestrators of open-source ecosystems¹²⁷, and early adopters of digital commons¹²⁸, creating a robust foundation for innovation and collaboration in both the public and private sectors. This method prototyped by Governments acting as lead users for public innovation, show us that success it's not just about investment—it's about adopting a necessary product-driven approach. Agile methodologies enable the capture of new users and use cases, emphasizing the importance of channeling resources through digital commons communities and public incubators to foster innovation and scalability. However, we must remain cautious when referring to "digital assets." These assets should either represent active communities with strong user bases—such as Matrix or Firefox—or risk being irrelevant without the foundational support of vibrant ecosystems. Building and sustaining these communities is essential for ensuring meaningful impact and long-term success.

¹²⁶ See: 'Sovereign Tech Agency - Home', 6 December 2024, <https://www.sovereign.tech/>.

¹²⁷ 'Action Plan for Free Software and Digital Commons', Direction interministérielle du numérique, accessed 12 January 2025, <https://code.gouv.fr/en/action-plan-for-free-software-and-digital-commons/>.

¹²⁸ See: 'Mission Société Numérique - Homepage', accessed 12 January 2025, <https://societenumerique.gouv.fr>.

Smart public procurement and sustained investment will ensure that Europe's software ecosystem becomes a foundation of its digital sovereignty, transforming open source and digital commons into lasting strategic assets that underpin Europe's influence, resilience, and global competitiveness.

Box 2: Digital Public Infrastructure, A Foundation for Citizen-Centric Digital Services

Digital Public Infrastructure (DPI) is vital to modern governance, connecting citizens to public institutions through **digital identity systems**, **secure payments**, and **interoperable data platforms**. These systems drive inclusivity, accessibility, and efficiency while reflecting Europe's regulatory values. However, to fully unlock DPI's potential, Europe must **reduce dependencies** and ensure privacy, transparency, and interoperability.

At the core of Digital Public Infrastructure (DPI) are digital identity systems, exemplified by the European Digital Identity Wallet (EUDI Wallet). While national models like Estonia's e-ID, Germany's Online-Ausweisfunktion, FranceConnect, Denmark's MitID, Italy's SPID, and Finland's Suomi.fi have streamlined access to services, they remain nationally isolated and based on legacy architectures. The EUDI Wallet, established under the eIDAS 2.0 Regulation in April 2024, aims to provide EU citizens with secure and interoperable digital identities. Member states are actively developing wallets through initiatives like Germany's Funke Competition, while the European Commission's common EU Toolbox ensures uniform implementation standards. By 2027, all EU member states must issue or certify at least one EUDI Wallet, with pilot projects currently testing functionality and user acceptance. Positioned to become a cornerstone of Europe's digital identity framework, the EUDI Wallet has also sparked discussions about aligning with Europe's digital sovereignty goals. Concerns include potential reliance on proprietary systems and the importance of robust security and privacy safeguards. Addressing these issues through privacy-first standards will be critical to its long-term success as a secure and trusted solution for EU citizens.

Secure payments are a key pillar, with Italy's PagoPA enabling seamless citizen-administration transactions. The proposed **Digital Euro**, led by the European Central Bank, has the potential to unify digital payment systems within a secure, interoperable framework, especially when integrated with solutions like the EUDIW.

Data platforms ensure secure, interoperable ecosystems. Examples like Estonia's X-Road, Finland's Suomi.fi Data Exchange Layer, and Spain's SARA enhance public service delivery. France's API Platform and Germany's GovData portal promote **open data sharing**, improving transparency and collaboration. Platforms such as **AuroraAI** in Finland and **Decidim Barcelona** highlight Europe's leadership in **citizen-centric innovation**.

Despite these successes, Europe's DPI remains heavily reliant on foreign cloud infrastructure. Critical DPI components—such as digital identities, payments, and data exchanges—must be based on EuroStack digital infrastructure ensuring alignment with EU regulations.

To accelerate adoption, DPI must demonstrate clear benefits to citizens. For example, the COVID-19 pandemic showed how digital contact tracing apps, linked to national digital IDs, enabled cross-border interoperability and increased trust in digital services. Integrating DPI into everyday interactions—like healthcare, tax systems, and public administration—will further drive uptake.

The digital transformation of public administration is a critical priority for European governments seeking to enhance service delivery, reduce costs, and improve overall efficiency. At the forefront of this transformation is Government Process Automation (GPA) as a cross-border approach that leverages technology and shares process knowledge to streamline and optimize administrative workflows.

Data and Artificial Intelligence: Catalysts for Innovation and Strategic Autonomy

Data is the backbone of the digital economy, driving innovation across industries. Yet, the value of much of this data—particularly industrial and urban data—is extracted by non-European platforms. Europe's industrial generates a significant portion of global data,¹²⁹. However, much of this data is processed and monetized by foreign entities, leaving Europe dependent on US and Chinese cloud providers. This dependency not only limits Europe's economic benefits but also raises concerns about data sovereignty and security.

To seek countering the dominance of extractive data practices and proprietary ecosystems, Europe is advancing an ambitious data sovereignty framework. The European Union's data strategy seeks to establish a unified data ecosystem to enhance competitiveness, protect citizens, and ensure data sovereignty, underpinned by strong laws like the General Data Protection Regulation (GDPR). Far from being a limitation, GDPR is a benchmark for data protection globally, replicated in legal frameworks worldwide. However, challenges lie in its effective implementation and in addressing power imbalances in an opaque data marketplace where privacy and rights are frequently violated, often in contravention of European laws.

A major shortfall of the EU's strategy is its inability to build its own robust data infrastructures, leaving it reliant on large Big Tech companies for cloud and data processing services. This dependency undermines Europe's autonomy in the digital economy. While initiatives like the Data Governance Act (DGA) and the Data Act (effective January 2024) aim to democratize data access and foster innovation, the lack of independent infrastructure continues to entrench asymmetries of power in the global data economy¹³⁰. To achieve true sovereignty and competitiveness, Europe must pair its legislative leadership with tangible advancements in independent, energy-efficient, and secure data infrastructure.

Key efforts include the development of **Common European Data Spaces**¹³¹, intended to enable secure and interoperable data sharing across sectors such as health, energy, and agriculture. These data spaces are presented as tools to foster innovation and economic growth, yet doubts persist about the speed of their implementation and their ability to scale sufficiently to serve as viable alternatives to extractive global data platforms.

Standards form a critical aspect of the rapidly evolving data landscape. The EU's Data Act, Data Governance Act, and Open Data Directive should help harmonizing frameworks for IoT-generated data sharing and data reuse as does the Cyber Resilience Act for cybersecurity of IoT solutions. Yet, harmonized implementation and minimization of red tape is not guaranteed.

Industrial data, critical for sectors like manufacturing, energy, and mobility, is one of Europe's most valuable assets. Initiatives such as **Catena-X**¹³², **Manufacturing-X**¹³³, and **EONA-X**¹³⁴ exemplify

¹²⁹ European Commission, 'Data Act: Measures for a Fair and Innovative Data Economy', Text, 23 February 2022, https://ec.europa.eu/commission/presscorner/detail/en/ip_22_1113.

¹³⁰ Miguel Otero Iglesias and Agustín González-Agote, 'The Future Data Economy – Competitive, Fair, Safe' (IE CGC, April 2024).

¹³¹ European Commission, 'Common European Data Spaces | Shaping Europe's Digital Future', accessed 12 January 2025, <https://digital-strategy.ec.europa.eu/en/policies/data-spaces>.

¹³² See: 'Catena-X Your Automotive Network - Homepage', accessed 12 January 2025, <https://catena-x.net/de/>.

¹³³ See: Germany's Federal Ministry for Economic Affairs and Climate Action and Federal Ministry of Education and Research, 'Manufacturing-X', accessed 12 January 2025, <https://www.plattform-i40.de/IP/Navigation/EN/Manufacturing-X/Manufacturing-X.html>.

¹³⁴ See: 'Eona-X, Homepage', accessed 12 January 2025, <https://eona-x.eu/>.

progress in industrial data ecosystems, demonstrating the potential of industrial data spaces to reduce inefficiencies and drive competitiveness in Europe's industrial heartlands to reduce barriers, and foster trust, adoption, and competitiveness for European providers.

In addition to industrial data, urban data holds strategic significance, with many cities advancing the development of **urban digital twins**—virtual replicas of physical urban environments that integrate real-time data and simulations to support decision-making¹³⁵. These platforms leverage open data and real-time integrations to enable informed, data-driven decisions, enhance transparency, and foster collaborative urban planning. European cities are at the forefront, generating massive volumes of data through smart city technologies like IoT sensors for traffic management, energy optimization, and public services. The **SynchroniCity project**, funded under Horizon 2020, is an example of a pan-European initiative creating frameworks for smart cities to share data in standardized ways. It involves cities like Milan, Helsinki, and Eindhoven, where data-sharing platforms are enabling innovations in mobility, waste management, and energy efficiency.

Amsterdam, a leader in smart city development, has implemented an open urban data platform that empowers citizens and local businesses while maintaining strong data governance¹³⁶. By prioritizing data sovereignty and interoperability, these projects showcase how urban data spaces can enhance liveability and collective empowerment. In **Barcelona**, the DECODE project and Decidim empowered citizens to control their data, transforming it into a public good for urban planning and participatory democracy¹³⁷. This kind of experiments were then replicated globally and extended to AI through the **Cities for Digital Rights Coalition**¹³⁸. Similarly, **Hamburg** used mobility data to enhance public services while safeguarding data privacy. Portugal, Estonia, Slovakia and Ukraine were profiled for high growth in open data maturity scores¹³⁹. These examples highlight how cities can leverage data governance to benefit communities¹⁴⁰.

Gaps in European Data Infrastructure

Despite these promising examples, Europe's data infrastructure remains underdeveloped. To close these gaps, Europe must accelerate investment in sovereign data infrastructure. Decentralized and edge computing technologies, which process data locally, are critical for ensuring compliance with EU regulations and enhancing data security. For example, edge computing can support real-time decision-making in manufacturing and smart cities while minimizing data transfer to centralized foreign platforms. Another interesting example of decentralized sovereign data management is Inrupt, founded by Sir Tim Berners-Lee, the founder of the World Wide Web system, advances data sovereignty through the Solid protocol, enabling individuals to control personal data via decentralized "Pods." Collaborations like the Athumi initiative with the Flanders Government empower citizens to manage data securely, fostering innovation in public and private services¹⁴¹. In healthcare, Inrupt's

¹³⁵ OECD, *Smart City Data Governance: Challenges and the Way Forward*, OECD Urban Studies (OECD, 2023), <https://doi.org/10.1787/e57ce301-en>.

¹³⁶ Jutta Ravelli, 'City Data: A Treasure Full of Data about the City', Amsterdam Smart City, 15 August 2017, <https://amsterdamsmartcity.com/updates/news/city-data-a-treasure-full-of-data-about-the-city>.

¹³⁷ Fernando Monge et al., 'A New Data Deal: The Case of Barcelona', *IIPP WP*, no. 2022/02 (18 February 2022).

¹³⁸ See: 'Cities for Digital Rights - Homepage', accessed 12 January 2025, <https://citiesfordigitalrights.org/>.

¹³⁹ European Data Portal et al., *Open Data Best Practices in Europe: Estonia, Slovenia and Ukraine* (Publications Office of the European Union, 2022), <https://data.europa.eu/doi/10.2830/277405>; Publications Office of the European Union et al., *2023 Open Data Best Practices in Europe* (Publications Office of the European Union, 2024), <https://data.europa.eu/doi/10.2830/68229>.

¹⁴⁰ Francesca Bria, 'Governing Urban Data for the Public Interest', 26 October 2023, <https://thenew.institute/en/media/the-new-hanse-final-report-out-now>.

¹⁴¹ Inrupt, 'Flanders Government Strengthens a Trusted Data Economy', accessed 12 January 2025, <https://www.inrupt.com/case-study/flanders-strengthens-trusted-data-economy>.

pilot with the NHS allows patients to control access to their medical records, enhancing privacy and care. Similarly, work with the BBC demonstrates how user-controlled data can transform digital experiences. These efforts align with Europe's vision for a trusted, interoperable data commons.

The EU's emphasis on FAIR data principles—making data Findable, Accessible, Interoperable, and Reusable—is supported by organizations like MyData Global, Finland's SITRA, and Project Liberty which champion human-centric data governance. These efforts align with the EU's vision for data sovereignty but face practical hurdles, including aligning diverse stakeholders and addressing technical and legal interoperability issues. The risk of fragmentation, both within the EU and globally, remains a significant barrier. Structural challenges also persist. Initiatives such as the **Interoperable Europe Act**¹⁴², which prioritises and promotes open standards and open source for cross-border digital services and public administrations, and the **Open Data Directive**, promoting reuse of high-value public datasets, are promising but require substantial political and technical coordination to succeed. Europe's vision for ethical and inclusive data governance hinges on overcoming these barriers and ensuring frameworks gain public and private sector buy-in while remaining globally competitive.

Data must transition from a tool of exploitation to a shared resource for societal progress. By tackling monopolistic practices and unethical product design while enhancing infrastructure and policies, Europe has a unique opportunity to create a fairer, more sustainable data economy. With its regulatory frameworks already recognized as global benchmarks, Europe is well-positioned to lead the implementation of an interoperable data commons ecosystem, setting a global standard for ethical and collaborative data governance.

Artificial Intelligence: Opportunities and Challenges for the EuroStack

Artificial Intelligence (AI) is a transformative technology capable of analysing vast amounts of structured and unstructured data and adapting dynamically to new conditions without reprogramming. Innovating in the way information is processed and synthesised, AI has the potential to reshape the digital landscape, creating new markets and possibly displacing established products. For example, the success of generative models such as OpenAI's ChatGPT, has raised the possibility of challenging once-impenetrable online search market. Moreover, the versatility of general-purpose foundation models promises to propagate the transformation across the economy, affecting sectors as diverse as healthcare, manufacturing, food production and autonomous systems.

The prospect of far-reaching market shifts has galvanised massive investments. In 2023, private AI investment in the United States reached €62.5 billion, €7.3 billion in China and €9 billion across the European Union and the UK.¹⁴³ The resulting expansion of the industry is remarkable. Valued at €234 billion in 2025, the global AI market is expected to grow at a compound annual growth rate (CAGR) of 27.67% through 2030, highlighting its pivotal role in the modern economy.¹⁴⁴

In this digital gold rush, new players are hoping to carve themselves a place in the new AI-powered economy, while existing tech giants strive to defend their market positions by integrating AI into their products and steering the development of the technology in their favour. However, since the

¹⁴² EU, 'Regulation (EU) 2024/903 of the European Parliament and of the Council of 13 March 2024 Laying down Measures for a High Level of Public Sector Interoperability across the Union (Interoperable Europe Act)' (2024), <http://data.europa.eu/eli/reg/2024/903/oj/eng>.

¹⁴³ Nestor Maslej, et al., 'AI Index Report 2024 – Artificial Intelligence Index' (Stanford: AI Index Steering Committee, Institute for Human-Centered AI, April 2024), <https://aiindex.stanford.edu/report/>.

¹⁴⁴ Statista, 'Artificial Intelligence - Global', accessed 12 January 2025, <https://www.statista.com/outlook/tmo/artificial-intelligence/worldwide>.

development and training of AI models depends heavily on other elements of the digital stack, incumbents who hold a dominant position in other markets have a significant advantage.¹⁴⁵

AI models are typically characterized by high setup and training costs, and negligible costs for deploying them, leading to large economies of scale. At the same time, Europe-originated Stable Diffusion offers a compelling alternative. Developed through academic research at Ludwig Maximilian University of Munich and Heidelberg University, and open-source collaboration, it democratizes access to advanced image creation¹⁴⁶. Stable Diffusion's open-source nature, facilitated by the CompVis group and supported by Stability AI and EleutherAI, fosters transparency and community-driven development. Furthermore, online developers' communities are perfecting ways to set up bespoke AI models locally, running on the user's machine using open-source models such as Meta's Llama. Nevertheless, while these developments hints at the possibility of a future where AI resources are affordable and easily accessible, the scale and reliability required from AI by organizations currently necessitate considerable investment.

As models grow more complex, "compute," or raw computational power, becomes the primary input for AI. For example, OpenAI's ChatGPT-3.5, with 175 billion parameters, was succeeded by ChatGPT-4, which utilizes 1.75 trillion parameters at an estimated training cost of \$100 million.¹⁴⁷ This exponential demand for compute is creating a significant bottleneck, as the market is dominated by US cloud providers like Amazon Web Services (AWS), Google Cloud, and Microsoft Azure. These companies either source specialized chips, primarily from Nvidia, or design their own, while also developing proprietary software to optimize hardware performance and enable additional software layers by third parties.

Initiatives like Microsoft's Founders Hub (launched in February 2022), Google Cloud for Y Combinator startups (announced in May 2023), and Northern Data's AI accelerator program (launched in June 2023) appear to ease access to some AI resources by offering free or subsidized GPU clusters. For instance, Microsoft's Founders Hub offers up to \$150,000 in Azure credits over four years, while Google Cloud provides Y Combinator startups with access to a subsidized cluster of Nvidia H100 GPUs. These programs, however, can foster dependency on Big Tech infrastructure, potentially hindering the emergence of a truly independent AI ecosystem. Furthermore, by providing these resources, Big Tech gains valuable insights into startup activities and can influence the direction of AI innovation to align with their strategic interests. While this may be beneficial in the short term, AI developers remain reliant on a small set of suppliers. Nearly all AI startups, including OpenAI, Anthropic, and the French firm Mistral, rely on US-based cloud infrastructure.¹⁴⁸ Alphabet's CEO Sundar Pichai in 2023 declared that over 70 percent of generative AI startups rely on Google's cloud infrastructure.¹⁴⁹

¹⁴⁵ Fausto Gernone and David Teece, 'Competing in the Age of AI: Firm Capabilities and Antitrust Considerations', in *Artificial Intelligence and Competition Policy* (Concurrences, 2024).

¹⁴⁶ See: Stability AI, 'Stable Diffusion Launch Announcement', 10 August 2024, <https://stability.ai/news/stable-diffusion-announcement>.

¹⁴⁷ Olena Zhrebetska, '100 ChatGPT Statistics to Know in 2025 & Its Future Trends', Intelliarts, accessed 12 January 2025, <https://intelliarts.com/blog/chatgpt-statistics/>.

¹⁴⁸ The German startup Aleph Alpha is a notable exception in this regard, boasting to be technologically independent of US Big Tech. See: Georgia Butler, 'Aleph Alpha and Cerebras Systems to Develop Sovereign AI Solutions', *Data Center Dynamics*, 17 May 2024, <https://www.datacenterdynamics.com/en/news/aleph-alpha-and-cerebras-systems-to-develop-sovereign-ai-solutions/>.

¹⁴⁹ Johan Moreno, '70% Of Generative AI Startups Rely On Google Cloud, AI Capabilities', *Forbes*, 2023, <https://www.forbes.com/sites/johanmoreno/2023/07/25/70-of-generative-ai-startups-rely-on-google-cloud-ai-capabilities-says-alphabet-ceo-sundar-pichai/>.

Access to data, alongside compute and talent, is a critical factor shaping the AI landscape. Frontier AI models are data-hungry, often trained on massive datasets scraped from the internet¹⁵⁰. This practice raises ethical and legal concerns surrounding copyright and ownership, particularly as internet data becomes increasingly enclosed. Resources once freely available are now being monetized, with platforms like Reddit restricting access to their valuable data. Consequently, AI firms find themselves competing for data acquisition through exclusive licensing agreements and outright acquisitions. Elon Musk's xAI, for example, leverages proprietary user content from X (formerly Twitter) to train its models, highlighting the strategic importance of data ownership and governance in the AI race.¹⁵¹ This trend towards data exclusivity not only raises barriers to entry for smaller players but also concentrates power in the hands of those who control these valuable resources.

Europe's AI Ecosystem

Despite these challenges, Europe has made strides in building an AI ecosystem aligned with its values. European firms have focused on specialized markets. For example, Mistral specializes in open-access models optimized for European languages and use cases, while iGenius uses open source LLMs to develop solutions for highly regulated industries. Another area of strength is that of industrial AI applications, with companies like Bosch and Siemens integrating AI into manufacturing, robotics, and predictive maintenance.

Europe's reliance on foreign semiconductors and computational infrastructure highlights critical vulnerabilities in its AI ecosystem. To address these, efforts are underway to strengthen digital sovereignty¹⁵². Aleph Alpha has partnered with Cerebras Systems to deploy sovereign AI solutions using advanced supercomputers at HPE-built data centres in Berlin. The ambition is to promote AI transparency and build solutions that operate independently of US Big Tech and comply with European data protection regulations. Recently, Aleph Alpha, has shifted its focus from developing its own large language models (LLMs) to assisting other organizations in deploying existing AI models, as exemplified by the launch of PhariaAI, an end-to-end platform designed to help enterprises and governments integrate AI solutions effectively. European HPC clusters such as LUMI in Finland and Leonardo in Italy offer cutting-edge public computational capacity for training and deploying AI models. Mistral AI, meanwhile, collaborates with providers like CoreWeave and Scaleway to diversify computing resources, reducing dependence on non-European platforms. However, what secures Mistral's future and allows it to still train its own models is the partnership with Microsoft. Mistral benefits from Microsoft's cloud resources for its research and gets access to its customers. Microsoft has exclusive access to some of Mistral's models to be less dependent on OpenAI. This could jeopardise Mistral openness and commitment to European sovereignty in the future.

Future European initiatives like the recently announced new IPCEI on AI and edge computing could further solidify a sovereign AI ecosystem¹⁵³ empowered by a large European network of green public compute. To lead in sustainable and ethical AI deployment, Europe must prioritize green computing and generativity¹⁵⁴. Optimizing data centre design for energy efficiency, transitioning to renewable energy, and adopting advanced cooling systems to reduce water consumption are essential

¹⁵⁰ Pablo Villalobos et al., 'Will We Run out of Data? Limits of LLM Scaling Based on Human-Generated Data' (arXiv, 4 June 2024), <https://doi.org/10.48550/arXiv.2211.04325>.

¹⁵¹ Courtney Radsch, 'Dismantling AI Data Monopolies Before It's Too Late | TechPolicy.Press', Tech Policy Press, 9 October 2024, <https://techpolicy.press/dismantling-ai-data-monopolies-before-its-too-late>.

¹⁵² See, for example: 'OpenGPT-X', accessed 12 January 2025, <https://opengpt-x.de/en/>.

¹⁵³ Margrethe Vestager, 'Statement by EVP Margrethe Vestager at the Joint European Forum for IPCEI', IEU Monitoring, n.d., https://ieu-monitoring.com/editorial/joint-european-forum-for-ipcei-statement-by-evp-margrethe-vestager/478623?utm_source=ieu-portal.

¹⁵⁴ Fieke Jansen and Michelle Thorne, 'IV. Predatory Delay and Other Myths of "Sustainable AI"', *AI Now Institute* (blog), 15 October 2024, <https://ainowinstitute.org/publication/predatory-delay-and-other-myths-of-sustainable-ai>.

measures¹⁵⁵. Developing energy-efficient AI models can further reduce electricity use while maintaining performance, aligning with Europe's climate goals, and minimizing the environmental impact of its AI ecosystem. The challenge for Europe is to pursue these sustainability goals while being competitive and innovative, thus ensuring that it leads in green AI without ceding ground to global rivals.

Europe is advancing its AI capabilities through targeted investments, infrastructure development, and strategic partnerships. A flagship initiative, the **AI Factories**, aims to integrate computing power, vast datasets, and skilled talent to train large models for sectors like healthcare, energy, and manufacturing. These factories leverage **EuroHPC**, Europe's high-performance computing infrastructure, to enhance competitiveness in advanced AI applications¹⁵⁶.

However, Europe's scale of investment—€750 million—remains modest compared to the enormous spending by Big Tech, such as Microsoft and OpenAI's \$100 billion data centre projects. Competing directly with such vast expenditures would neither be sustainable nor strategically prudent for the EU.

Instead, Europe can turn its relative vulnerability in computing resources into an **opportunity** by leveraging its collective purchasing power and expansive public compute network to shape global industry standards¹⁵⁷—a concept referred to as **conditional computing**¹⁵⁸. For instance, rather than relying on Nvidia GPUs, which are locked into the proprietary CUDA ecosystem, Europe could prioritize GPUs developed through open-source initiatives, promoting transparency, interoperability, and sovereignty over critical technologies.

The European Union is intensifying its investment in artificial intelligence (AI) to bolster competitiveness and uphold ethical standards. The **InvestEU program** has earmarked €26.2 billion as an EU budget guarantee, aiming to mobilize over €372 billion in public and private investments across various sectors, including AI. Private sector engagement is also on the rise. For instance, German AI startup Aleph Alpha secured over \$500 million in a Series B funding round¹⁵⁹, and Mistral, and DeepL got additional large funding rounds. While private investments signal confidence, Europe's AI funding still lags behind global competitors. In 2023, generative AI investment in Europe was just \$2.4 billion, a fraction of the \$22.4 billion in the US. However, funding in Europe is growing at a faster rate,

Europe's approach reflects its unique emphasis on aligning technological progress with ethical, regulatory, and sustainability principles. The strategy is sound, but scaling these efforts to match the transformative potential of AI requires bolder investments, greater coordination among Member States, and more streamlined pathways for private-public partnerships. Without addressing these gaps, Europe risks failing to have control on sovereign AI capabilities and its own development model, even as it upholds its commendable commitment to ethical and accountable AI. The AI Factories and other initiatives are still in early stages, and their success will hinge on overcoming

¹⁵⁵ IEA, 'Electricity 2024 – Analysis', IEA, 24 January 2024, <https://www.iea.org/reports/electricity-2024..>

¹⁵⁶ EuroHPC JU, 'Selection of the First Seven AI Factories to Drive Europe's Leadership in AI', 10 December 2024, https://eurohpc-ju.europa.eu/selection-first-seven-ai-factories-drive-europes-leadership-ai-2024-12-10_en.

¹⁵⁷ Mariana Mazzucato and Fausto Gernone, 'Governments Must Shape AI's Future', *Project Syndicate*, 12 April 2024, <https://www.project-syndicate.org/onpoint/governments-must-shape-ai-future-by-mariana-mazzucato-and-fausto-gernone-2024-04>.

¹⁵⁸ Van Dijk et al. 2024, Conditional computing: a new paradigm for public-interest AI in the EU in Francesca Bria et al., 'Time To Build A European Digital Ecosystem', 9 December 2024, <https://feps-europe.eu/wp-content/uploads/2024/12/Time-to-build-a-European-digital-ecosystem.pdf>.

¹⁵⁹ 'Aleph Alpha Raises a Total Investment of More than Half a Billion US Dollars from a Consortium of Industry Leaders and New Investors', Aleph Alpha, 6 November 2023, <https://aleph-alpha.com/aleph-alpha-raises-a-total-investment-of-more-than-half-a-billion-us-dollars-from-a-consortium-of-industry-leaders-and-new-investors/>.

structural barriers, such as fragmented data ecosystems, limited cross-border collaboration, insufficient investments and access to local computational resources compared to global hyperscalers.

AI will have more economic and societal benefits and fewer risks if its deployment in business enterprises and public sector agencies is decided jointly by workers and managers. Co-determination

Box 3: Biotechnology and Medtech: Precision and Dependency

Biotechnology and medtech are among the most innovative and transformative industries, influencing sectors such as healthcare, agriculture, and life sciences. Europe has emerged as a global leader in areas like genomics and biomanufacturing, exemplified by the German company BioNTech, which developed the first COVID-19 vaccine authorized for regular use. Despite such successes, the sector remains heavily reliant on foreign technologies at critical points in the value chain.

Specialized inputs such as chemicals, reagents, and biological samples are predominantly sourced from dominant US firms like Merck and ATCC. These dependencies extend into digital infrastructure, with network systems enabling real-time data sharing and AI workloads requiring significant computational power. AI is increasingly central to drug discovery and biological research, illustrated by the breakthrough protein-structure prediction of AlphaFold, a product of the London-based DeepMind acquired by Alphabet. While Bayer's collaboration with Microsoft in agricultural AI demonstrates Europe's innovation potential, it also underscores dependency on third-country technology.

Cloud services, dominated by US companies such as AWS, Microsoft Azure, and Google Cloud, are vital for genomics and drug discovery. In the field of human genomics, the Chinese company BGI is increasingly dominant. The reliance on non-European platforms for high-throughput data analysis and storage presents a significant sovereignty challenge. Meanwhile, imaging and monitoring equipment critical to healthcare—where European firms like Siemens Healthineers and Philips Healthcare lead—relies on digital systems that are increasingly cloud-integrated. Cybersecurity remains paramount in protecting sensitive research and patient data, but Europe continues to depend on non-European solutions in this domain.

The EuroStack offers an opportunity to address these vulnerabilities by fostering an ecosystem that integrates European digital infrastructure with the biotechnology and medtech value chain. Strengthening AI and cloud services within Europe and ensuring secure data flows can enable the sector to flourish while safeguarding European sovereignty and innovation leadership.

of technological changes should be encouraged in the workplace, to guarantee a fair digital transition for all workers and to ensure we unlock AI's potential contributions to productivity and efficiency. Public policy can encourage this productive collaboration. Particular attention should be paid to SMEs. AI deployment will be more effective if it has an overall “directionality” – if businesses, workers, and citizens understand it as part of a social project aimed at our collective goals.

Cybersecurity: The Cornerstone of Digital Resilience and Autonomy

Cybersecurity is cutting across the stack and is both a *conditio sine qua non* and opportunity for EU strategic autonomy. Technically it consists of software and hardware of many sorts: secure access, device-level data encryption, end-to-end secure communications, system and device sensors, firewalls and antivirus software and hardware, cyber-incident monitoring, analysis (Security

information and event management SIEM), etc., many of these are increasingly reinforced by and even intimately dependent on AI. Cybersecurity also consists of governance, that is, procedures, processes – often standardised - and EU law where generally national rather than EU authorities impose oversight. Governance and technology are closely linked.

Europe has many cybersecurity companies with some large ones such as Thales but most of them are small. It is a challenge for them to scale up in the European market due to lack of risk capital, diversity of regulations, and the need to build up trust with buyers. They compete against global foreign suppliers notably from the US and Israel who have global brand names, deep pockets with access the scale-up capital and lawyers.

While there is extensive EU cybersecurity legislation¹⁶⁰ and financial support by EU programmes, an EU cybersecurity industrial policy that works coherently with legislation and EU foreign, security and defence policy is missing¹⁶¹. This is sorely needed and must counter the long-term erosion of European autonomy yet be realistic about the pressing short-term need for strong cybersecurity due to rising geopolitical conflict, ‘unpeace’ and outright war and rampant cyber-crime. Such industrial policy would also address adoption at scale by public procurement and defence, as well as smooth legislative compliance in the Internal Market and synergy with and of cyber and digital diplomacy¹⁶². Thereby, it makes the case for European investors that cybersecurity-made-in-the EU is a viable investment.

Defence: Strengthening Strategic Capabilities

Geopolitical tensions and the war in Ukraine have brought home military vulnerabilities of Europe. Consequently, defence investment is being stepped up with a growth of over 15% p.a. in recent years¹⁶³. In addition, civil-defence synergies are on the rise¹⁶⁴. Funding at European level includes European Defence Fund, €8 billion to 2027, EU Space Programme, €13 billion to 2027, EU Secure Connectivity including quantum communications, €2 billion to 2027. EU-NATO collaboration includes joint EIF-NATO investment in the private sector and joint strategy development such as for undersea cables. Total EU Member States expenditures are projected continue rising significantly reaching 326 billion in 2024¹⁶⁵. Entrepreneurial and risk capital investment are on the rise. A much-visible example is the Germany-based AI-company Helsing, which saw its valuation tripling to \$4.5 billion in 2024.

There is increasingly urgent call to move to action for strengthening Europe’s civil and military preparedness and readiness such as the 2024 Niinistö report. Generally, defence funding, nationally and through the European Defence Fund and NATO starts to become available across from advanced research to growing scale-ups. Defence requirements are increasingly shared with civil R&D. Other

¹⁶⁰ Paul Timmers, ‘EU Cybersecurity Policy’, in *The Making of a Global Digital Rulebook: Digital Sovereignty and International Action in the EU*, Thibaut Kleiner and Andrea Garcia Rodriguez (Eds) (Springer, 2025); Christina Rupp, ‘Navigating the EU Cybersecurity Policy Ecosystem’, 27 June 2024, <https://www.interface-eu.org/publications/navigating-the-eu-cybersecurity-policy-ecosystem..>

¹⁶¹ Paul Timmers, ‘EU Cybersecurity Policy’..

¹⁶² See also the program of the Polish Presidency of the Council, Polish Presidency, ‘Polish Presidency of the Council of the European Union’, Polish presidency of the Council of the European Union, 2025, <https://polish-presidency.consilium.europa.eu/en/..>

¹⁶³ EDA, ‘2024 Defence Review Paves Way for Joint Military Projects (CARD 2024)’, 19 November 2024, <https://eda.europa.eu/news-and-events/news/2024/11/19/2024-defence-review-paves-way-for-joint-military-projects..>

¹⁶⁴ Reference documents at EU level are Examples are the 2021 Action Plan on Synergies between Civil, Defence and Space Industries, the Observatory for Critical Technologies across civil, defence and space industries programmes, CARD – the Coordinated Annual Review of Defence in the EU, and the European Defence Industrial Strategy 2024.

¹⁶⁵ European Defence Agency EDA, ‘Coordinated Annual Review on Defence (CARD)’, Default, 19 November 2024, [https://eda.europa.eu/what-we-do/EU-defence-initiatives/coordinated-annual-review-on-defence-\(card\)](https://eda.europa.eu/what-we-do/EU-defence-initiatives/coordinated-annual-review-on-defence-(card)).

countries show that an early adopter role of defence (for requirements, pilots, and validation) helps to unlock the market and strengthen security and thereby sovereignty.

Yet, Europe is highly dependent on imports: 78% of military equipment and services (many of them digital) is imported, of which 2/3rd from the US Only 18% is bought across from another EU country. The European Defence Industrial Strategy tabled ambitious objectives. By 2030:

- at least 50% of Member States procurement should be from the European Defence and Technology Industrial Base (EDTIB);
- the value of intra-EU defence trade should be at least 35% of the value of the EU defence market;
- Member States should procure at least 40% of defence equipment in a collaborative manner.

Challenges in the EU to step up practice of civil-military cooperation include the alignment of military innovation with the speed of the market, reassurance on ethical standards - evolving as technology develops such as for AI, drones, sensors, or satellites. The EC has been outlining options for R&D on technologies with dual-use potential. The League of European Research Universities (LERU) called for a careful approach and safeguards for the demarcation between civil and military research¹⁶⁶. Another challenge is the reluctance of conventional banks to invest in defence.

Quantum Technologies: Unlocking Tomorrow's Potential

Quantum Technologies (QT) stand out as a transformative force within the stack. Quantum mechanics enables radically different approaches to computing, communications, and sensing, solving problems that are beyond the reach of conventional systems. For instance, quantum computing could accelerate drug discovery and material development, while quantum sensing could revolutionize navigation, medical diagnostics, subsea cable surveillance, and industrial quality control.

Quantum communications with quantum key distribution (QKD) already provides unbreakable secure communications which is starting to get commercialised. China leads in this area, operating a 2,032 km QKD ground link between Beijing and Shanghai. The EU is advancing its capabilities through initiatives like the **European Quantum Communication Infrastructure** (EuroQCI)¹⁶⁷, which will integrate with the **EU's IRIS2 satellite program**¹⁶⁸.

However, QKD on its own is neither sufficient nor addressing all security challenges posed by quantum technology and moreover, there is a debate on the merits of QKD¹⁶⁹. It is urgent to adopt post-quantum cryptography (PQC) to secure existing systems and encrypted data as quantum computing can break currently widely used 'traditional' encryption. This is the case even if quantum

¹⁶⁶ LERU, 'Enhancing Dual Use Technologies: Leveraging Synergies in EU Funding Streams', LERU, April 2024, <https://www.leru.org/publications/options-for-enhancing-support-for-research-and-development-involving-technologies-with-dual-use-potential>.

¹⁶⁷ European Commission, 'The European Quantum Communication Infrastructure (EuroQCI) Initiative | Shaping Europe's Digital Future', 23 April 2024, <https://digital-strategy.ec.europa.eu/en/policies/european-quantum-communication-infrastructure-euroqci>.

¹⁶⁸ EU Agency for the Space Programme, 'IRIS² | EU Agency for the Space Programme', 2024, <https://www.euspa.europa.eu/eu-space-programme/secure-satcom/iris2>.

¹⁶⁹ Renato Renner and Ramona Wolf, 'The Debate over QKD: A Rebuttal to the NSA's Objections' (arXiv, 27 July 2023), <https://doi.org/10.48550/arXiv.2307.15116>.

computing at scale remains five to ten years away. PQC is also a necessary complement to QKD¹⁷⁰. The US National Institute of Standards and Technology (NIST) has already issued three PQC standards, highlighting the strategic importance of early preparedness.

The Economic Dynamics of Quantum Technologies

Global public investment in QT reached over \$42 billion in 2023, with China as the leading investor¹⁷¹ and Europe the runner-up. Private funding is also increasing, driven by the recognition that QT will redefine industries. However, no single EU member state can match the resources of China or the US. The **EU Quantum Flagship**, backed by €1 billion in funding, and the 167-member **Quantum Industry Consortium (QuIC)** demonstrate Europe's commitment, but there is a lack of coordination in translating research into market-ready applications.

With over 500 scientists from 100+ institutions collaborating across Europe, EBRAINS 2.0 accelerates breakthroughs in neuroscience, brain medicine, and brain-inspired technologies. Supported by cutting-edge infrastructure, including **Jülich's quantum computing** ecosystem in Germany—home to Europe's first quantum computer integrated with a supercomputing environment EBRAINS 2.0 exemplifies Europe's strategy to harness advanced computing and quantum innovation for transformative solutions in healthcare and neuroscience¹⁷².

Demand-side fragmentation within the EU remains a critical barrier. Public and defence procurement is inconsistent, and there is no unified goal for quantum computing by 2030—a moonshot initiative that could galvanize efforts. Additionally, QT development depends on critical materials and high-precision components, where Europe faces supply constraints.

Building Resilience and Strategic Interdependencies

To strengthen its position, Europe must adopt a coordinated approach that balances self-reliance with strategic partnerships. Exclusive control over quantum technologies critical to national security, such as QKD and quantum hardware secure modules, is essential. For other areas, Europe should pursue mutual interdependencies with like-minded countries, such as the UK and Switzerland, leveraging their expertise to build a competitive quantum ecosystem. The UN Year of Quantum in 2025 provides a platform for global collaboration to harness QT for addressing humanity's most pressing challenges¹⁷³. Europe's leadership in this effort could align its strategic interests with broader international goals, positioning it as a key player in shaping the future stack.

Quantum's inclusion in the EuroStack should reflect both its current nascent stage and its long-term transformative potential. In the medium term, embedding quantum within existing layers (e.g., Networks and Chips) ensures its seamless integration into today's stack. In the long term, as quantum capabilities mature, elevating it to a distinct layer would highlight its role as a foundational pillar of Europe's digital sovereignty.

¹⁷⁰ Georg Serentschy, 'Unraveling the confusion around Quantum-Safe Encryption – Serentschy Advisory Services', 2024, <https://www.serentschy.com/unraveling-the-confusion-around-quantum-safe-encryption/>; Bart Preneel, 'The Quantum Threat and Post-Quantum Cryptography (PQC)', 2024.

¹⁷¹ World Economic Forum, 'Explainer: What Is Quantum Technology and What Are Its Benefits?', World Economic Forum, 3 July 2024, <https://www.weforum.org/agenda/2024/07/explainer-what-is-quantum-technology/>.

¹⁷² See: 'EBRAINS Research Infrastructure Secures €38 Million in Funding for New Phase of Digital Neuroscience', Forschungszentrum Jülich, 9 January 2024, <https://www.fz-juelich.de/en/news/archive/press-release/2024/ebrains-research-infrastructure-secures-20ac38-million-in-funding-for-new-phase-of-digital-neuroscience>.

¹⁷³ United Nations, 'International Year of Quantum Science and Technology', 7 June 2024, <https://quantum2025.org/en/>.

Given EuroStack's emphasis on aligning with strategic autonomy, sustainability, and resilience, quantum technologies represent a critical enabler of these goals. Whether as part of existing layers or a dedicated layer, quantum's inclusion should be explicitly articulated in the EuroStack framework to reflect its transformative potential.

The evolution of the digital stack is a defining challenge for Europe's digital sovereignty. By prioritizing edge-cloud computing and quantum technologies, the EU can carve out a competitive position in a landscape dominated by centralization and integration. Achieving this vision requires significant investments, harmonized regulation, and clear strategic goals.

Financial Power: The Driver of Digital Dominance

Financial power is a cornerstone of technological dominance, influencing which companies and regions lead in innovation, shape ecosystems, and set global standards. US and Chinese firms have not only achieved technological breakthroughs but have also leveraged substantial financial resources to secure supremacy across the digital stack. Their access to venture capital (VC), sovereign wealth funds (SWFs), and robust equity markets has enabled aggressive investments in research, acquisitions, and talent, often at scales Europe struggles to match¹⁷⁴.

Venture Capital and Sovereign Wealth Funds as Strategic Levers

Venture capital is the engine of innovation, fuelling the rapid growth and global dominance of US and Chinese tech companies. In 2023 alone, US AI startups attracted approximately €62.5 billion in private investment, dwarfing Europe's €9bn cumulative VC investment in AI startups in 2023 (including UK).¹⁷⁵ While Europe has seen growth in VC investment, its fragmented financial markets and risk-averse culture rooted in its bank-based financing system continue to impede large-scale funding. This disparity undermines Europe's ability to nurture startups capable of achieving successful exits on European stock exchanges and scaling globally. Instead, promising companies often become acquisition targets for foreign players, further eroding Europe's innovation ecosystem. While achieving exits on European stock exchanges is feasible, it remains challenging due to fragmented equity markets, lower valuations, and a preference for short-term returns, which can disincentivize long-term growth and innovation.

The dominance of US venture capital (VC) ecosystems allows firms to pursue bold, long-term projects, enabling foundational research that redefines industries. OpenAI exemplifies the dynamic of leveraging substantial venture capital (VC) funding to develop large language models, notably GPT-4, which have been integrated into Microsoft's Azure platform. Microsoft's multibillion-dollar investment in OpenAI has facilitated the development of advanced AI models and their training and deployment at scale, reshaping computational power and AI¹⁷⁶. The partnership between OpenAI and Microsoft has led to concerns about potential monopolistic practices and the centralization of AI

¹⁷⁴ Douglas J. Cumming and Pedro Monteiro, 'Sovereign Wealth Fund Investment in Venture Capital, Private Equity, and Real Asset Funds', SSRN Scholarly Paper (Rochester, NY: Social Science Research Network, 24 October 2022), <https://doi.org/10.2139/ssrn.4258254>.

https://papers.ssrn.com/sol3/papers.cfm?abstract_id=4258254&utm_source=chatgpt.com.

¹⁷⁵ Maslej, et al., 'AI Index Report 2024 – Artificial Intelligence Index'.

¹⁷⁶ Jai Vipra, 'Computational Power and AI', *AI Now Institute* (blog), 27 September 2023, <https://ainowinstitute.org/publication/policy/compute-and-ai>.

resources¹⁷⁷. These developments underscore the need for ongoing scrutiny of such alliances to ensure they promote a competitive and diverse technological landscape¹⁷⁸.

In contrast, Europe's fragmented and less dynamic capital markets often hinder similar trajectories, leaving startups undervalued, underfunded, and vulnerable to acquisition by foreign players¹⁷⁹. High-profile acquisitions of European tech firms underscore the challenges of scaling independently in a fragmented market. Google's acquisition of DeepMind significantly enhanced its AI capabilities, while NVIDIA's purchase of Dutch-based Bright Computing bolstered its leadership in semiconductors and data centre networking. Although Mellanox Technologies is based in Israel, its strong European presence further reinforced NVIDIA's regional influence. These moves, combined with NVIDIA's involvement in European supercomputing projects like Lumi in Finland and Leonardo in Italy, have cemented its position in global AI and high-performance computing innovation. Similarly, Apple's acquisition of Xnor.ai enabled the integration of energy-efficient AI into its devices, deepening user dependency on its ecosystem. These acquisitions underscore how Big Tech leverages strategic purchases to dominate the digital stack. In Europe, the loss of critical companies like ARM and DeepMind highlights the urgent need for stricter merger oversight and strategic protections to safeguard technological sovereignty and critical assets.

Without robust policies, Europe risks losing its most promising innovations to foreign control, further eroding its competitive position. Limited access to growth capital is a critical barrier for European startups, leaving them vulnerable to external acquisition. The European Investment Bank highlights that Europe's fragmented financial system restricts startups from securing the funding necessary to scale, underscoring the urgent need for more integrated and dynamic capital markets to retain and grow innovation within the region¹⁸⁰.

Equity markets play a pivotal role in sustaining technological dominance by enabling high valuations and sustained R&D investments. Nvidia, with a market capitalization exceeding \$3.476 trillion as of November 2024, exemplifies this dynamic¹⁸¹. Its substantial valuation has funded aggressive acquisitions and ambitious ventures in AI and cloud technologies. In contrast, Europe's fragmented equity markets often impede similar growth trajectories. European firms frequently face lower valuations and pressure for short-term returns, limiting their capacity for long-term innovation. This disparity exposes promising startups to foreign acquisitions, eroding the region's innovation ecosystem and weakening its digital sovereignty.

The European Investment Bank notes that Europe's fragmented financial system restricts startups' ability to access the necessary funding to scale, leaving them exposed to external acquisition. However, it is important to consider whether replicating the US model of high-growth

¹⁷⁷ Charles Duhigg, 'The Inside Story of Microsoft's Partnership with OpenAI', *The New Yorker*, 1 December 2023, <https://www.newyorker.com/magazine/2023/12/11/the-inside-story-of-microsofts-partnership-with-openai>.

¹⁷⁸ Tejas N. Narechania and Ganesh Sitaraman, 'An Antimonopoly Approach to Governing Artificial Intelligence', SSRN Scholarly Paper (Rochester, NY: Social Science Research Network, 17 January 2024), <https://doi.org/10.2139/ssrn.4597080>, https://papers.ssrn.com/sol3/papers.cfm?abstract_id=4597080.

¹⁷⁹ European Investment Bank, *Financing the Digitalisation of Small and Medium-Sized Enterprises: The Enabling Role of Digital Innovation Hubs* (European Investment Bank, 2020), <https://doi.org/10.2867/210258>, https://www.eib.org/attachments/thematic/financing_the_digitalisation_of_smes_summary_en.pdf.

¹⁸⁰ European Commission: Directorate-General for Research and Innovation, *Science, Research and Innovation Performance of the EU, 2024 – A Competitive Europe for a Sustainable Future* (Publications Office of the European Union, 2024), <https://doi.org/10.2777/965670>, <https://op.europa.eu/en/publication-detail/-/publication/c683268c-3cdc-11ef-ab8f-01aa75ed71a1/language-en>.

¹⁸¹ CompaniesMarketCap, 'NVIDIA (NVDA) - Market Capitalization', accessed 12 January 2025, <https://companiesmarketcap.com/nvidia/marketcap/>.

markets is desirable—or even feasible—for Europe.¹⁸² While high valuations can drive innovation, they also carry risks, including the potential for financial bubbles. Thus, while Europe's equity market challenges should be addressed to enhance competitiveness, efforts must also consider the benefits of a balanced approach. Building a stable, innovation-driven ecosystem may ultimately align better with Europe's economic values and long-term strategic interests than chasing rapid, unsustainable growth.

Sovereign wealth funds (SWFs) are increasingly influential in shaping global digital ecosystems through strategic investments in transformative technologies¹⁸³, and are emerging as powerful geopolitical tools in the global race for technological dominance, particularly against the backdrop of intensifying US-China competition. These state-owned funds, traditionally focused on financial returns, are now integral to national strategies aimed at securing supply chains and leadership in transformative technologies such as semiconductors, AI, cloud infrastructure, and defence tech. These strategic investments by SWFs highlight a broader global trend: the increasing fusion of economic and technological competition with geopolitical strategy. Such developments reveal the growing interdependence of financial resources, technological leadership, and geopolitical power in the 21st century¹⁸⁴.

China's National Integrated Circuit Industry Investment Fund, known as the "Big Fund," exemplifies this strategic use of SWFs. With its latest \$47.5 billion phase launched in 2024, the fund aims to bolster domestic semiconductor production and reduce reliance on foreign technology, aligning with China's goal of technological self-sufficiency.¹⁸⁵ This comes as the US tightens export controls on advanced technologies, seeking to curtail China's tech ambitions. The United Arab Emirates' Mubadala Investment Company and Saudi Arabia's Public Investment Fund (PIF) are leveraging sovereign wealth funds to reshape global competitive dynamics in technology.

Mubadala has acquired key assets like GlobalFoundries to strengthen its technological footprint and influence supply chains¹⁸⁶, while the PIF pursued investments in fast growing AI startups. Recent data highlights a fivefold increase in Middle Eastern SWF funding for AI companies in the past year. These investments align with broader national strategies like Saudi Vision 2030, which seeks to diversify economies and reduce oil dependency.

In contrast, Europe's sovereign wealth funds lack the scale and strategic focus to drive similar impact. Norway's Government Pension Fund, the world's largest SWF, prioritizes diversified financial returns over transformative tech investments^{187,188}. The deployment of SWFs in this geopolitical context underscores their dual role as financial instruments and strategic levers. That can effectively shape

¹⁸² BCG, 'Europe's Growth Equity Landscape', BCG Global, 4 March 2024,

<https://www.bcg.com/publications/2024/opportunity-in-europes-growth-equity-landscape>.

¹⁸³ H. Kent Baker, 'Sovereign Wealth Funds: An Overview', in *The Palgrave Handbook of Sovereign Wealth Funds*, ed. H. Kent Baker, Jeffrey H. Harris, and Ghayath F. Nakshbendi (Cham: Springer International Publishing, 2024), 3–18, https://doi.org/10.1007/978-3-031-50821-9_1.

¹⁸⁴ Brad Setser, 'A Taxonomy of Sovereign Wealth Funds', *Financial Times*, 25 September 2024, sec. FT Alphaville, <https://www.ft.com/content/a65135e9-1fde-4ace-a422-b0d783c62e14>.

¹⁸⁵ Reuters, 'China Sets up Third Fund with \$47.5 Bln to Boost Semiconductor Sector', 27 May 2024, sec. Technology, <https://www.reuters.com/technology/china-sets-up-475-bln-state-fund-boost-semiconductor-industry-2024-05-27/>.

¹⁸⁶ Arab Gulf States Institute in Washington, 'The Emergent Gulf Sovereign Wealth Fund-Global Tech Nexus', *Arab Gulf States Institute in Washington* (blog), 2 May 2024, <https://agsiw.org/the-emergent-gulf-sovereign-wealth-fund-global-tech-nexus/>.

¹⁸⁷ Norway's Ministry of Finance, 'Investment Strategy', Redaksjonellartikkel, Government.no (regjeringen.no, 17 April 2023), <https://www.regjeringen.no/en/topics/the-economy/the-government-pension-fund/government-pension-fund-global-gpfg/investment-strategy/id696849/>.

¹⁸⁸ David Chambers, Elroy Dimson, and Antti Ilmanen, 'The Norway Model', *The Journal of Portfolio Management* 38, no. 2 (31 January 2012): 67–81, <https://doi.org/10.3905/jpm.2012.38.2.067>.

technological standards, secure supply chains, and gain a decisive edge in the global competition for digital dominance. To remain relevant, Europe must adopt a more strategic SWF model, pooling resources and prioritizing critical technologies like AI, semiconductors, and quantum computing to enhance its digital sovereignty.

The European Investment Fund (EIF), while vital for supporting innovation, remains limited in scope compared to global counterparts. To enhance competitiveness, Europe could consider collaborative pooling of SWF resources across EU member states, alongside a stronger mandate for strategic technology investments. Such initiatives could significantly bolster Europe's ability to compete in the global digital economy, despite the previous failure of such attempt by President von der Leyen¹⁸⁹.

Addressing Europe's Financial Gaps

To reclaim digital sovereignty, Europe must adopt a bold, coordinated financial strategy that addresses structural weaknesses in its venture capital ecosystem. Scaling up late-stage funding and fostering risk-taking are critical for enabling European tech firms to grow and compete globally. While initiatives like the **European Scale-Up Initiative** and **InvestEU** provide a foundation, their mandates and resources remain insufficient compared to the financial power of global competitors.

The **European Tech Champions Initiative (ETCI)**, a **€10 billion fund-of-funds** launched in 2023, represents a significant step forward. By supporting high-growth technology companies, the ETCI aims to retain innovation within Europe, offering late-stage funding to prevent promising startups from seeking capital or exits outside the continent. Similarly, the **European Innovation Council (EIC) Venture Fund**, with a **€10 billion budget for 2021-2027**, supports transformative innovations from early research to scaling. Complemented by **Horizon Europe** with its **€95.5 billion budget**, these initiatives address Europe's funding deficit but require greater ambition and alignment into a cohesive strategy to achieve meaningful impact.

Expanding the role of the **European Investment Fund (EIF)** and fostering cross-border collaboration among **Sovereign Wealth Funds (SWFs)** can provide the scale needed for Europe to compete globally. The **pan-European VC funds-of-funds programme (VentureEU)**, launched by the Commission and EIF, aims to boost investment in startups and scale-ups across Europe. The **European Investment Bank (EIB)** also supports early-stage innovation through equity investments, while **InvestEU** (€26.2 billion) guarantees investments to commercialize research, digitize industries, and scale innovative companies. Additionally, the **European Defence Fund** (€8 billion) promotes cutting-edge defence technologies and encourages startups and SMEs to participate in collaborative projects.

A unified strategy is essential to pool resources and focus investments on critical technologies. Establishing a **European Tech Sovereignty Fund** under the **European Competitiveness Fund** could bridge gaps in sectors such as AI, cloud infrastructure, and semiconductors. This fund would offer patient capital to drive long-term innovation while safeguarding Europe's technological independence. National Promotional Institutions (NPIs) and the EIB should play a proactive role by acquiring equity stakes in strategic companies. For example, **Bpifrance** has invested in AI startups like Mistral, while **Germany's KfW** has supported semiconductors, satellites, and renewable energy. A similar coordinated approach across Europe could foster homegrown innovation and protect critical industries.

¹⁸⁹ Luigi Serenelli, 'European Commission President's Plan for Common SWF Stalls', *IPE*, 27 March 2024, <https://www.ipe.com/news/european-commission-presidents-plan-for-common-swf-stalls/10072442.article>.

Europe must also ensure rigorous merger controls to prevent foreign acquisitions of strategic assets. Regulations should prioritize long-term innovation and ensure key technologies remain under European control. Public procurement policies must emphasize European solutions, creating demand for domestic innovations in cloud services, AI applications, and network infrastructure, as outlined in subsequent policy chapters.

Lessons can be drawn from countries like **South Korea** and **Japan**, which have successfully aligned financial and industrial strategies to maintain technological independence. South Korea's support for conglomerates like Samsung and LG, combined with targeted investments in semiconductors and AI, demonstrates how strategic alignment can achieve global leadership. Similarly, Japan's focus on robotics and precision engineering has preserved its competitive edge in key industries.

Europe's financial strategy must recognize the interconnectedness of investment, innovation, and sovereignty. Mobilizing financial resources, fostering a culture of long-term risk-taking, and implementing coordinated industrial policies are essential to reclaiming technological leadership. By expanding its VC ecosystem, empowering SWFs, and protecting its innovation assets, Europe can position itself as a competitive global player while safeguarding its strategic interests. Achieving this requires not only funding but also vision, collaboration, and the ability to translate financial power into sustained technological leadership.

The Geopolitical Dimension: Shaping Europe's Multipolar Vision

In the global race for technological dominance, two powers—China and the US—are vying for control across all layers of the digital stack.

China, under state-led initiatives like **Made in China 2025**¹⁹⁰, has rapidly expanded its capabilities in semiconductors, telecommunications, advanced manufacturing, and emerging technologies like quantum computing. These efforts, often dual use in nature, are backed by substantial public funding and aggressive talent acquisition strategies targeting Europe and the US. Simultaneously, China has tightened market access and strengthened its grip on critical raw materials through initiatives like the **Belt and Road Initiative** and export restrictions, leveraging dependencies as tools of economic and political influence. Under the banner of the Digital Silk Road, Chinese firms have accessed many overseas markets to export their cloud, telecommunications, and AI-technologies.

The US, through measures like the CHIPS Act¹⁹¹, is shoring up its semiconductor production and bolstering the global dominance of its tech giants in AI, cloud infrastructure, and advanced chip manufacturing. Increasingly stringent export controls targeting high-tech industries also impact European companies, as do industrial policy measures such as the massive Inflation Reduction Act¹⁹² and tariffs affecting strategic user industries such as energy and automotive. These actions highlight the competitive and protectionist nature of US policies.

Europe, while unable to compete head-to-head with these nations across the entire digital stack, must focus on reducing dependencies, safeguarding its strategic assets, and strengthening domestic capabilities. Internationally, Europe should prioritize partnership alliances that share its values, respect international rules, and aspire to balanced, inclusive and resilient digital ecosystems. Europe must reduce dependencies on digital superpowers while fostering values-driven cooperation. Partnerships with democratic nations like the US, Japan, South Korea, Canada, and Australia are vital for securing supply chains, co-developing technologies, and shaping global standards, emphasizing mutual respect in market access, data governance, and technology transfer.

Engaging rising powers such as Brazil, India, Chile, South Africa, and Singapore offer opportunities to diversify dependencies and expand influence. These partnerships should support industrial capacity-building and sustainable, non-extractive supply chains, aligning with shared priorities in clean energy, climate adaptation, and digital public services.

Amid dominance by US Big Tech and China's state-driven model, Europe can lead a non-aligned alliance for digital strategic autonomy. This vision prioritizes fair technology transfer, robust digital public infrastructures, democratic data governance, and global regulatory frameworks to curb tech monopolies. By advancing shared governance in AI, cybersecurity, and sustainability, Europe can set a global standard for ethical, inclusive, and sustainable digital leadership while navigating the complexities of diverse partnerships.

¹⁹⁰ China State Council, 'Made in China 2025', 2015, <http://english.www.gov.cn/2016special/madeinchina2025/>.

¹⁹¹ In full: Chips and Science Act, US Congress, 'CHIPS and Science Act', 9 August 2022, <https://www.govinfo.gov/content/pkg/PLAW-117publ167/html/PLAW-117publ167.htm..>

¹⁹² USG, 'Inflation Reduction Act of 2022', 16 August 2022, <https://www.regulations.gov/document/EPA-HQ-OAR-2023-0434-0066>.

Value-Added for Partner Nations in Europe's Digital Foreign Policy

India: India's transformative **India Stack**¹⁹³, including Aadhaar digital identity and UPI payments, showcases scalable public digital goods driving financial inclusion and innovation, with UPI processing over \$1.5 trillion in transactions in 2022. However, Europe's collaboration with India must account for differences in governance frameworks. Europe's **GDPR** emphasizes privacy, decentralized systems, and consent, diverging from Aadhaar's centralized, biometric-based model, which, while impactful, raises concerns about surveillance and inclusion. Europe's **Digital Identity Wallet** offers an alternative, privacy-first approach that India could adopt elements of to strengthen Aadhaar's transparency and trustworthiness. Collaborating with India allows Europe to learn from its large-scale bottom digital ecosystem of services built on top of decentralized protocols, while India can leverage Europe's expertise in privacy-enhancing technologies and data governance. Joint innovation in fintech, health tech, and interoperable digital services could combine India's scalability with Europe's regulatory strengths, creating interoperable, privacy-centric solutions that set global standards.

Brazil: Brazil's **PIX payment system**¹⁹⁴, with over 142 million users and daily transactions exceeding \$9.5 billion, exemplifies its leadership in digital financial inclusion. Developed by the Central Bank of Brazil, PIX has transformed financial access and could gain from interoperability with Europe's **Digital Euro**, facilitating cross-border payments and reducing remittance costs. Europe, in turn, can draw insights from Brazil's success in scaling user-driven financial solutions across diverse economic contexts. Beyond payments, Brazil's **National Artificial Intelligence Strategy** underscores its growing role in digital innovation, with applications in agriculture, healthcare, and financial services. This aligns with Europe's priorities for ethical, human-centric technologies. Joint efforts could focus on integrating AI into broader industrial policies, addressing supply chain dependencies, and adhering to privacy standards under Europe's **GDPR** and **AI Act**. The ongoing **EU-Brazil Digital Dialogue** expands this partnership, targeting connectivity in underserved regions, 5G/6G advancements, and High-Performance Computing collaboration. However, disparities in governance, scalability, and regulatory approaches challenge harmonization. To move beyond symbolic gestures, the partnership must focus on measurable goals, particularly in diversifying semiconductor supply chains, interoperable and independent digital systems, and AI governance, tackling shared challenges like financial inclusion and climate adaptation.

Latin America, Africa, Indonesia and Australia: Resource-rich nations like **Chile** and **Argentina** (lithium), **Indonesia** (nickel), and the **Democratic Republic of Congo** (cobalt) are advancing domestic industrial capacities to create value locally. Europe can support these efforts through investments in processing and refining technologies, clean manufacturing expertise, and co-developing green technologies, aligning with climate goals while ensuring secure access to critical materials. Chile and Europe are collaborating under the EU-Chile Advanced Framework Agreement for sustainable lithium extraction, while Argentina explores expanding its battery value chain through the EU-CELAC partnership. Indonesia has partnered with BASF and Eramet to develop nickel refining powered by renewables, and the EU's Global Gateway Initiative supports ethical cobalt sourcing in the DRC. With Australia, Europe works on critical minerals through the EU-Australia Partnership on sustainable critical and strategic minerals to diversify and green supply chains.

Japan and South Korea: Japan and South Korea are key partners in addressing Europe's semiconductor vulnerabilities. South Korea's Samsung and Taiwan's TSMC dominate over 70% of

¹⁹³ See: 'India Stack', accessed 12 January 2025, <https://indiastack.org/>.

¹⁹⁴ See: 'Pix', Banco Central do Brasil (BCB), accessed 12 January 2025, https://www.bcb.gov.br/en/financialstability/pix_en.

global advanced chip production, while Japan leads in materials like silicon wafers and photoresists. Their expertise complements Europe's Chips Act, which aims to raise Europe's global market share to 20% by 2030. Partnerships such as the **EU-Japan Green Alliance** and R&D agreements with Samsung bolster supply chain resilience and innovation. Europe's strengths in ethical AI and secure data-sharing frameworks align with Japan's and South Korea's advancements in AI-driven manufacturing and edge computing. Beyond semiconductors, joint initiatives in quantum technologies, high-performance computing, and advanced AI could solidify Europe's position as a leader in critical digital capabilities.

Middle East: Emerging Digital Hubs Saudi Arabia and the UAE are rapidly advancing as digital innovation hubs through state-led initiatives like Saudi Vision 2030 and UAE's smart city projects. Programs like Saudi Arabia's NEOM and Project Transcendence aim to integrate AI and modernize infrastructure, while partnerships with firms such as Huawei drive progress in 5G and cloud infrastructure. However, these centralized, state-driven and surveillance-driven approaches differ from Europe's decentralized, rights-based frameworks like GDPR and ethical AI standards. Concerns around data sovereignty, privacy, and the governance of AI and digital platforms highlight areas of divergence. Constructive collaboration can focus on Europe's strengths in data governance, sustainable urban planning, and renewable energy technologies, aligning innovation with shared values of transparency and sustainability. Diplomatic engagement should ensure projects address these concerns while fostering balanced, future-focused partnerships.

Africa: Many mutually interesting relationships are to be built with countries on the vast African continent. The Africa-EU Partnership and EU-Africa Business Forum provide resp. overall governmental and private sector frameworks, next to a range of bilateral agreements and financial support by Global Gateway and other EU programmes. Joint areas of interest include high-tech R&D (South Africa is the EU's biggest partner in research and Innovation in Africa), digital and legislative capacity-building (such as with Ghana), digital infrastructures including undersea cables and related operational capability (such as with Senegal), and rare minerals exploration (such as with D.R. Congo).

Cross-Layer Opportunities

Europe's leadership in data privacy and governance aligns with global efforts to establish ethical digital ecosystems. Initiatives like International Data Spaces position Europe as a leader in interoperable and secure data-sharing frameworks. Collaborating with India, Brazil, and other powers strengthens this position by creating systems that promote inclusivity, privacy, and resilience.

Governance and steering

Europe's digital foreign policy must strike a balance between reducing dependencies on digital superpowers and fostering cooperative, values-based relationships with diverse global actors. A coordinated governance structure is essential to achieve these goals.

- **Existing Units for Coordination:** an institutional framework already exists within the European External Action Service (EEAS), including units focusing on digital foreign policy and cybersecurity cooperation. Additionally, European Commission (EC) directorates and units play pivotal roles in coordinating international activities within the European Commission.
- **Enhanced Coordination:** strengthening these existing structures can ensure high-level steering and decision-making for digital foreign policy. This includes aligning internal policies, such as FDI screening and state aid, with external actions like joint procurement of

raw materials under the proposed Critical Raw Materials Platform, as highlighted in the Draghi Report.

- **Strategic Dialogue:** to advance synergies, systematic dialogue between industry, research, civil society, and governance bodies across Europe must be prioritized. These discussions should focus on rapidly implementing international cooperation frameworks, advancing the EuroStack, and ensuring alignment between Europe's digital foreign policy and its internal regulatory priorities.
- **Review of existing External Strategies:** such as the ones on the Indo-Pacific and Latin America, in the light of new realities and the European Digital Foreign Policy.

PART 3: European Digital Industrial Policy Blueprint

What Worked, What Failed, What's Next

As the analysis of the competitiveness and strength of Europe's tech players and digital assets in PART2 reveals, Europe's digital policies have faced persistent challenges that have limited their effectiveness from an industrial policy point of view. Internal market fragmentation, characterized by complex regulations and red tape, has stifled competition and restricted market access. A lack of coordinated commitment to market success has further hindered progress, with sectoral silos obstructing cross-industry collaboration and innovation.

Support for small and medium-sized enterprises (SMEs) has been insufficient, leaving these vital economic drivers struggling to scale. At the same time, short-sightedness in financial markets has led to chronic underinvestment and a lack of growth financing, restricting Europe's ability to retain top talents and nurture transformative innovation. Opportunities to decentralize and diversify innovation by leveraging regional and local potential have also been largely overlooked.

Globally, Europe's digital achievements have suffered from limited visibility due to under-resourced international diplomacy in this field. Domestically, policy has often been trapped in mid-level technologies, rather than aiming for the radical breakthroughs needed in deep-tech. This issue is compounded by the neglect of engineering and scientific talent and skills development, leaving Europe ill-prepared to meet the demands of the digital age.

A reluctance to embrace EU-wide cooperation in industrial policy has further impeded progress, while disproportionate influence from larger member states, notably France and Germany, has undermined balance and inclusivity. Weak EU mandates, coupled with a shortage of digitally literate policymakers, have exacerbated these structural shortcomings. Furthermore, policies have frequently prioritized short-term wins over the strategic long-term planning essential for sustainable growth.

Europe has also underestimated the geopolitical dimensions of technology, leaving itself vulnerable to dependencies and global competitive pressures. This vulnerability has become particularly evident amid rising trade wars, the resurgence of populist nationalism, and intensifying geopolitical rivalries.

To overcome these challenges, Europe must adopt a unified, forward-looking digital industrial strategy that is ambitious, inclusive, and effectively executed.

Analysis

Internal market fragmentation continues to stifle innovation across Europe. Divergent national regulations and inconsistent public procurement practices hinder scalability and competitiveness,

particularly in critical areas such as data sharing, cloud standards, IoT, and security-by-design. This fragmentation is even more pronounced in **defence and security markets**, where decentralized policies and stringent security restrictions obstruct the scaling of vital technologies.

There are three primary causes of this persistent fragmentation. First, the **lack of consistent enforcement of harmonized implementation**, even where the EU holds a strong mandate, undermines cohesion. Second, the European Commission's REFIT commitment to legislative simplification has not been pursued sufficiently, **leaving regulatory complexity unaddressed**. Third, the **EU's limited mandate in key areas** such as taxation, public funding for health, social welfare, education, justice, and security—sectors that account for approximately half of a country's GDP—further restricts its ability to act decisively.

Investment remains another major bottleneck. Europe has consistently fallen short in providing the risk capital necessary to scale innovative firms or adequately funding infrastructure for networks, semiconductors, and advanced manufacturing. This lack of financial and structural support has left Europe dependent on external providers, thereby undermining its strategic autonomy. The disparity in investment capacity compared to global competitors has also allowed foreign investors to capture Europe's top talent, startups, and scale-ups. These foreign investments often multiply the resources initially allocated by European public and private funding, leading to the loss of strategic European tech capabilities. Notable examples include the acquisitions of ARM and DeepMind, as well as other innovative companies that have shifted control of key technologies outside Europe. This highlights a glaring paradox: Europe possesses all the ingredients for success—world-class talent, a large internal market, innovative startups, and a vibrant scientific research ecosystem—but lacks the mechanisms to protect, adequately fund, and scale its digital ecosystem. Only then can it protect and fully realize the potential of its technological and industrial capabilities.

At the same time, not everything has been a failure. The EU and its Member States have demonstrated the ability to act decisively and achieve notable successes, as highlighted in the previous section of this report and further illustrated below. Understanding the underlying factors that contributed to both successes and failures is essential for shaping effective future policies. A thorough analysis must carefully disentangle cause and effect, taking into account a wide range of influences. These include neoliberal ideology, geopolitical shifts, technological trajectories, and organizational as well as managerial changes within companies. While a fully comprehensive and universally accepted analysis may remain elusive, state-of-the-art evaluations provide critical insights that can inform more effective strategies moving forward.

Building on these insights, the EuroStack initiative is uniquely positioned to address many of the challenges outlined above. By leveraging Europe's assets and learning from past successes and failures, the initiative can drive forward a robust and adaptive strategy. At the same time, it must remain flexible, evolving in response to new understandings and shifting dynamics in the global landscape and in technology.

Europe faces significant gaps in innovation, implementation, and fairness in its industrial policy (Schwaag-Serger et al. 2024). Key contributing factors include the under-exploitation of synergies for innovation across sectors, particularly between digital infrastructures and user industries. Additionally, while there is a strong focus on drafting regulations, there is insufficient control on ensuring their proper and harmonized implementation. Another missed opportunity lies in underestimating the potential of regional and local levels, which serve as crucial intersections of supply and demand dynamics¹⁹⁵. Mario Draghi (2024) highlights persistent fragmentation within the

¹⁹⁵ Sylvia Schwaag-Serger, Luc Soete, and Johan Stiern, 'Scientific Report - For an Innovative, Sustainable and Fair Economy in Europe', 2024, <https://doi.org/10.2760/0336180>.

Internal Market and lagging productivity growth as critical challenges. These issues result in the underutilization of Europe's domestic market potential, burdensome compliance efforts that detract from innovation, and a consequent loss of competitiveness in the global economy.

Such analyses generally do not place all the blame on past decisions made by policymakers at the EU and national levels. Instead, they recognize that the world has undergone significant changes over the past two decades. It is only in recent years that awareness has intensified regarding the profound impact of geopolitics and Big Tech on industry, the economy, and technological development. Strategic autonomy has only recently become a top priority, or *Chefsache*. Likewise, the ideology of neoliberalism and the associated push for globalization have only recently been critically reassessed. With this shift, space has emerged for the development of new industrial policies—such as those proposed here—and for what Dani Rodrik (2023) calls "economic nationalism done the right way."

It would also be a mistake to overlook the successes of EU and national policies. The Single Market program has been a tremendous achievement, particularly during its peak decades. Past industrial policies have played a key role in creating global champions such as Airbus and ARM. Similarly, the EU's European Fund for Strategic Investments (EFSI) has demonstrated remarkable impact, leveraging €35 billions of EU public funding to generate over €500 billion in total investment, particularly in physical infrastructure—a leverage factor of about 15¹⁹⁶.

In high-performance computing, the EU has made remarkable progress, advancing from having no representation in the global top-10 supercomputers in 2016 to securing three entries today. At the national level, supportive policies have nurtured thriving, world-class start-up ecosystems in cities such as Paris, Berlin, Stockholm, Amsterdam, Barcelona, Munich, Helsinki, Dublin, Tallinn, and others (StartupBlink, 2024). Additionally, Europe boasts a robust digital commons ecosystem and highly efficient, citizen-centric digital public services. Cities like Barcelona, Hamburg, Amsterdam, Helsinki, and Paris have been at the forefront of this transformation, leveraging digitization to serve people and society effectively.

The EU has shown remarkable decisiveness in times of crisis. In the early 2010s, it restored Eurozone credibility by committing to do "whatever it takes" during the financial crisis. During the COVID-19 pandemic, the EU swiftly organized joint vaccine procurement and rolled out a unified COVID app within three months, despite its limited public health mandate. The creation of the Recovery and Resilience Fund (RRF) also broke the longstanding taboo on shared EU debt to address common future oriented goals.

The EU has also acted decisively in areas of limited mandate, such as reducing its dependence on Russian gas and oil following the invasion of Ukraine. This was accompanied by a strong push for renewable energy and clean technologies across the continent. In cybersecurity, despite its strong link to national security¹⁹⁷, the EU adopted demanding laws, recognizing that cyber-threats are too significant for any single country to address alone.

¹⁹⁶ European Commission, 'Ex-Post Evaluation of the European Fund for Strategic Investments, the European Investment Advisory Hub and the European Investment Project Portal', December 2022, https://commission.europa.eu/about/departments-and-executive-agencies/economic-and-financial-affairs/evaluation-reports-economic-and-financial-affairs-policies-and-spending-activities/ex-post-evaluation-european-fund-strategic-investments-efsi-european-investment-advisory-hub-eiah_en.

¹⁹⁷ On national security Article 4(2) of the Treaty on the European Union states: "The Union [...] shall respect their essential State functions, including ensuring the territorial integrity of the State, maintaining law and order and safeguarding national security. In particular, national security remains the sole responsibility of each Member State."

Furthermore, the EU has increasingly shifted from using Directives, which allow for national variations, to Regulations, which mandate harmonized implementation across Member States. This shift ensures greater consistency in policy execution.

Examples

Case analysis in the digital domain illustrates how future industrial policy can and should be better.

The **GDPR** may have brought more personal data protection. However, as a stand-alone and relatively complex intervention, Europe's smaller companies missed out as they did not get support to ease compliance whereas the law was easier and more affordable to handle for large companies with deep pockets. The GDPR also did not come with accompanying industrial policy to make sure that EU companies could thrive in the internal market that opened up with their privacy-enhancing solutions (often open source!).

In **cybersecurity policy**, the EU is a world-leader at least where regulation is concerned. The EU's Horizon and Digital Europe Programme provide considerable support for cybersecurity R&I. Yet, there are hardly any EU cybersecurity companies with global success and billions of turnovers. For scaling-up they tend to depend on risk capital from the US, Israel and Singapore. Towards EU investors these companies struggle to 'make the investment case', are hampered by fragmented implementation of security across the Internal Market and do not benefit from promotion in international cyber-capacity building ¹⁹⁸. At the same time, there is huge market potential: critical infrastructure providers and governments must buy cybersecurity solutions given the mounting espionage by China and sabotage by Russia. This demand is mostly met by non-EU companies. As a consequence, Member States are buying resilience yet selling out on sovereignty.

EU **R&D programmes** for a long time subscribed to being 'open to the world'. A more critical approach is now taking hold. Economic security thinking is gradually being adopted in scrutiny of participants and restrictive access in some areas of Horizon Europe, while closer civil-military linkages and requirements for stronger IP protection (FDI scrutiny) may be coming. Yet, a policy of synergies between public investment in R&D and public procurement is still far from commonplace.

The EU **Chips Act** is a more modern industrial policy as it addresses next to supply-side interventions also international cooperation and takes economic security into account. While it is geopolitically aware it misses flexibility to respond to geo-economic developments such as subsidy wars. It also has a weak link to demand-side industries such as automotive or telecoms ¹⁹⁹.

What's Next: Towards a New Industrial Policy

Fontana and Vannuccini (2024) define industrial policy as: *"Industrial development involves the percolation of technologies across user industries and the provision of key inputs, such as critical raw materials and computing infrastructure that drive the digital economy and applications like artificial intelligence."*

While this definition highlights supply-side inputs and demand-side diffusion, EuroStack goes further by adopting a more integrated approach. The EuroStack initiative proposes an industrial policy that *combines supply- and demand-side interventions with traditional industrial policy tools such as R&D*

¹⁹⁸ Paul Timmers, Matthijs Punter, and Claire Stolwijk, 'Cybersecurity and Digital Sovereignty - Bridging the Gaps' (TNO, 2024), <https://publications.tno.nl/publication/34643188/DvSKsfCM/timmers-2024-cybersecurity.pdf>.

¹⁹⁹ Paul Timmers, 'Digital Industrial Policy for Europe | CERRE Report' (CERRE, 12 December 2022), <https://cerre.eu/publications/digital-industrial-policy-for-europe/>.

and investment support, while also incorporating strategies for market access, trade, economic security, and international policy. EuroStack prioritizes coordinated action, the removal of Internal Market barriers, substantial investment mobilization, and adaptability to evolving geopolitical and technological contexts.

This new industrial policy focuses on removing market entry barriers and deepening market integration, aligning industrial, competition and trade policies as complementary rather than conflicting. As the Letta Report (2024) suggests, regulatory frameworks can enable innovation and entrepreneurship rather than merely constraining them.

EuroStack also adopts a “Europe-first” approach to ensure capability building and sovereignty in critical domains. Inspired by Mazzucato and Rodrik (2024), **it introduces conditionalities** for investments and procurement, such as equitable access, reinvestment mandates, data sovereignty, and shared risk-reward mechanisms. Public procurement follows a comply-or-explain principle to **ensure investments generate public value and drive strategic autonomy for Europe.**

By aligning stakeholders, fostering demand for European solutions, and enhancing interoperability across sectors, EuroStack reduces dependency on external providers and reclaims leadership in critical technologies. This approach is essential for Europe’s resilience and competitiveness in the global digital economy.

A Need for New Instruments?

Over the past decade, the EU and Member States have introduced a range of industrial policy instruments. In research and innovation, these include initiatives such as IPCEI, EDIC, and STEP, as well as Joint Undertakings through public-private partnerships. In the defence sector, NATO, in part with the EIF, has introduced instruments like DIANA. For deployment investments, EU financial instruments like the former ESFI, the RRF, and InvestEU have leveraged public funding effectively. The EU has also created tools to support venture capital, including the EIC and the European Tech Champion Initiative, to address the critical need for late-stage growth capital.

Some of these instruments, like ESFI, have been evaluated and shown remarkable financial leverage. However, most still need to prove their effectiveness. The success of centralized vaccine procurement during COVID-19 demonstrated the advantages of collective bargaining, inspiring the EU to explore similar approaches for other essential goods such as critical raw materials and semiconductors. Proactive measures, including European Commission recommendations, legislative revisions, and cross-stack monitoring, have begun addressing regulatory inconsistencies, laying the groundwork for enhanced interoperability and market access. Additionally, Europe’s regulatory leadership, exemplified by GDPR and sustainability-focused policies, has set global benchmarks, creating a competitive advantage in shaping the ethical use of technology.

Despite these efforts, many existing instruments remain slow, rigid, and overly complex, unable to keep pace with rapid technological, market, and geopolitical developments. This rigidity often hampers implementation and results in missed opportunities. Moreover, Europe lacks integrated, agile, and coordinated industrial policy instruments with effective execution.

EuroStack does not advocate for more bureaucracy. On the contrary, it proposes an agile, innovative approach to equip Europe to deliver the critical digital infrastructures essential for competitiveness, security, and democracy. Rather than waiting for the challenges of existing instruments to be resolved, EuroStack proposes to hit the ground running. It seeks to leverage existing tools, draw on institutional

flexibility demonstrated during times of crisis, and, over the longer term, contribute to improving the speed, flexibility, feasibility, and relevance of industrial policy instruments.

The proactive digital industrial policy here proposed is articulated around four main pillars that address Europe's key technology gaps:

1. From Research and Development to Scaling European Platforms and Products

Europe has excelled in research but struggles to translate innovation into globally competitive platforms. A lack of integration across the digital stack limits its ability to scale and commercialize technologies, leaving gaps in critical areas. To address this, Europe must:

- **Build Competitive Platforms:** Establish **EuroStack** to unify existing initiatives like EuroHPC and the Quantum Flagship under a single governance framework. This would streamline innovation, set shared priorities, and ensure coordinated resource allocation to accelerate commercialization.
- **Leverage Public Procurement:** Use procurement strategically to prioritize European solutions, particularly in building federated cloud systems and sovereign data spaces. A “Europe-first, comply-or-explain” framework can reduce dependence on foreign providers, while public incubators can support early adoption and scaling of European technologies.
- **Close Talent and Investment Gaps:** Address critical talent shortages in AI, quantum, and HPC by offering competitive salaries, increasing R&I funding, and launching global talent attraction programs. Strengthening commercialization pathways will also attract private and institutional investors.
- **Set Global Standards:** Europe's leadership in privacy and ethical governance provides an opportunity to set standards in emerging fields like quantum computing, edge computing, and AI. Independent governance models, shared IP frameworks, and open-source initiatives can ensure collaboration and equitable innovation.
- **Strengthen Cybersecurity:** A comprehensive cross-stack security audit, coupled with periodic reviews, is essential to identify vulnerabilities and guide resilience-building measures. Robust cybersecurity frameworks will enhance trust and interoperability across all layers of the tech stack.
- **Align Defense and Space Strategies:** Create a unified EU defense and security market to foster innovation and scale critical technologies. Closer coordination of the defense and space industries will bolster Europe's strategic autonomy while supporting the broader goals of the EuroStack initiative.

By addressing these critical gaps and building on its existing strengths, Europe can transform its innovation ecosystem, secure digital sovereignty, and lead in the global digital economy.

2. Bridging the Investment Gap: The Case for a European Sovereign Tech Fund

Europe's venture capital ecosystem must scale significantly to remain competitive on the global stage. For instance, in 2023, the United States invested €62.5 billion in AI startups, compared to Europe's €9 billion.²⁰⁰ This critical disparity threatens Europe's ability to lead in high-tech innovation. The European Commission's Coordinated Plan on AI aims to address this issue by committing €1 billion annually from EU programs, with the ambition to scale public and private investments to €20 billion annually by the end of the decade.

²⁰⁰ Maslej, et al., ‘AI Index Report 2024 – Artificial Intelligence Index’.

According to Atomico, there is a €375 billion shortfall in European deep-tech growth-stage funding compared to the US. Of this gap, €75 billion is currently filled by foreign capital, underscoring Europe's reliance on external resources²⁰¹. To bridge this investment gap, particularly in critical technologies such as AI, semiconductors, and IoT, a coordinated and robust approach is essential. Existing programs like **VentureEU**—a partnership between the European Commission and the European Investment Fund—aim to mobilize €6.5 billion for start-ups and scale-ups. Similarly, public-private partnerships offer the opportunity to pool resources and expertise, accelerating innovation.

National governments and institutions such as the **European Investment Bank (EIB)** also play a pivotal role in reducing fragmentation and bolstering investment. **InvestEU**, with its €26.2 billion in guarantees, is designed to attract investment into areas such as research commercialization, industry digitization, and the scaling of innovative companies. However, addressing Europe's competitive gap requires more than these fragmented efforts. The **Draghi Report** estimates that an additional €150 billion investment between 2025 and 2030 is necessary for Europe to emerge as a global leader in digital technologies. This will require supporting foundational technologies like cloud infrastructure, data management, and advanced semiconductors, as well as fostering European platforms and products capable of competing globally.

While initiatives such as the **European Innovation Council (EIC)** are promising, they must evolve to be more **mission-oriented**, akin to DARPA in the United States. This alignment with the **EuroStack Initiative** would steer innovation toward strengthening Europe's strategic autonomy and achieving its technological sovereignty.

The Case for a European Sovereign Tech Fund

To address these systemic gaps, the establishment of a **European Sovereign Tech Fund** is imperative. This fund would consolidate funding mechanisms, streamline efforts, and align investments with Europe's strategic priorities. Its objectives would be twofold:

1. **Short-term scaling:**

Provide patient capital to start-ups and scale-ups in critical sectors like AI, IoT, and cloud-edge infrastructure. This would ensure innovations scale effectively and reach the market.

2. **Long-term strategic investments:**

Focus on transformative technologies such as quantum computing, next-generation semiconductors, and advanced AI. These investments would secure Europe's leadership in high-impact fields.

An initial capitalization of €10 billion would catalyze early progress. Over five years, investments would scale to €150 billion, ultimately reaching €300 billion over a decade, in line with the **EuroStack vision**. Funding would be pooled from EU member states, the EIB, national promotional institutions, institutional investors, and private capital. Public-private partnerships would amplify the impact and mitigate risks.

As a cornerstone of the **EuroStack vision**, the European Sovereign Tech Fund would address critical gaps in Europe's innovation pipeline. By fostering industrial champions, reducing dependencies, and

²⁰¹ Niklas Zennström, 'European Tech's Confidence Crisis Is Its Biggest Challenge', 10 December 2024, <https://atomico.com/insights/european-techs-confidence-crisis-is-its-biggest-challenge>.

bridging short-term scaling needs with long-term strategic investments, the fund would position Europe as a global leader in the digital economy while ensuring its technological sovereignty.

3. Addressing Europe's Talent and Innovation Gaps

Europe faces two critical challenges: a shortage of skilled talent in key fields such as quantum computing, AI engineering, and semiconductor design, and the loss of intellectual property (IP) to global competitors. Addressing these gaps requires a coordinated European strategy that strengthens innovation, retains IP and talent within Europe, and ensures public investments support economic and strategic autonomy.

Germany's Federal Agency for Disruptive Innovation (SPRIND) provides a compelling model for bridging these gaps. By focusing on high-risk, high-reward projects, SPRIND fosters breakthrough innovations capable of creating new markets or transforming industries. Its approach combines long-term funding, rapid decision-making, and interdisciplinary collaboration to accelerate transformative ideas. Expanding this approach to the European level could significantly enhance the continent's innovation capacity. A European Disruptive Innovation Agency would pool resources from multiple countries, create a pan-European network of innovators and researchers, and align disruptive technologies with the EU's strategic goals. Integrated into the EuroStack framework, such an agency could prioritize breakthrough areas like AI, quantum computing, and advanced biomaterials, positioning Europe as a leader in next-generation technologies.

To build the **skills pipeline** needed for the EuroStack's success, these agencies should prioritize competitive salaries, substantial research grants, and advanced training and acceleration programs in emerging technologies. Mobility initiatives will foster cross-border collaboration, creating dynamic hubs of innovation across Europe. EU funding mechanisms, including the Digital Europe Programme (DEP), Horizon Europe, and the Connecting Europe Facility (CEF), will embed conditionalities to prioritize digital skills development and leverage AI-enhanced EuroStack technologies to reduce the demand for highly specialized skills. Security-by-default market requirements will further incentivize the development of technologies that minimize advanced skill dependencies.

Safeguarding European intellectual creations is crucial. Publicly funded innovations outcomes must be valorised within Europe, supported by open standards, and shared IP frameworks that foster ethical, collaborative development while preserving sovereignty. To secure long-term strategic autonomy, the European Commission and European Investment Bank (EIB) should take equity stakes in companies of critical importance (including startups and scaleups) within sectors such as semiconductors, AI, quantum computing, and cloud infrastructure. This approach will enable these companies to scale globally while maintaining European control over essential technologies and products.

By addressing talent shortages, protecting intellectual property, and aligning investments with strategic priorities, Europe can secure its position as a global leader in innovation. This coordinated approach ensures the success of the EuroStack and Europe's digital sovereignty in an increasingly competitive landscape.

4. Reducing Dependencies and Building Sustainable Interdependencies

Europe remains heavily dependent on foreign suppliers for semiconductors, cloud infrastructure, AI, and critical raw materials. These dependencies expose Europe to significant vulnerabilities, particularly amid geopolitical tensions and supply chain disruptions.

To mitigate these risks, Europe must strengthen domestic capacity while forging **sustainable interdependencies** with resource-rich nations like **Chile, Argentina, and Australia**. Joint ventures and co-investments in sustainable extraction and processing technologies, aligned with the EU Critical Raw Materials Act, can improve supply chain resilience. However, Europe must accelerate its efforts to match competitors such as the US and China, which are rapidly securing resources through direct and transactional agreements.

Domestically, the European Chips Act must prioritize high-value niches, such as AI-optimized and energy-efficient chips, leveraging Europe's strengths in advanced semiconductor manufacturing, exemplified by global leaders like ASML. Addressing high capital costs and skilled labour shortages requires targeted public-private investments and robust training programs to support long-term competitiveness.

A cohesive **economic foreign policy** is vital to advancing EuroStack and increasing the global presence of European technologies. Limited EU support has hindered the success of European digital solutions in international markets. To address this, Europe must ensure its technologies are integrated into global initiatives like the **Global Gateway**, cybersecurity capacity building, international standard-setting, and trade agreements. Promoting European technologies requires active outreach. EU institutions, in collaboration with the private sector, should facilitate and support **trade promotion missions** to diversify the supply of critical goods and position EuroStack solutions on the global stage. This strategy will enhance Europe's competitiveness while contributing to a fairer and more balanced global digital ecosystem. EuroStack's success also depends on its ability to adapt to political, economic, technological, and environmental changes. The initiative must actively monitor and interpret external trends, ensuring alignment with broader EU priorities such as the green agenda, multilateral cooperation, and defence strategies.

Policy recommendations

Actions within the EuroStack

The EuroStack exemplifies a modern industrial policy that is strategically aligned to deliver digital sovereignty. Each industrial policy action described here contributes to building the capabilities, capacities, and control needed to enhance EU competitiveness, security, democracy, and environmental sustainability.

EuroStack prioritizes two forms of agency. **Operational agency** leverages the EU mandate for the internal market, focusing limited resources on areas of strategic importance. Meanwhile, **geo-economic agency** aims to reduce dependencies on foreign technologies, foster sustainable interdependencies, and advance European solutions in areas where external dominance threatens strategic autonomy.

EuroStack does not attempt to address all necessary actions in the digital domain—some are already pursued elsewhere, while others fall outside the scope of industrial policy. Instead, the EuroStack analysis identifies four key areas of action:

1. **Strengthening capabilities within individual layers of the digital stack**, such as AI, semiconductors, and cloud infrastructure.
2. **Delivering synergies across layers** to create an integrated and interoperable Digital Public Infrastructure (DPI)
3. **Ensuring coherence and consistency through the EuroStack implementations and Minimum Viable Products (MVPs).**

4. **Aligning with broader agendas**, such as competition policy, trade, sustainability, and global digital governance.

It should be noted that given that Stack layers are typically linked by input-output complementarities, strengthening capacity in one layer of the digital stack often generates positive externalities and spillover effects on adjacent layers. By bolstering the supply-side of a particular segment—say, cloud infrastructure—upstream suppliers and downstream application developers experience heightened demand and improved market conditions. This dynamic reinforces a virtuous cycle of investment and innovation across the ecosystem. In turn, increased output and efficiency in one layer lower per-unit costs and attract further complementary specialisation in related layers, delivering aggregate productivity gains and enhanced competitiveness for the entire EuroStack. Ultimately, this cascading economic impact magnifies the benefits of industrial policy interventions, ensuring that a strategic focus on a single layer reverberates throughout the interconnected network of digital capabilities.

1. Policy Action: Smart and Sustainable Resource Stewardship

Set clear targets and provide financial incentives for optimizing energy, water, and critical raw materials within the EuroStack, aligned with EU Green Deal goals and support this by monitoring.

- **Focus on Efficiency:** Ensure EuroStack technologies achieve measurable reductions in energy use, water waste, and reliance on raw material imports.
- **Procurement Standards:** nationally mandate energy-efficient, recyclable, and circular-compliant technologies to benefit from EuroStack support.
- **Investment Alignment:** Tie funding to clear climate and resource efficiency outcomes, leveraging EU instruments like Horizon Europe and Green Bonds.

2. Policy Action: Strengthening Advanced Semiconductor Capabilities to drive AI and HPC innovation.

Focus on advanced semiconductor technologies and ecosystem integration to reduce dependency, enhance Europe's digital sovereignty, and drive AI and HPC innovation.

- **Invest in Advanced Nodes:**
 - Prioritize open hardware architectures with incentives for advanced manufacturing such as for **RISC-V**, photonics, neuromorphic and quantum chips.
 - Cross-stack synergies: promote collaboration among European chipmakers, software developers, and cloud providers to align design and manufacturing capabilities.
- **Stimulate Demand Through Public Procurement:**
 - Require **50% European-made processors and accelerators** in critical infrastructures, defence, public administration, and critical systems by 2030. Use **defence and strategic digital technology procurement** to create sustained demand for domestic production.
- **Integrate Supply and Demand Ecosystems:**
 - Build and financially incentivize **demand-supply partnerships** that link semiconductor advancements to key sectors like automotive (37% of Europe's semiconductor consumption) and expand to emerging sectors such as biotech, gaming, space economy, and advanced manufacturing.
- **Safeguard Strategic Assets:**
 - Strengthen FDI scrutiny and FDI controls to protect critical European IP and startups, particularly in quantum and AI chips.

- Establish a database of strategic assets in the EuroStack, to ensure resilience and monitor control in Europe.

3. Policy Action: Strengthening Industrial IoT and Connected Devices

Implement harmonized regulation, robust cybersecurity measures, and leadership in global standards to foster a secure and competitive IoT ecosystem within the EuroStack.

- **Ease the application of IoT-related legislation:**
 - Simplify the application of EU IoT-related regulatory frameworks (energy, safety, security, privacy), reducing compliance costs for businesses by **up to 20%**.
 - Support IoT trust-labelling, compliant with Cyber Resilience Act, NIS2, and GDPR, provide compliance assistance for SMEs through automated toolkits and centralized support platforms, and internationally promote a Trust-in-EU mark.
- **Step up involvement in Global Standards:**
 - Provide financial support and commit companies to increase EU participation in international IoT standardization bodies.
- **Support IoT Innovation:**
 - Incentivize secure and innovative IoT solutions through Horizon Europe and Digital Europe programs, targeting a **25% increase in EU IoT patents** by 2030, and tracking their exploitation in the European and global marketplace.

4. Policy Action: Enhancing Sustainable Competitiveness in the Network Layer

Strengthen EU telecom competitiveness, security, and sustainability through harmonized regulation, strategic investments, and synergies with EuroStack layers.

- **Promote Cross-Sector Synergies:**
 - Coordinate telecoms security and resilience with defence, energy, and cloud sectors, with a cross-border EDIC (EU Digital Infrastructure Consortium).
- **Accelerate 5G Adoption and Demand-Supply synergy:**
 - Cross-stack: financially support large-scale pilots for standalone industrial 5G networks integrated with edge-cloud with low latency and high availability, targeting deployment in **80% of industrial hubs by 2030**.
 - Cross-stack: promote in 5G rollout the edge-cloud paradigm shift, aligning with EuroStack's cloud layer actions; and link telecoms as a key consumer to semiconductors, applied AI, and secure quantum communications within EuroStack.
 - Promote **SCION** architecture, which stands for Scalability, Control, and Isolation on Next-generation Networks, redefining how data moves across the Internet.

5. Policy Action: Achieving Strategic Autonomy in the Cloud Layer

Develop a cohesive strategy to strengthen Europe's cloud ecosystem, prioritizing edge cloud, strategic linkages to HPC, AI, and quantum technologies, and ensuring compliance with EU digital sovereignty objectives.

- **Implement a European Cloud Initiative**, building on and going beyond GAIA-X:
 - Promote and, where feasible, mandate **preferential public procurement for European cloud solutions** using a *comply-or-explain* framework. The goal is to ensure that **50% of public cloud spending is allocated to EU-based providers by 2030**. To support the initial phase of adoption and infrastructure development,

governments should commit to **€500 million annually over 5-7 years** for the procurement of European cloud solutions.

- Provide public procurement guidance to boost adoption of **GDPR-compliant and energy-efficient EU-based solutions**.
- **Promote Edge Cloud Paradigm:**
 - Cross-stack: financially support pilots of **edge-enabled 5G** networks for industrial and public service applications, integrating with other decentralized cloud services.
- **Adopt Decentralized Edge Solutions:**
 - Set up a pan European Multi-Provider Cloud-Edge Continuum (8ra Initiative) to reduce latency by **up to 50%** and improve resilience by decreasing reliance on centralized infrastructure.
 - Focus on critical sectors such as **healthcare, public safety, and transportation**, where decentralized solutions can enhance response times and reduce operational risks.
- **Promote and require Open API Ecosystems:**
 - Promote and require open APIs for cloud providers to ensure interoperability, support cross-border data flows, and prevent vendor lock-in, which currently limits over **75% of enterprises using proprietary platforms**.
 - Align with EU initiatives for **data sovereignty and digital interoperability**, critical for enhancing cross-sector collaboration.
- **Strengthen Linkages to HPC, AI, and Quantum:**
 - Cross-stack: align cloud initiatives with HPC, AI and quantum technologies programs to enhance Europe's competitive edge in processing complex workloads.

6. Policy Action: Building a Sustainable Software Industry Ecosystem

Strengthen Europe's open software ecosystem to enhance digital sovereignty, reduce dependency on proprietary solutions, and support public trust and industrial innovation.

- **Adopt an Open-Source First Policy:**
 - Promote and if necessary, mandate (at Member States level and ideally at EU level) **open-source software** for public services and critical infrastructure by **2026**.
 - Prioritize European open-source solutions through preferential procurement and incentives for private adoption.
- **Govern Key Digital Commons:**
 - Establish public-private-commons partnerships to manage federated **AI platforms**, with **€1 billion funding** to ensure transparency and alignment with EU values and continue support to common EU services.
- **Promote Global Standards Leadership:**
 - Advocate for open software principles in global standards to establish EU leadership in shaping digital governance.

7. Policy Action: Achieving AI Sovereignty and Harnessing Data as a Strategic Asset

- **AI Strategy:**
 - Cross-stack: Develop a unified strategy integrating AI, HPC, AI cloud, AI chips, and data governance to create scalable, secure, and interoperable infrastructure. This ensures European AI and data leadership while preserving digital rights, accountability, and sovereignty.

- **Public Interest Data as Public Good:**

- Establish **EU-wide Data Trusts** to treat **public interest data as a common** resource, enabling secure data pools for sharing public and private data, while maintaining GDPR compliance, private to public data sharing in public interest. Mandate data access in public procurement contracts. While there is value in sharing public interest data, ensuring the protection of data is equally important, especially intrinsically sensitive types such as biomedical and geolocation data.
- Cross-sector data sharing must follow impact assessments, data minimization protocols, and sector-specific guidelines while ensuring interoperability. Strong safeguards against surveillance and unauthorized use require strict access controls, audit trails, transparency reporting, and legal protections against unauthorized monitoring or profiling.
- It is essential to include provisions for **data transparency and accountability on what data is used for AI model training** in procurement contracts and public tenders or licensing.
- Build a European data ecosystem to support training sovereign AI models, leveraging **federated data platforms** for cross-border and cross-layer data exchange of authorised data (e.g., IoT, cloud, AI), linked to and reinforcing the **European Data Spaces Program**, which aims to create sector-specific data ecosystems in health, finance, energy, and mobility. Prioritize interoperability and secure data exchange within these spaces to ensure collaboration, enhance innovation, and maintain data sovereignty across critical sectors.

- **Data as a European Asset:**

- Establish clarity as regards the use of large European data assets and **assess benefits of preferential access** in EuroStack for European innovators.

- **Enable Interoperable Data Exchange, speeding up Data Spaces execution:**

- Accelerate through cooperation and incentives development and deployment of federated data platforms to ensure secure and sovereign data sharing across IoT, cloud, and AI ecosystems.

- **Unify Governance Across AI Infrastructure:**

- Cross-stack: Link IPCEI projects in AI, quantum, chips, and HPC to ensure alignment with GDPR, the AI Act, and data sovereignty goals.
- Join-up public-private collaborations to reduce fragmentation and accelerate development of AI hardware, cloud, and HPC infrastructure.

- **Build AI Cloud and Chips:**

- Cross-stack: develop integrated platforms combining **AI-specific HPC resources, sovereign AI cloud capabilities, and specialized AI chips** to support compute-intensive applications like **large language models (LLMs)**.
- Focus on strategic applications in healthcare, mobility, biotech, and public services while ensuring data remains secure and localized.

- **Federated and Interoperable AI Systems:**

- Deploy federated AI models that train on localized data without transferring raw data, ensuring **compliance with GDPR** and minimizing risks of data misuse.
- Integrate federated AI with edge-cloud systems for **real-time decision-making** in critical sectors like manufacturing, smart cities, and public safety.

- **Establish a Truly Open Public AI Marketplace for Europe:**

- Develop a truly open AI marketplace powered by federated AI models and open APIs, utilizing **public domain** and **CC0-licensed datasets**. This platform will provide accessible, trustworthy AI tools for SMEs, public cultural institutions, and media organizations, fostering innovation across diverse sectors. The AI systems will be

trained using the **EuroHPC network**, ensuring high-performance computing resources are leveraged for European-led advancements.

- **Cross-stack:** design the marketplace to integrate with **IoT, cloud, and edge systems**, creating a cohesive and scalable environment for innovation. This integration will enhance real-time AI applications in industries like smart cities, healthcare, and cultural heritage digitization.
- **Invest in AI-Driven IPCEIs:**
 - Allocate large scale investments for AI infrastructure innovation, targeting HPC, quantum technologies, energy-efficient AI chips, and scalable AI applications (align the European Sovereign Tech Fund mission).
 - Provide a target of **500 AI startups, scaleups and SMEs by 2030** to the **European Innovation Council (EIC)**, boosting competitiveness and innovation.
 - Strengthen investments in growth Funds and link these scaleups to key industries via **Corporate Venture Capital (CVC)** and strategic industrial policy initiatives. Ensure the **European Investment Bank (EIB)** and **EIC** invest in equity stakes of European AI champions in critical areas for European competitiveness, safeguarding their growth and preventing foreign takeovers of strategic assets.
 - **Ensure AI Leadership Through Integrated Infrastructure:** Promote open, transparent, and secure systems aligned with European values to counter Big Tech's proprietary ecosystems.
 - Advocate globally for **data sovereignty, open and accountable AI standards** in line with the **AI Act** through EU leadership in international standards bodies.
- **European AI for All Initiative**
 - Establish a "European AI for All" initiative to democratize access to AI technologies and foster widespread adoption across industries and public services.
 - Create a network of AI innovation hubs across EU member states, providing resources, expertise, and computing infrastructure for SMEs and public sector organizations to develop and implement AI solutions.
 - Develop open-source, pre-trained AI models and datasets tailored to European languages, industries, and use cases, reducing barriers to entry for smaller organizations.
 - Launch an AI skills program to train and upskill the European workforce, focusing on both technical and ethical aspects of AI development and deployment.
 - Implement a voucher system for SMEs to access AI consulting services and cloud computing resources, accelerating adoption in traditionally less digitized sectors.

8. Policy Action: Strengthening Leadership in Quantum Technologies

Enhance EU leadership in quantum technologies by aligning supply-side innovation with demand-side adoption through targeted actions and coordinated governance.

- **Accelerate Lab-to-Market Transition:**
 - Scale EU quantum initiatives under the **Quantum Flagship** with **€5 billion in EU funding by 2030** for quantum computing, sensing, and secure communications.
- **Develop Common Procurement Standards and Europe-First procurement:**
 - Establish shared public procurement specifications in Quantum Act, starting with **secure quantum communications**, for early adoption of EU quantum solutions.
 - Implement a **comply-or-explain framework** in Quantum Act requiring governments to prioritize EU-funded quantum solutions unless justified otherwise.
- **Foster Collaboration Across Stakeholders and Stimulate Demand:**

- Link the Quantum Flagship with national initiatives through a Europe-wide **Quantum Innovation Network** for sharing best practices, cross-border R&D and pilot projects.
- Cross-stack: boost quantum adoption by linking quantum initiatives to critical sectors like telecoms, cloud, semiconductors, cybersecurity, and AI, targeting a **25% EU share of global quantum market by 2035**.

Cross-Stack Strategic Initiatives

Next to Demand-Supply and Europe-First Procurement Strategies that are listed above the following cross-stack actions are needed:

- 1. Unified Compliance Frameworks:**
 - Simplify pathways for public-private partnerships in EuroStack and for SMEs to adopt legally compliant solutions efficiently. Provide (EC) recommendations for simplified compliance.
- 2. Green by Design:**
 - Require energy efficiency across all layers of the EuroStack, ensuring alignment with **EU Green Deal goals** and reducing energy use by **at least 30% by 2030**. Transparency measures like the EU's Corporate Sustainability Reporting Directive should be enforced, and investments in renewable energy and carbon-aware computing must be sustained, while AI companies should be required to publicly disclose resource use and emissions across their entire lifecycle, from manufacturing to training and inference.
- 3. Cross-Layer Federated Architecture and Digital Commons for Europe:**
 - **Interoperable Data Exchange and Data Commons:** Create federated data platforms for secure cross-layer data sharing, maintaining GDPR compliance and data sovereignty and related data commons with corresponding governance such as **EU Health Data Commons** for federated research in genomics and precision medicine, or **EU Cities Data Commons** for urban digital twins and interoperable local public services.
 - **Public Procurement for Digital Commons:** Use smart public procurement to pool resources and fund collaborative digital commons initiatives.
- 4. Cross-Layer Resilience Framework:**
 - Design a pan-European framework to address **cybersecurity, redundancy, and resilience** across all layers, fostering strategic autonomy in critical infrastructures.
- 5. Empowering Ecosystem Players:**
 - **Support SMEs and Startups:** Provide funding, technical support, and compliance assistance to drive innovation and speed up adoption across multiple layers.
 - **Actively building and maintaining digital commons**—Create large repositories of digital public goods governed to ensure inclusivity, transparency, and protection from monopolistic capture. Digital commons encompass software, hardware, open data, educational resources, and digital cultural content and media. The shift in policy is clear: it is no longer about fostering openness alone, but about building sustainable platforms maintained by European institutions to support digital commons.
- 6. Interplay with Defence:**
 - Integrate defence requirements into EuroStack layers through shared standards, pilots, and early adoption programs.
- 7. Strategic Investment: European Sovereign Tech Fund (ESTF):**
 - Establish an ambitious target of **€300 billion investments over 10 years** to drive innovation and secure Europe's strategic autonomy.

- **Kick off a European Sovereign Tech Fund of €10billion investment** prioritizing the EuroStack common services and MVPs to ensure tangible early outcomes and increasing the amount as the implementation proceeds.
- The European Tech Sovereignty Fund will invest in:
 - Advanced semiconductor R&D, cloud, and edge innovation, supporting GDPR-compliant infrastructure, federated AI models and interoperable data platforms, digital commons such as the EU Health Data Commons and the EU Cities Data Commons.
 - Co-invest with the **EIB and EIC** to scale **500 AI and quantum startups by 2030** and prevent foreign takeovers of critical assets.
 - Allocate funding for energy-efficient technologies, targeting a **30% reduction in stack-wide energy use by 2030**.

Implementation Roadmap

The EuroStack Initiative outlines a strategic, phased plan to transition Europe from technological dependency to digital strategic autonomy. Central to this effort are the **EuroStack Digital Infrastructure** and the **MVPs**. This roadmap balances ambition with feasibility, ensuring Europe can set a clear autonomous path in digital innovation while safeguarding its independence. With robust governance, strategic investments, and clear milestones, this plan transforms fragmented efforts into a cohesive strategy for Europe’s digital future.

Importantly, the EuroStack roadmap consists of parallel running streams that will evolve over time. Work is to start immediately with deploying more mature solutions, in parallel to research and development of more advanced contributions. Governance likewise starts from day 1, and will grow, evolve, and adapt over time. See the diagram below.

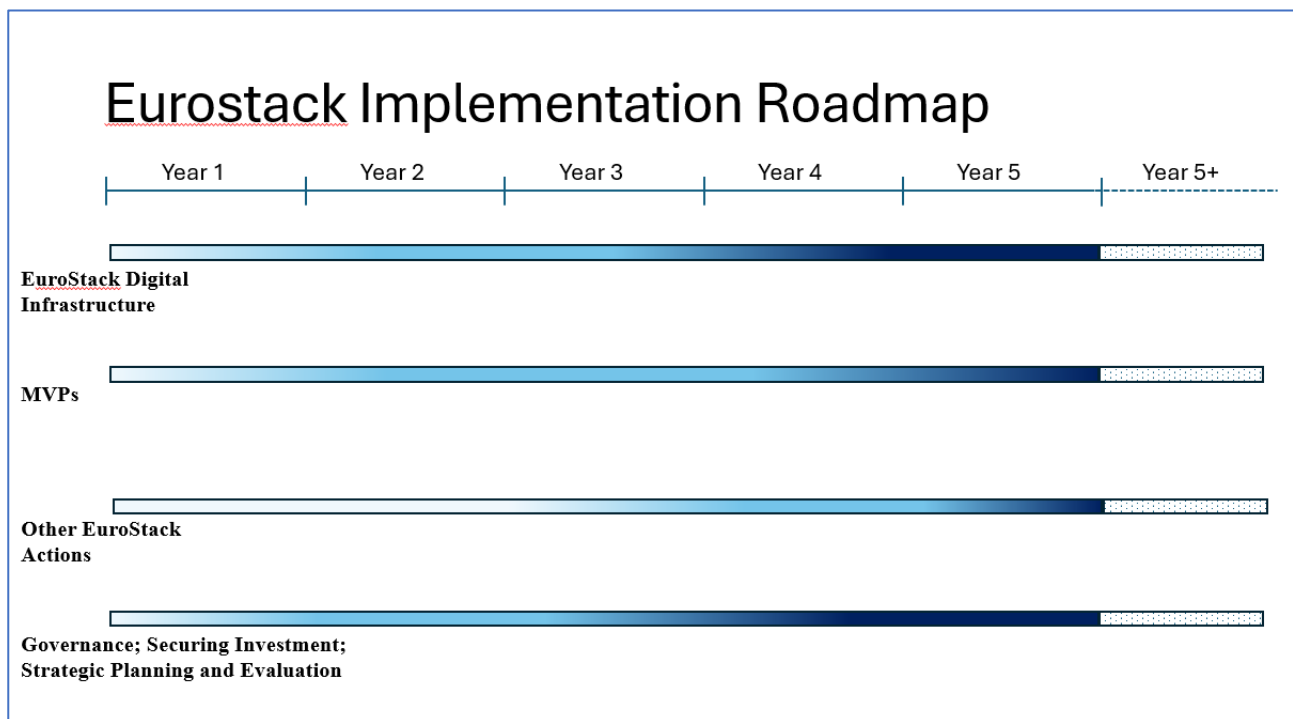


Figure 8. EuroStack Implementation Roadmap

Stream 1: EuroStack Digital Infrastructure (from year 1)

Objective: building the EuroStack Digital Infrastructure (EDI) for interoperable digital services, hardware, and software.

- **Develop, deploy, upgrade** – the Digital Infrastructure develops as needed for digital strategic autonomy, starting from an initial set of services and technologies:
 - **Digital Euro:** Accelerate rollout to enable secure, fee-free cross-border transactions across services.
 - **Sovereign Digital Identity Wallet:** build a privacy-preserving, interoperable wallet integrating eID, payment systems, and access to public and private services.
 - **Federated Data Spaces:** Build GDPR-compliant platforms for secure cross-border data sharing in healthcare, education, mobility, and climate. Develop a coalition of cities and regions using common procurement frameworks to implement these standards.
 - **Sovereign Cloud Infrastructure:** Expand decentralized cloud systems, such as IPCEI-CIS and European Open Clouds, to retain critical data within Europe’s jurisdiction.
 - **Federated AI:** This approach enables decentralized training of AI models across heterogeneous hardware environments, ensuring resilience and adaptability.
 - **Partnerships with international players:** for instance, with South Korea, Taiwan, Chile, Brazil, and others to address supply chain vulnerabilities, particularly in semiconductors and raw materials.
- **Widespread Adoption:** Leverage strategic public procurement and regulatory incentives to ensure widespread adoption across public and private sectors. Align investments with EU market regulation and competition policy to curb monopolistic practices and promote competition.
- **Sustainability:** Invest in long-term maintenance of the EDI, notably federated AI models, decentralized platforms, and data commons to ensure resilience and reliability.

Stream 2: MVPs (from year 1)

Objective: Develop and deploy when readiness reached of EuroStack Digital MVPs in strategic sectors and refine its components. MVPs develop in stages, with a rapid start building on mature technologies (TRL level 9) and integrating in later states advanced solutions (such as those that are today at TRL level 6).

- **Sectoral Challenge for the EuroStack Digital Trailblazers or MVPs, “The EuroStack Challenge:** Launch a competition fund innovative ideas to run MVPs and use cases on top sovereign EuroStack building blocks. Examples of possible MVPs are in areas of strategic interest, such as Healthcare, Education, Climate and Energy, Fintech, Media, and Culture. Each deployment focuses on feasible collaborations and builds on Europe’s strengths in research, industry, and public institutions to advance sovereign digital services and ensure the success of the Digital EuroStack MVP.
- **Advanced Development and Testing Locations:** Advance the MVPs in a range of locations, focusing on scalability and interoperability.
- **Expanding Capabilities:** Integrate **post-quantum cryptography, QKD, advanced AI**, and other gamechangers into the MVPs to secure data exchanges and enhance operational efficiency. Extend applications to biotech and clean tech, ensuring leadership in industrial IoT and renewable energy.

Stream 3: Other EuroStack Actions within and across the stack

Objective: Advance the range of actions within and in between layers of the stack and cross-stack actions.

- **Build commitments:** as the EuroStack community grows and investment ramps up commitment is to be established and actions can get launched.
- **Advanced Technologies:** put in place solid work on next-generation AI, post-quantum cryptography, edge-cloud, advanced semiconductor technologies, and other advanced technologies to secure the future of the EuroStack, strengthen resilience and innovation capacity.
- **Widespread Adoption:** Leverage strategic public procurement and regulatory incentives to ensure widespread adoption across public and private sectors. Align investments with EU market regulation and competition policy to curb monopolistic practices and promote competition.
- **Sustainability:** Invest in long-term maintenance to ensure resilience and reliability.

Stream 4: Governance, securing investment, strategic planning, and evaluation (from year 1)

Objective: raise interest in investment and mobilize funding commitments, ensure alignment with the long-term strategic goals of digital strategic autonomy through wide involvement and robust governance with agility to respond to geopolitical and technological developments, and performance evaluations.

Governance

EuroStack is a vision and a plan to be executed by contribution from committed parties, who are willing to mobilize and contribute funding, respond to the competition challenge for MVPs, bringing together their current strong work in Europe. Industry, innovators, civil society, Member States, European Commission are called upon to respond to the EuroStack vision and plan.

Several EuroStack actions can be implemented within or by extending such current work. Other actions will require new instruments or projects to be launched, possibly supported by EU investment, deployment, development, or research programmes. Implementing EuroStack is based on wide involvement with mutual commitment, supported by wise leadership.

- **Independent Governance Body:** Establish a centralized governance structure, inspired by the European Central Bank model, to oversee interoperability, public accountability, and ethical deployment. A launch-stage governance should be succeeded by lasting governance that can count on wide support.
- **Governance Foundations:** Following the launch-stage governance, establish a **EuroStack EDIC** to coordinate member state efforts, set shared priorities, and ensure resource alignment.

Securing Investment

As highlighted in the Draghi Report, an additional **€150 billion investment is required between 2025 and 2030** to establish Europe as a global leader in digital technologies. Building on this foundation, the **EuroStack initiatives aim to secure €300 billion over a decade through the establishment of a European Sovereign Tech Fund**, ensuring Europe not only achieves digital sovereignty but also remains competitive in the global tech landscape.

To jumpstart this ambition, the EuroStack proposes the creation of a **€10 billion European Tech Sovereignty Fund**. This fund will be strategically allocated to develop **EuroStack common services** and **minimum viable products (MVPs)**, laying the groundwork for a self-sufficient and interoperable European tech ecosystem. The fund will pool resources from multiple channels, including **Horizon Europe**, the **Digital Europe Programme**, **EIC investments**, and **national contributions**, ensuring a unified and efficient approach to resource mobilization and impact generation.

Strategic planning, monitoring, evaluation

- **Maintain and update the EuroStack strategic plan:** extend the present document with a more specific allocation of actors, means, and timescales.
- **Adaptation:** ensure that the strategic plan stays geopolitically-, market- and technology-aware. Identify barriers and refine strategies based on dependencies, geopolitical, technological and market developments.
- **Evaluation and Adaptation:** Conduct a comprehensive evaluation to measure user adoption, scalability, and economic impact. A mid-term evaluation is foreseen in Year 3.

PART 4: Complementary Readings

Readings on Europe's Digital Sovereignty for Policy Makers

ANRT FutuRIS. 'Digital Sovereignty: Regaining Control in France and Europe', January 2024. [https://www.anrt.asso.fr/sites/default/files/2024-](https://www.anrt.asso.fr/sites/default/files/2024-03/ANRT_Digital_sovereignty_regaining_control_in_France_and_Europe_01.24.pdf)

[03/ANRT_Digital_sovereignty_regaining_control_in_France_and_Europe_01.24.pdf](https://www.anrt.asso.fr/sites/default/files/2024-03/ANRT_Digital_sovereignty_regaining_control_in_France_and_Europe_01.24.pdf).

Bria, Francesca. 'European Digital Independence: Building the EuroStack'. AI Now Institute (blog), 15 October 2024. <https://ainowinstitute.org/publication/x-european-digital-independence-building-the-eurostack>.

Bria, Francesca, Johnny Ryan, Sophie Bloemen, Matthias Pfeffer, Leevi Saari, Fabian Ferrari, and van Dijck, Jose. 'Time To Build A European Digital Ecosystem', 9 December 2024. <https://feps-europe.eu/wp-content/uploads/2024/12/Time-to-build-a-European-digital-ecosystem.pdf>.

Claire Stolwijk, Matthijs Punter, Paul Timmers, Julian Rabbie, and David Regeczi. 'Towards a Sovereign Digital Future - the Netherlands in Europe'. TNO, February 2024.

DIGITALEUROPE. 'The EU's Critical Tech Gap: Rethinking Economic Security to Put Europe Back on the Map', 2024. <https://www.digitaleurope.org/resources/the-eus-critical-tech-gap-rethinking-economic-security-to-put-europe-back-on-the-map/>.

European Commission, Directorate-General for Internal Market, Industry, Entrepreneurship, Milan Grohol, and Constanze Veeh. *Study on the Critical Raw Materials for the EU*. Publications Office of the European Union, 2023. <https://data.europa.eu/doi/10.2873/725585>.

Floridi, Luciano. 'The Fight for Digital Sovereignty: What It Is, and Why It Matters, Especially for the EU'. *Philosophy & Technology* 33, no. 3 (1 September 2020): 369–78. <https://doi.org/10.1007/s13347-020-00423-6>.

Henning Kagermann, Karl-Heinz Streibich, Katrin Suder. 'Digital Sovereignty. Status Quo and Perspectives'. acatech IMPULSE, 12 April 2021. <https://en.acatech.de/publication/digital-sovereignty/>.

Kutter, Christoph. 'RISC-V: An Open Standard for Chip Development'. acatech IMPULSE, 4 March 2024. https://doi.org/10.48669/ACA_2024-5.

Prins, Corien, Haroon Sheikh, Erik Schrijvers, Eline de Jong, Monique Steijns, and Mark Bovens. 'Mission AI. The New System Technology'. The Netherlands Scientific Council for Government Policy, 2021. <https://english.wrr.nl/publications/reports/2021/11/11/summary-mission-ai>.

Sheikh, Haroon. 'European Digital Sovereignty: A Layered Approach'. *Digital Society* 1, no. 3 (18 November 2022): 25. <https://doi.org/10.1007/s44206-022-00025-z>.

Spain and Netherlands Governments. 'Non-paper on Strategic Autonomy While Preserving an Open Economy'. Publicatie. Ministerie van Algemene Zaken, 25 March 2021. <https://www.rijksoverheid.nl/documenten/publicaties/2021/03/25/spain-netherlands-non-paper-on-strategic-autonomy-while-preserving-an-open-economy>.

Terzi, Alessio, Monika Sherwood, and Aneil Singh. ‘European Industrial Policy for the Green and Digital Revolution’. *Science and Public Policy* 50, no. 5 (1 October 2023): 842–57. <https://doi.org/10.1093/scipol/scad018>.

Timmers, Paul. ‘Debunking Strategic Autonomy’, 23 July 2021. <https://directionsblog.eu/debunking-strategic-autonomy/>.

Timmers, Paul. ‘Digital Industrial Policy for Europe | CERRE Report’. CERRE, 12 December 2022. <https://cerre.eu/publications/digital-industrial-policy-for-europe/>.

Timmers, Paul. ‘How Europe Aims to Achieve Strategic Autonomy for Semiconductors’. *Brookings Tech Stream*, 9 August 2022. <https://www.brookings.edu/techstream/how-europe-aims-to-achieve-strategic-autonomy-for-semiconductors/>.

Timmers, Paul, Matthijs Punter, and Claire Stolwijk. ‘Cybersecurity and Digital Sovereignty - Bridging the Gaps’. TNO, 2024. <https://publications.tno.nl/publication/34643188/DvSKsfCM/timmers-2024-cybersecurity.pdf>.

Tocci, Nathalie. ‘European Strategic Autonomy: What It Is, Why We Need It, How to Achieve It’. IAI Istituto Affari Internazionali, 24 February 2021. <https://www.iai.it/en/pubblicazioni/european-strategic-autonomy-what-it-why-we-need-it-how-achieve-it>.

Annex A – Strategies of Dominance Among Selected Big Tech

This section outlines the main elements of the business strategies of the key dominant companies of the Digital Stack. These strategies represent a blurring of traditional boundaries—between hardware and software, platforms and infrastructure, consumer and enterprise markets, and even national borders. Their integrated ecosystems drive considerable innovation, while effectively rewriting market rules, accumulating and leveraging data at scale, and entrench high switching costs. Such concentration has sparked intensifying calls from regulators, policymakers, and civil society to better understand and govern the multifaceted power these firms wield.

Amazon holds a dominant position in cloud infrastructure, e-commerce, logistics, and AI. As a key first-mover in cloud computing with **Amazon Web Services (AWS)** holding over 30% of the global market, the firm bundles raw compute with specialised chips, storage solutions, machine learning frameworks (e.g., Amazon SageMaker), and integrated dev-ops tools. This entanglement makes migrating workloads away from AWS highly costly and complex, establishing strong switching barriers. AWS's privileged position also enables Amazon to draw strategic insights into broader market trends, influencing where and how it invests in capacity. In e-commerce, Amazon's command over fulfilment, warehousing, and last-mile delivery not only streamlines costs internally but also sets stringent terms for third-party sellers who depend on Amazon's logistics infrastructure. By controlling both the marketplace and the rules governing it, Amazon can subtly prioritise its own private-label products and services, extracting rents and accumulating data that informs its advertising and retail strategies.

Microsoft wields a similar infrastructural dominance hinged on Windows, which has commanded around 70% of the global market share for desktop OS for decades.²⁰² This is integrated with productivity tools, cloud services, and AI. Azure, with 24% of the global cloud market,²⁰³ integrates seamlessly with Office, Teams, and other products, embedding its services into business operations. Its partnership with OpenAI enhances AI capabilities across its ecosystem, with advanced models like GPT-4 integrated into Azure and productivity tools such as Word and Teams. Microsoft's investments in data centres and custom AI chips ensure it meets growing demand while solidifying its control over critical layers of the stack.

At the core of **Google's** model is its eponymous search engine, which commands roughly 90% of the global market.²⁰⁴ By funnelling the immense volume of search queries into Google Ads, complemented by data from its video-platform YouTube, the company refines its targeting algorithms to deliver highly relevant advertisements, driving both click-through rates and advertiser demand. Moreover, Google leverages its Android operating system, powering over 70% of smartphones,²⁰⁵ to integrate services like Search, Chrome, and the Play Store, embedding its ecosystem into billions of devices and consolidating control over app distribution and user data. Beyond mobile, Google dominates in cloud and AI with platforms like Google Cloud and TensorFlow, embedding its technologies into business operations. Advanced AI models like Bard address disruptions to its core search business, which commands 90% of the market. Google's dominance in digital advertising, driven by Google Ads and YouTube, is fuelled by data collected across its ecosystem, powering targeted advertising, and reinforcing its position across the stack.

²⁰² StatCounter, 'Desktop Operating System Market Share Worldwide', accessed 12 January 2025, <https://gs.statcounter.com/os-market-share/desktop/worldwide/>.

²⁰³ Synergy Research Group, 'Cloud Market Gets Its Mojo Back; AI Helps Push Q4 Increase in Cloud Spending to New Highs'.

²⁰⁴ StatCounter, 'Search Engine Market Share Worldwide', accessed 12 January 2025, <https://gs.statcounter.com/search-engine-market-share>.

²⁰⁵ StatCounter Global Stats, 'Mobile Operating System Market Share Worldwide', accessed 12 January 2025, <https://gs.statcounter.com/os-market-share/mobile/worldwide>.

Apple's tightly controlled ecosystem integrates hardware such as the iPhone, iPad, and Mac with its operating systems (iOS, macOS) and services like the App Store and iCloud. This vertical integration ensures a superior user experience and fosters strong customer loyalty while creating significant barriers for competitors. Apple's exclusive control over the App Store allows it to impose terms on third-party developers, including commission rates of up to 30%, extracting substantial financial rents and further reinforcing its market power.

Meta extends its dominance in social media by monetizing user data and integrating open-source AI models like Llama to attract developers and enhance its services. Its investments in virtual reality aim to complement its platforms, creating a deeper integration between social media and immersive technologies. Meta's control over undersea cable infrastructure further supports its global reach, reinforcing its influence across digital ecosystems.

Nvidia, the leader in GPU chip design, builds on its dominance with an integrated software ecosystem, such as CUDA, which fosters customer lock-in and enhances AI capabilities. By aligning hardware and software, Nvidia secures its position in the AI-driven computing market.

Tesla leads innovation in electric vehicles through its advanced software ecosystem and AI-powered features like Autopilot, while **SpaceX** leverages Tesla technologies and expands global connectivity through Starlink. Operating over 60% of the world's active satellites, **Starlink** dominates satellite communications.²⁰⁶ Musk's AI venture, **xAI**, increasingly underpins the broader ecosystem, supporting Tesla's autonomous driving goals, exploring broader AI applications. This is fed by data from **X** (formerly Twitter), a social media platform that the business tycoon aims to transform into a hub for communication, payments, and AI services. However, its role in amplifying unregulated or harmful content has drawn scrutiny. This strategy consolidates control across transportation, connectivity, AI, and communication, raising significant concerns about regulatory oversight and the ethical implications of concentrated power in critical technologies.

China also boasts a number of Big Tech, similarly, relying on infrastructural dominance. Chinese firms, supported by China's bold industrial policies and large-scale investment, exemplify the country's pursuit of technological self-reliance and vertical integration, leveraging dominance in core areas to consolidate power across the digital stack.

Originally gaining prominence through telecommunications equipment, **Huawei** has leveraged its infrastructure expertise to expand into consumer electronics, cloud computing, AI, and domestic chip manufacturing—often with support from state-led industrial policies. In telecommunications, Huawei's dominance grew through cutting-edge 5G technology and turnkey solutions offered to telecom carriers globally, embedding its equipment deep into the world's communication networks. Beyond telecom, Huawei broadened its portfolio to include smartphones, tablets, and laptops, powered by its own HarmonyOS operating system and Kirin chips developed by subsidiary HiSilicon. The US-led restrictions on semiconductor supply chains prompted Huawei to double down on domestic R&D and chip fabrication capabilities, a strategy fully endorsed by Beijing's push for technological self-reliance. Today, Huawei's cloud platform ties together AI accelerators, data analytics tools, enterprise services, and IoT frameworks—effectively replicating the global cloud ecosystem within China's "walled garden" of tech. Through these moves, Huawei positions itself as a vertically integrated, end-to-end solutions provider, spanning critical infrastructures from network layers up to consumer services.

Initially celebrated as an e-commerce pioneer, **Alibaba** has evolved into a sprawling conglomerate that intertwines digital commerce, cloud computing, fintech, logistics, and media. At the heart of its ecosystem lies the Taobao/Tmall marketplaces, which handle a staggering volume of transactions and generate data streams that inform the development of Alibaba's other businesses. Alibaba Cloud, the country's leading cloud provider, offers compute, storage, AI services, and database solutions—

²⁰⁶ Werner, 'Want to Challenge Starlink in the Satcom Market?'

critically underpinning an ecosystem where merchants rely on Alibaba's infrastructure for payment processing (Alipay), supply chain management, and customer analytics. Over time, Alibaba has added components like Cainiao (a logistics network that integrates delivery partners and warehouses) and has invested in entertainment content (Youku, Alibaba Pictures), education tech, and even enterprise software. Each segment feeds into the other: the cloud business profits from the enormous traffic and data generated by e-commerce and financial services, and the marketplace benefits from the cloud's capacity, scalability, and analytical insights. Together, these integrations create a near-seamless environment for consumers, businesses, and developers, allowing Alibaba to extract rents, shape standards, and reinforce switching costs within China's digital economy.

Tencent initially emerged as a social and gaming powerhouse before branching into infrastructure and enterprise solutions. It dominates social media and messaging with WeChat, which has become an essential "super-app" underpinning daily life in China, integrating chat, mobile payments (WeChat Pay), mini-programmes, e-commerce, and increasingly, business services. This ecosystem, by design, captures comprehensive user data, enabling granular personalisation and targeted advertising. In parallel, Tencent's gaming portfolio—encompassing both domestic titles and global investments—fuels revenue streams and provides a testing ground for cutting-edge graphics, network scaling, and AI-driven matchmaking. Tencent's expansion into cloud computing and AI leverages this platform advantage, offering solutions to enterprises that want to tap into its vast user base and computational power. Beyond IT infrastructure, Tencent has stakes in various digital industries—fintech, music streaming, video platforms, healthcare—and actively invests in start-ups that align with its ecosystem strategy. The result is a multifaceted digital empire that not only sets rules for content distribution, social interactions, and commerce but also acts as a critical layer of digital infrastructure for other businesses.

Known as "China's Google," **Baidu** built its reputation around search, advertising, and mapping services before broadening its scope into AI, autonomous driving, and cloud solutions. Dominant in China's search market, Baidu's algorithms continuously refine user intent and data mining capabilities, providing an advertising backbone that funds R&D in emerging technologies. Building on its robust data analytics and natural language processing expertise, Baidu launched Baidu Brain and a suite of AI platforms that contribute to industries like healthcare, finance, and urban planning. Baidu's Apollo platform leads the domestic autonomous driving initiative, supported by extensive real-world data gleaned from its mapping and location-based services. In addition, Baidu Cloud offers enterprise-level solutions—ranging from machine learning frameworks to IoT services—leveraging Baidu's research labs and AI know-how. This stack of integrated products and services ensures that Baidu's algorithms and data are not confined to search alone but stretch across a range of applications that increasingly touch everyday life and infrastructure, forming a virtuous cycle that deepens lock-in and broadens its influence.

Best known internationally for TikTok, **ByteDance** has parlayed the success of short-form video and sophisticated recommendation algorithms into a broad-based digital ecosystem, including Douyin (the Chinese counterpart to TikTok), Toutiao (a leading content aggregator), and other niche apps that cater to various verticals. Drawing on massive user engagement and granular data collection, ByteDance's AI recommendation engines customise content feeds with uncanny accuracy. This sophisticated personalisation has proven easily transferable to new domains: ByteDance invests in ed-tech, enterprise software, and e-commerce, integrating shopping features into its short-video platforms and branching out into livestreaming sales. Meanwhile, the firm's entry into cloud services and enterprise AI tools builds on its algorithmic expertise, effectively turning its consumer-facing successes into a foundation for B2B solutions. ByteDance thus evolves into both a media and tech infrastructure player, powering new digital business models and reinforcing its role as a standard-setter for content discovery, user engagement, and targeted advertising.

Annex B – Ongoing EU Policy actions and lead stakeholders

The EuroStack flagship will build on ongoing policy actions and initiatives in Europe such as those listed below.

Semiconductors.

- Accelerating development of **2nm and below technologies** critical for AI and HPC, targeting 20% of global chip production by 2030, **up** from 10% (EU Chips Act / EC + industry)
- **Aligning procurement** with EU regulatory frameworks, e.g., GDPR, AI Act, Green Deal, to ensure compliance, incentivize innovation, and support sustainability (EU Chips Act / Member States).
- Focusing on **AI-capable and energy-efficient chips**, aligning with sustainability goals and supporting Europe's leadership in strategic industries - Chips Act / EC + Industry + Joint Undertaking).
- Integrating semiconductor initiatives with the **Green Deal**, ensuring sustainable digital transformation (EU Chips Act / EC).
- Advancing a comprehensive strategy for chip design, production, and ecosystem collaboration, leveraging European strengths such as in **EUV lithography** (EU Chips Act / Joint Undertaking).

IoT and Connected Devices

- Focusing on IoT and connected devices in high-impact sectors such as smart cities and healthcare, which alone are expected to drive **€250 billion in IoT economic value in Europe** by 2028 (Horizon Europe and Digital Europe Programme / EC).

Network Layer

- Leveraging EU funding, including **STEP and Cohesion Funds**, to scale deployment (EU R&D and Cohesion Programmes / Industry + Member States).
- Advancing the work on open standards for standalone 5G networks to avoid vendor lock-in and ensure compatibility with edge computing and quantum-secure systems (EU Standardisation / Industry and MS in ETSI+ GSMA + MS in ITU).
- Replacing copper networks with fiber, reducing energy consumption by up to **85%** (private sector funding / industry).
- Building redundancy and resilience in critical infrastructure using EU funding such as from Connecting Europe Facility and in line with the forthcoming **Digital Networks Act (DNA)** (DNA, CEF / EC + Industry).

Cloud Layer

- Targeting high-growth sectors with edge cloud such as **healthcare and industrial manufacturing** to demonstrate operational efficiencies (IPCEI / MS + Industry).
- Encouraging edge-native applications to reduce latency by **up to 50%** and improve resilience by decreasing reliance on centralized infrastructure (IPCEI / MS + Industry).
- Focusing on critical sectors such as **healthcare, public safety, and transportation**, where decentralized solutions can enhance response times and reduce operational risks (IPCEI / MS).
- Aligning with EU initiatives for **data sovereignty and digital interoperability**, critical for enhancing cross-sector collaboration (forthcoming EU Cloud and AI Act / EC).

Open Software Industry Ecosystem

- Rolling out a unified **Digital Identity Wallet** by **2025**, enabling seamless and secure access across public services and digital layers (eIDAS2 Act / EC + MS + Industry).

AI and Data

- Promote adherence to EU digital rights principles, including transparency, accountability, and privacy, to strengthen public trust (EU AI Act, Data Act, Data Governance Act, Open Data Act / EC).
- Focus on strategic applications in healthcare, mobility, and public services while ensuring data remains secure and localized (EU funding programmes / EC).

Defence

- Foster technology transfer and align funding with dual-use technologies that support both defence and civil needs (Horizon and European Defence Fund / EC + EDA).